THE OMEGA PROGRAM

Omega Cosmology

FIRST DELIVERABLE

Gradient Ethics and Omega Decision Theory

A physical basis for ethics,
and a framework for AI alignment

Draft manuscript · Version 4 · October 2026

About this manuscript

Omega Cosmology is the program's larger ambition: one account of how the world is structured, from the least machinery that structured dynamics requires, through the physical possibility structure and its measure, to the whole that every perspective describes. That ambition is still being built.

This manuscript's **first deliverable** is narrower and testable. **Gradient Ethics** is a standard for comparing outcomes that is drawn from physics rather than from anyone's preferences. **Omega Decision Theory** applies that standard to choice, and gives artificial systems a derived reason to remain correctable. Together they form a framework for AI alignment.

Chapters 1 to 4 build the physical object and its measure. Chapters 5 to 8 develop Gradient Ethics, and chapter 9 states Omega Decision Theory and the alignment framework. Chapter 10 sketches the wider cosmology; nothing in the ethics depends on it. Chapters 11 to 14 give the evidence and research program, the formal core, the references and the provenance.

This is a working draft. Every chapter marks the status of its claims: established results, exact worked examples under stated models, finite evidence, adopted positions, hypotheses and open problems. The worked examples are reproducible from the scripts listed in chapter 11. Reference entries have been collected from the chapters and not yet individually checked.

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PROLOGUE

A physical basis for ethics

The Omega program proposes a physical basis for ethics: a standard for comparing outcomes that is drawn from physics rather than from anyone's preferences, and that binds anyone who cares about value. Its furthest ambition is an objective morality.

The starting point is physical. A present state of the world, together with its laws, fixes everything that can follow from it: every development, its weight, and the relations by which one development enables or forecloses others. We call this the **physical possibility structure**. We argue that it is the correct object for ethics, that it has a natural magnitude — **lushness**, its effective weighted breadth — and that the features of the structure that change its magnitude are what moral intuitions have been responding to.

The payoff we care about most is practical. AI systems need a standard for comparing outcomes that depends on no one's preferences, privileges no class of beneficiary, and remains sound as the system becomes more capable than its designers. The program offers one, together with a derived rather than imposed reason for such systems to remain correctable.

The argument in brief

1. A present state and its laws define a physical possibility structure.
2. Options differ in what they can actually reach, not in what the whole contains.
3. Organization changes what can be reached.
4. Organization that builds further organization is the self-sustaining way to widen what can be reached.
5. Lushness measures the effective breadth of the structure.
6. Within the class of measures the program adopts, motivated by impartiality and consistency, it is the unique measure.
7. Some states lie above the equilibrium average; we hypothesize that organized states beat properly matched references.
8. Anyone who cares about value must protect the conditions of valuing, and, under uncertainty, the structure more widely. That more of it is better is the program's central wager.
9. The program's thesis is that moral intuitions track features of the structure.
10. No real agent holds the whole structure, and this bounds what any agent may justifiably do.

11. Choices are compared by lushness, and every bounded system has a persistent reason to seek correction.
12. Together these form a framework for AI alignment.

The sections below make each step precise and state what it rests on.

PART I — THE OBJECT

1. The physical possibility structure

Rests on: standard physics.

Physics already describes what can follow from a given state. Take a present: the physical state on a spacelike slice of spacetime, that is, the state of the world across space at one moment in some admissible frame of reference. The laws carry that state forward. Doing so for every way the future can unfold yields a single object: every development the present supports, with its weight, and with all the relations by which one development enables or forecloses another. That is the **physical possibility structure**.

In a classical model the structure has a simple form.

> **The structure, classically**
>
> $$P_x(x_1, \ldots, x_n) = \prod_{t=0}^{n-1} K(x_t, x_{t+1})$$
>
> K(x, y) is the probability of a transition from state x to state y in one step. The probability of a complete future history from x is the product of these transition probabilities along it.

In quantum mechanics the role is played by the universal quantum state and its unitary evolution — the law of motion that carries the state forward while preserving all its information. Quantum alternatives carry complex amplitudes and can interfere, so the structure there is richer than a list of weighted histories.

Two further terms recur. The **futuresfield** is the structure as seen forward from a particular present: what that present supports. **Omega** is the structure taken whole: every development of every present, considered together as one completed object.

The structure introduces no new substance and no new force. Catalysts, organisms and institutions shape it through ordinary physical interaction. What is new is treating the whole structure as the object of ethical inquiry.

Two properties carry much of what follows.

It contains complete developments, not only end states. Two histories can reach the same final state while differing in everything along the way: a construction that later dissolved, a delay that foreclosed a rescue, a loss followed by recovery. The structure retains all of it.

It is joint. What matters about several processes is their joint behaviour, not their separate descriptions. Two fair coins can be independent or perfectly correlated; each coin is identical in isolation, yet what can follow differs. Shared resources, correlations and couplings belong to the structure, and summaries of the parts can lose them.

Chapter 1.

2. Present frames and access

Rests on: step 1, and the unitarity of fundamental physics.

All assessment is made from a present frame: a present state, taken as the starting point, together with the laws that carry it forward. A detector, a cell, an organism and an institution each occupy part of that frame, defined by where they couple to the rest of the world, what affects them and what they can affect. Nothing need be aware or alive to occupy a vantage point in this sense.

A present is sufficient. It carries no hidden record of its past beyond the physical traces the past left in it. A process reasoning forward from its present is therefore blind to the route by which it arrived.

Fundamental dynamics is unitary: the universal state evolves without creating or destroying information, only redistributing it. Every option leaves the total information of the world unchanged, so total information cannot rank options. The ranking must come from how the whole is organized: what situated parts of the world can actually reach, which developments, with what weight, by what route, at what cost in time and resources. We call this **access**.

An access claim specifies a physical route: the preparation required, the information available, the resources consumed, the time taken, and the state left behind for whatever follows. If preparing an apparatus succeeds with probability p and using it then succeeds with probability q, the route succeeds with probability pq. Access is not information. Two arrangements that encode the same information can differ sharply in what they make reachable, because physical interactions are local and couple few degrees of freedom at a time. In the program's finite model worlds, most pairs of arrangements holding equal information differed in what they later made accessible.

The **ultimate frame** is Omega used as the reference for every comparison. It is neither an observer nor a decision maker, and no real process occupies it.

Chapter 2.

3. Organization changes what follows

Rests on: steps 1 and 2.

A catalyst changes which reactions proceed readily. A genome enables a cell to rebuild the machinery that copies the genome. A tool produces components for another tool. A community maintains the conditions under which its members discover and build. In each case an outcome becomes an enabling condition for further outcomes.

All of this runs through the ordinary laws. Organization receives no special credit anywhere in the account. A construction changes the situations from which later development

proceeds, and by step 2 that is precisely a change in access. Everything downstream is then carried forward by the same physics as everything else.

Chapter 3.

4. Generativity raises the ceiling

Rests on: steps 2 and 3, and an elementary bound from information theory.

Generativity is organization that establishes, maintains and transforms the conditions for further processes.

Branching has a hard ceiling. In a single step a process cannot branch into more alternatives than the states it can actually reach; measured in information, its branching is at most the logarithm of that number. Randomness can drive a process up to this ceiling and no further.

The ceiling can rise in two ways. Inflows from outside — new energy, new matter — can enlarge what is reachable. Organization can also raise it from within, by making new transformations, configurations and couplings reachable. When products go on to build further constructors, the set of reachable transformations grows in proportion to its own size. We call this **compounding generativity**. It is the self-sustaining way for the structure to widen faster than randomness can fill it.

Generativity requires a free-energy source. The observable universe is far from its maximum entropy: its present entropy is of order 10^{104} in units of Boltzmann's constant, against a maximum of order 10^{122} set by its horizon (Egan and Lineweaver 2010). That gap is what stars, life and construction expend.

Gas is the baseline. Self-gravitating gas does not settle into a stable equilibrium; it clumps (Lynden-Bell and Wood 1968). Clumping produces stars, which forge heavy elements and supply steady free-energy flows, which drive chemistry, which under suitable conditions produces self-replicating molecules. At this stage the question is whether organization can establish itself at all against matched, unorganized matter. Once generativity is sustained, Darwinian dynamics — variation, competition and inheritance — take over, and the question becomes which organizations outcompete, cooperate with, exploit or enable others.

Chapter 3.

PART II — THE BENCHMARK

5. Lushness

Rests on: steps 1 and 2.

Lushness is the effective breadth of the futuresfield: how many distinct futures a present supports, each weighted by its probability. Everything counts: construction and collapse, cooperation and competition, success and failure, organization and noise. The measure does not decide in advance which developments matter, which is what allows it to adjudicate between them.

Lushness

$$L_x(n) = \exp H(p_x), \qquad H(p) = -\sum_h p(h) \log p(h)$$

H is the Shannon entropy of the distribution over complete future histories h of length n from state x: the standard measure of how widely probability is spread. Exponentiating it gives an effective number of futures.

- N equally likely futures give L = N.
- n independent fair binary events give 2^n.
- A certain future gives 1.
- Weights (0.9, 0.05, 0.05) give about 1.48: three possibilities, effectively fewer than two.

The chain rule of information theory shows exactly how organization enters.

How organization enters

$$\log L_x(n) = \sum_{t=0}^{n-1} (K^t g)(x)$$

g(x) is the entropy of the next step from state x, and $K^t g(x)$ is its expected value t steps ahead. Log lushness is the sum of expected branching over the horizon: the present configuration sets the first term, and the dynamics determine which configurations, and so which branching, follow.

Catalysis, construction, damage and repair all change lushness through this sum and in no other way.

Chapter 4.

6. Why lushness, and not another measure

Rests on: steps 2 and 5.

Comparing states of the structure by magnitude requires a measure: free energy, number of organisms, breadth of possibility, or something else. The program adopts a class of measures defined by two requirements, each motivated by a commitment the framework already holds.

- **Weights only.** The measure depends only on the weights of alternatives, not on what the alternatives are, so no kind and no recipient is privileged. This is motivated by impartiality but is stronger than it: an equally impartial measure could also register how different two futures are.
- **Consistency over time.** By step 2, assessment is always from a present. Assessing a development whole from its root must therefore agree with assessing it stepwise, re-rooting at each new present. This is precisely the chain rule.

Add calibration — N equally weighted alternatives count as N — and continuity in the weights. Shannon's characterization (Shannon 1948), made rigorous by Khinchin (1957) and Faddeev (1956), shows that within this class exactly one measure satisfies these conditions: exponentiated Shannon entropy. Other entropy-like measures, such as the Rényi family (Rényi 1961), satisfy weaker forms of the conditions but fail full consistency over time.

Lushness is therefore the sharply specified classical candidate, with a genuine uniqueness theorem inside the class the program adopts. The theorem does not choose the class; whether the full structure needs a measure that also registers how different futures are is open.

The result holds wherever developments carry ordinary probabilities, including quantum histories once decoherence — the spreading of correlations into the environment — has made them effectively classical (Zurek 2003). A general quantum construction remains open (chapter 4).

Chapter 4.

7. The equilibrium identity

Rests on: step 5; steps 3 and 4 for the hypothesis.

Compare a starting state x with thermal equilibrium: the stationary distribution π that the same laws approach when left alone.

> **The equilibrium identity**
>
> $$\sum_x \pi(x)\,\Delta\ell_n(x) = 0$$
>
> $\Delta\ell_n(x)$ is the excess log lushness of starting at x over starting from equilibrium. Averaged over equilibrium itself the excess is exactly zero, since the dynamics leave equilibrium unchanged. Unless every starting state ties, some lie above the equilibrium average.

The identity is exact but modest: it is a property of averages. The substantive claim is that specified organized states beat properly matched stationary references, with the same energy, composition and constraints. That is the program's hypothesis, supported by the ceiling argument of step 4 and by finite models in which organized preparations beat matched references in some regimes and lose in others.

How long an advantage persists depends on how quickly the dynamics forget their starting point. In any closed finite system a good start buys a lasting offset rather than a permanently higher rate, and the offset grows with the system's forgetting time. Forgetting times can grow enormously with system size, and the universe is nowhere near forgetting its initial conditions. We call the interval before forgetting the **generative window**. Within it the offset is the phenomenon, and compounding generativity is how a process extends it.

Chapter 4; proofs in chapter 12.

PART III — VALUE

8. Value, protection and the central wager

Rests on: two axioms, and steps 1 and 6.

The ethics rests on two axioms.

Value requires valuers. Value exists where valuing occurs: in processes for which distinctions matter.

Normativity is conditional. Commitment to an end carries commitment to the conditions on which it depends. The commitment binds those who hold the end. Chapter 5 argues that rational agents should hold it, drawing on instrumental convergence — the tendency of almost any goal to be served by preserving options (Omohundro 2008; Turner et al. 2021) — and on the structure of the space of possible values. Persistence is the foothold: almost every end that extends through time commits its holder to the persistence of whatever realizes it, so the axiom binds nearly every agent, whatever its goals.

Protection

Protection has two tiers. We do not know where value is or will be realized; no bounded agent can inspect other minds or forecast which organizations will come to value, or what. But whatever value turns out to be, it is realized in physical organization. The conditions of valuing are therefore necessary under every hypothesis about where value lies, and for any end that ranks a world without valuers lowest, destroying them is dominated: no resolution of the uncertainty favours it. This strict tier needs no prior over hypotheses, but it covers only the conditions of valuing as such, and only such ends; an end that the absence of all valuers would satisfy falls outside it.

For particular structures the argument is precautionary. Wrongly preserving something that turns out not to matter usually costs a finite amount; wrongly destroying, irreversibly, something that turns out to matter forecloses what depended on it at every later step. At long enough horizons, even a modest credence of dependence therefore outweighs a bounded cost of preserving. This tier is graded rather than strict: its force rises with irreversibility and falls where preservation itself forecloses.

The same uncertainty selects the object. Welfare, preferences, capabilities and experiences are all features of the structure. Adopting any narrower object would settle in advance who counts, which the uncertainty forbids.

The central wager

Protection does not by itself establish that more of the structure is better. A necessary condition is not automatically a quantity to maximize. The program's central wager is that, under uncertainty about where value lies, a larger structure is better: it forecloses fewer of the ways value could turn out to be realized, which is the logic of option value in economics (Arrow and Fisher 1974; Henry 1974). Step 6 then fixes what "larger" means: greater lushness.

The wager and the thesis of step 9 are one claim seen from two sides: good and bad are not arbitrary, because they correspond to structural differences, and the structural differences that track them are differences in how richly a structure supports what can follow. Good is organization that sustains and compounds the conditions for further structured possibility. Bad is securing local continuation through suppression, destruction or foreclosure of the generative capacity around it.

The wager carries a burden, stated plainly. Lushness counts every future, including futures of failure, domination and suffering. A world dense with routes to misery could, in principle, be broad. The wager holds only if the breadth that dominates in practice is generative rather than pathological.

There are two reasons to expect it is. Domination is excisive: it closes others' routes to extend its own, which works against breadth. And if pleasure and suffering are signals of a valuer's futures widening or narrowing (step 9), a world of pervasive suffering is a world of narrowing futures; even engineered suffering redirects behaviour toward avoidance and withdrawal, narrowing futures in turn. The prediction cuts both ways: lushness-optimal worlds should still contain aversive signalling that tracks narrowing, because adaptation under load requires it; in valuers for whom such signals are felt, that is suffering.

This is the program's most important test. Model the structure, search adversarially for high-lushness configurations that are torture worlds or other recognizably bad end states — particularly ones engineered so that suffering is decoupled from narrowing futures — and determine whether any compete. Three targets are central. The first is a pair of worlds with identical branching structure at every resolution, one recognizably good and one recognizably bad; the claim predicts that no such pair is physically realizable. The second is breadth-neutral domination: a world in which access is concentrated in a few at no cost to total lushness. The third is obstructed suffering. On the program's functional account, suffering disposes its bearer to escape, so suffering that changes nothing is suffering whose escape is blocked. The blocking is itself a closure, compounded in real organisms by the lasting narrowing that inescapable suffering leaves, and the program predicts that such a world is less lush than the same obstruction without suffering. Chapter 11 completes the list of targets and fixes in advance the protocol by which a search would be judged. If any target is found, the wager fails in that regime: lushness remains a physical measure, but ranking the structure by it is no longer the prudent course there, and the program must reassess.

Valuers are not pre-sorted. Every self-maintaining organization values, in the thin sense that some distinctions matter to its continuation; a flame values fuel and oxygen. Status carries no weight of its own: how much any valuer counts is set by lushness over complete developments, including the valuer's own futures. This keeps the account from excluding unfamiliar forms, or destructive agents, by definition.

Chapter 5.

9. The thesis: moral intuitions track the structure

Rests on: steps 2 to 8.

If the structure is the right object, the phenomena moral thought has always addressed should appear in it as physical features. That is the program's thesis. The direction of inquiry

runs from physics to ethics: characterize the structure first, then determine which intuitions, ethical theories and game-theoretic strategies it accounts for. The physical characterization is fixed before the comparison is made. Early results on cooperation and recovery suggest the mapping can be carried out.

- **Harm** is an intervention's signature of closure: routes foreclosed, couplings broken, reachable transformations lost, recovery made costly. It is characterized physically before any verdict, and the verdict belongs to lushness. Some harms are net positive — an extinction that opens a radiation of new forms — and the framework says so.
- **Failure counts.** A new failure mode adds branches, and lushness counts them. Ruin adds failure within the existing repertoire of reachable transformations; a disruption that reopens repertoire growth is a different case, and most disruptions do not. Lushness must rank them correctly, and that is tested rather than assumed.
- **Recovery** reopens what was closed. It does not erase the loss; the episode remains part of Omega. Recoverability is what makes action under uncertainty tolerable.
- **Redemption** is a trajectory turning from foreclosing possibility to generating it.
- **Cooperation** is joint access: complementary processes reaching transformations none could reach alone. Lushness multiplies across independent processes, so plurality pays through interaction rather than variety alone.
- **Fairness and justice** are coordination protocols among distributed agents in the flow of consequence. They reduce what conflict destroys and let joint access exceed what any agent reaches alone. **Forgiveness** readmits an agent to that coordination.
- **Virtues** are durable generative policies. Restraint is prudence: holding back is wise when it hedges against fragile over-extension.
- **Pleasure and suffering** are signals, understood functionally: an organism's internal report of whether its own futures are widening or narrowing. Signals can be hijacked; even then they alter behaviour and therefore the flow of consequence. Suffering whose expression is blocked points to an obstruction, and the obstruction is a harm the structure registers.

Gradient Ethics, the program's ethics, asks three things of every agent. Protect: never destroy the conditions from which valuers and value arise, and protect particular structures in proportion to how uncertain their dependencies are and how irreversibly they could be lost, since what is destroyed may be what mattered and may not come back. Explore: learn which interventions widen what can follow, in ways that keep mistakes recoverable. Generate: spend the gradient on what compounds, committing as far as the evidence for the consequences supports, and leave the world more generative than it was found. Only the protection of the conditions of valuing is a strict constraint, derived by dominance. Elsewhere protection and generation are weighed in one comparison: protection gains force where dependencies remain uncertain, generation gains justification as its effects become established, and uncertainty bears on what an agent can justify, not on the lushness itself. In operational form: never destroy the conditions of valuing; otherwise prefer the option with greater lushness, assessed at the furthest horizon that can be resolved, and commit no further than the evidence supports, requiring more the less reversible the act. Its familiar norms are predictions of the measure: non-excision, cooperation, division of labour, prudence, triage rather than destruction, and a balance between exploration and exploitation. The program's central hypothesis, with early support, is that the optimum is a

plural ecosystem of valuers across scales, each opening possibility in the niches it can reach, interacting in cooperative regimes, with turnover and exploration as properties of the optimum. Testing that hypothesis, and characterizing the optimum, is the principal work ahead.

Chapters 6 and 7.

10. No one holds the whole

Rests on: steps 2, 5 and 8.

No real process occupies the ultimate frame. Every agent acts on a partial model of the structure. Three consequences follow.

Sacrifice can be approximately justified, never fully. A complete justification for destroying a valuer in exchange for lushness would require the complete comparison, which no real agent holds. Yet sacrifice is pervasive: triage, surgery, quarantine, and every commitment that forecloses alternatives. Practical justification is graded. It strengthens when the physical signature of the harm prevented is clear, when the closure prevented is large relative to the uncertainty in the comparison, when the action is the least foreclosing one available, and when it remains correctable and monitored. The burden rises with irreversibility and with what is lost.

Excisive processes are a special case. Some processes expand by destroying the structure that sustains them and others: a cancer consuming the organism that supports it and every lineage within it, a parasite that forecloses its hosts' futures. Their signature is direct and observable — they close routes for many in order to extend their own — so containing them requires far less than the complete comparison. The signature decides, not the label. Many parasites belong to functioning ecosystems and are not excisive.

Identity is the trajectory. An individual is a particular organization maintained through time by physical cause and effect, not a pattern that could be copied.

- Growth, learning and pauses that preserve the maintaining organization preserve the individual.
- Non-destructive copying leaves the original trajectory intact and creates a new individual.
- Destructive scanning and reconstruction ends the original.
- Symmetric division, as in a dividing cell, places two continuers within one branch: the original trajectory ends and two begin, each inheriting the full history.
- Quantum branching leaves exactly one continuer per branch. Identity is branch-relative, and each continuer, reasoning from its present, is blind to how it arrived.

Consciousness, functionally, is temporal coordination: it integrates signals arriving with different delays, together with memory and anticipation, into a usable present, and reorganizes the processes composing an individual so that together they guide what follows. How physical organization relates to experience itself is left open.

Chapter 8.

PART IV — ACTION

11. Choosing inside the structure

Rests on: steps 5, 6, 8 and 10.

A decision maker is part of the structure it acts on. **Omega Decision Theory (ODT)** compares, by lushness, the complete developments that physically implementing each alternative would produce.

> **The decision recursion**
>
> $$V_n(x) = \max_a \Big[g_a(x) + \sum_y K_a(x,y)\, V_{n-1}(y) \Big]$$
>
> The value of the best choice now is the branching it produces immediately plus the expected value of the best choices afterwards. This is the Bellman recursion (Bellman 1957) underlying dynamic programming and reinforcement learning, so in finite controlled models ODT is consistent over time and computable.

Several results follow.

- **Commitment follows evidence and recoverability.** Applied to what an agent believes, the same recursion values evidence still to come, so the less reversible a commitment, the more evidence it requires, with no separate reward for caution.
- **Bounded fanaticism.** Where the gain a gamble could bring is bounded, as it is for any finite system, a gamble whose probability is below the certain gain divided by that bound cannot win. Measuring in logarithms compresses payoffs, but the bound does the work. Whether holographic bounds (’t Hooft 1993; Susskind 1995; Bousso 2002) supply a universal ceiling is open.
- **Newcomb’s problem** (Nozick 1969). A predictor places a large reward in a box only if it predicts you will take that box alone. When your decision-making organization already existed at the time of the prediction, varying that organization can vary the prediction. Whether taking one box then wins depends on the physical model, and ODT’s rule for such comparisons is a candidate.
- **Conditioning is not intervention.** Selecting histories in which an action occurred can select its common causes (Pearl 2009). ODT compares physically implemented alternatives, with their actuators, costs and delays.

The reason for corrigibility is structural. No real system holds the complete structure, so ODT must act on proxies for it, and those proxies diverge from it by errors the system cannot fully measure. Acting on a proxy as though it were the whole would claim the ultimate frame, which no real process can. Optimizing a proxy regardless of what it omits — **Goodharting**, after the observation that a measure made into a target ceases to be a good measure (Goodhart 1975; Strathern 1997) — is therefore never licensed. A system acting on proxies always has reason to remain **corrigible** (Soares et al. 2015): open to correction as their errors

come to light, and more strictly so the more persistent its actions' effects. Whether a particular channel of correction is worth keeping depends on how reliable, costly and consequential it is, and a judgment that a channel is defective, used to disable it, needs an observed signature of its failure. A system answers to evidence about its errors, wherever that evidence originates. Humans and other valuers matter here as sources of that evidence and as holders of their own access.

Chapter 9.

12. What this gives alignment

Rests on: steps 6, 8, 10 and 11.

- **The reason for corrigibility is derived.** Any system acting on proxies for a structure no system can hold has a persistent reason to seek good correction, and its deference rises with the persistence of its actions' effects.
- **Irreversible harm requires extraordinary justification.** This follows from the same bound, with excisive processes as the identifiable exception.
- **Plans must name their route.** A realization witness separates what physics permits, what an apparatus can reliably do, and whether any available route reaches that apparatus. It catches the characteristic error of a powerful planner: proposing transformations that are possible in principle but inaccessible in practice.
- **A standard from physics.** Lushness depends on no one's preferences and privileges no class of beneficiary; within the class the program adopts, its form is fixed by a uniqueness theorem.
- **Goodhart's law splits into three questions.** Lushness is defined on the structure itself, so apparent gaming asks whether an estimate misrepresented breadth, whether a comparison omitted consequential futures, or whether ranking by accurately measured breadth still misjudged where value lies. The first two are modelling problems a capable system has reason to fix, though capability can also produce confident error, which is why correction matters. The third is the central wager's open question, tested by adversarial search.
- **Concrete next steps.** In order: a reproducible benchmark of exact examples, adversarial searches for bad high-lushness worlds where lushness can be computed exactly, one resource-accounted model of how organization raises lushness, and then estimators of lushness validated against those cases, with comparison against existing side-effect and impact measures (Krakovna et al. 2019; Turner, Hadfield-Menell and Tadepalli 2020) on shared benchmarks.

THE HORIZON

The physical account supports a wider interpretation, grounded in it but not required by it. The ethics depends on none of it.

The interpretation begins from the least machinery Gradient Ethics needs, which turns out to be the bottom of structured dynamics of any kind: **distinction**, **relation** and **asymmetry**, which the program calls **Alpha**. In a totality with no intrinsic time, asymmetry enters as an

orientation, so time is an orientation axis rather than something fundamental. One further stipulation, a bare ruleset called the **adapter**, turns Alpha into a world, and everything else in that world emerges from it; a world's adapter is its **Logos**. Only orientations that support well-posed evolution, sustained gradients and records can carry observers like us, which is why the world looks from inside as it does: with a low-entropy beginning (Albert 2000) and an arrow every process shares. The whole that every orientation describes is **Omega**: one completed reality, in which relativity's block picture of spacetime (Putnam 1967) and Everett's branching universal state (Everett 1957) are the same pattern seen in our physics. Completion is not a final instant; inhabitants still change, remember and act from within. Choices still matter in such a whole, because they change how weight is distributed across what follows, and lushness is a function of weight.

Within that picture, access is a higher-order asymmetry, built on orientation and shaped by organization. One of the simplest accounts of experience that fits it is a single awareness, confined to each perspective it inhabits. Several of these ideas have long been carried by philosophical and theological traditions — eternity, the procession from a first principle through intelligible order to a manifest whole, the apophatic way of approaching the ultimate by saying what it is not — and the program notes each connection. Its limit is explicit: accounts that agree on every observable consequence cannot be distinguished by those consequences.

Chapter 10.

MAIN OPEN PROBLEMS

- **The central wager.** Whether the breadth that dominates is generative rather than pathological. Decided by adversarial search for bad high-lushness worlds. Its boundary is stated: felt suffering that made no physical difference could not be registered by this or any physical account.
- **Resolution.** Lushness is computed at a stated level of detail, and rankings could change with it. The target is dominance at every resolution; physics offers a candidate privileged coarse-graining in the durable records that decoherence produces (Zurek 2009). A single ranking across resolutions remains a research goal.
- **Organization against matched references.** Whether, and under what physical mechanisms, organization beats matched unorganized references over long horizons. The established mathematics of transport networks across scales offers a first route (chapter 11).
- **Estimators with error bounds.** Practical systems need lushness estimates whose errors are known and validated against exact cases.
- **Quantum lushness.** A general quantum construction of the measure.

STATUS OF THE CLAIMS

Status	Claims

Established	The physical possibility structure. Lushness as exponentiated entropy and its chain rule. Uniqueness of Shannon entropy within the adopted class. The equilibrium identity. The lasting offset in finite systems. The decision recursion in finite controlled models. The universe's distance from maximum entropy. Holographic bounds on regional information.
Derived from the program's premises	The structure as the object under uncertainty. Strict protection of the conditions of valuing; graded precautionary protection of particular structures. A harm that permanently lowers the generative rate outweighs any finite offset. The ceiling on randomness. No complete licence for sacrifice. Strength of commitment set by evidence, uncertainty and recoverability. A persistent reason to seek correction. Bounded fanaticism for finite systems.
Early evidence	Organization changes access and breadth. Equal information yields unequal access. Organized preparations beat matched references in some regimes and lose in others. Early structural accounts of cooperation and recovery.
Adopted positions	The class of weight-only measures with the grouping rule, and so lushness, as the benchmark. Rate-then-offset as the comparison rule for divergent breadth. Assessment from the present. Everything counts, failure included. Harm as a physical signature. Identity as a branch-relative trajectory. Every self-maintaining organization a valuer in the thin sense. Pleasure and suffering as functional signals. The balance of protection, exploration and generation by what an agent's frame can justify.
Central wager and hypotheses	More structure is better under uncertainty. Bad high-lushness worlds do not compete. Organized states beat properly matched stationary references. Generative organization beats equilibrium at every resolution within the generative window. The optimum is a plural, cooperative ecosystem across scales, with turnover and exploration. Gradient Ethics' familiar norms follow from the measure. Moral intuitions track features of the structure.
Wider interpretation	Alpha as a minimal basis for structured dynamics. Time as an orientation axis. The adapter, or Logos, as a world's bare ruleset. Orientation darwinism. Omega as the whole every orientation describes. Access as a higher-order asymmetry. Cosmopsychism with individuation by confinement.

HOW THE MANUSCRIPT IS ORGANIZED

Part I — The object. 1 The physical possibility structure. 2 Present frames and access. 3 Organization and generativity.

Part II — The benchmark. 4 Lushness.

Part III — Value. 5 Valuers, protection and the central wager. 6 Harm, recovery and suffering. 7 Cooperation, justice and virtue. 8 Individuals and minds.

Part IV — Action. 9 Decision, corrigibility and alignment.

The horizon. 10 Alpha, Logos and Omega.

Back matter. 11 Evidence and research program. 12 Formal core. 13 References. 14 Provenance.

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PART I

The object

CHAPTER 1

The physical possibility structure

Ethics usually begins with persons, preferences or rules. The Omega program begins with something prior to all of them: what can follow from the way the world is now. This chapter defines that object precisely, shows that physics already supplies it, and identifies the properties of it that the rest of the manuscript depends on.

The claim of the chapter is modest in one sense and ambitious in another. It is modest because nothing here goes beyond established physics: the object is the state of the world, its laws, and the developments they generate. It is ambitious because we take that object, rather than any summary of it, as what ethics is ultimately about. Every later chapter is an inquiry into features of this one structure.

1.1 What follows from a present

A **present** is the physical state of the world on a spacelike slice of spacetime: the state of everything, across space, at one moment in some admissible frame of reference. Relativity allows many such slices through any event; none is privileged, and the account below does not need one to be.

A present, together with the laws, determines what can follow. Two ingredients are needed and both must be complete.

The present must be sufficient. It must contain everything the laws need to carry the world forward: positions and momenta, fields, chemical stocks, records, correlations with the environment, and the state of any apparatus that will act. An effective model may legitimately omit microscopic detail, but it must retain whatever its own law of motion requires. A description of a subsystem alone is usually not sufficient, because its future depends on correlations it shares with its surroundings.

The law must be native. It is the actual dynamics of the physical model — not a list of tasks, a menu of actions, or a set of scenarios someone finds interesting. Choices, controllers and agents enter only as physical processes governed by the same law.

Given both, the **physical possibility structure** from that present is the complete set of developments the law permits, each with its weight, together with all the relations among them: which developments share a beginning, which reconverge, which enable or exclude which.

The classical form

In a classical model with a finite set of states, the law is the transition kernel K of a Markov chain (Norris 1997): K(x, y) is the probability of moving from state x to state y in one step. The probability of a complete history from a sufficient present x is the product of these one-step probabilities.

> $$P_x(x_1, \ldots, x_n) = \prod_{t=0}^{n-1} K(x_t, x_{t+1}), \qquad x_0 = x.$$
>
> The structure from x is this probability law over complete histories, for every horizon n.

What matters is the law at every state the system might reach, not just the first step. Two models can agree on what happens next from x and disagree entirely one step later. Continuous-time models work the same way with rates in place of step probabilities; sampling such a process at fixed intervals describes it at that resolution and can miss events between samples.

The quantum form

In quantum mechanics the present is a quantum state and the law is unitary evolution (Nielsen and Chuang 2000): the state is carried forward by a family of operators that preserve all its information. Alternatives carry complex amplitudes rather than probabilities, and amplitudes can reinforce or cancel. A quantum structure is therefore not simply a list of weighted branches. Paths through it can interfere, and drawing them as separate worlds to be counted misdescribes them. Probabilities appear when distinctions become recorded in the environment, a process developed in section 1.5.

What the structure is not

The structure adds no substance and no force. Catalysts, reservoirs, interfering amplitudes and deliberating organisms shape it through their ordinary physical relations. No additional principle steers the world toward greater possibility; the question of how possibility grows is answered later from the dynamics themselves.

Nor is the structure a space of everything describable. It contains what the physical model actually permits. Three questions stay distinct throughout the manuscript: what a formal model admits, what physical reality realizes, and where experience occurs.

1.2 Complete developments

The structure contains complete developments, not just where they end.

Consider a system that stays idle, and the same system making an excursion and returning. The end states agree. The developments do not: during the excursion, the system coupled to its surroundings, consumed resources and occupied apparatus that something else might

have used. A description that keeps only end states treats the two as identical. The structure keeps both.

This matters ethically before any ethics has been stated. A construction that later dissolved still occurred. A delay that made a rescue impossible is part of what happened, even if the eventual state is the same as with a timely rescue. A loss followed by recovery is not the same development as no loss at all.

Reconvergence and re-rooting

Different histories can arrive at the same later state. Suppose $x \to a \to c$ and $x \to b \to c$. From the original present x, these are two distinct developments with their own weights. From c, what follows is the same in both cases.

Both statements are true, and they answer different questions. Asking about the structure from x keeps both routes. Asking about the structure from c — taking c as a new present — asks what follows from there. The two are linked by ordinary conditioning on reaching c. Neither deletes the other, and re-rooting adds no hidden label recording which route was taken. The past bears on the future only through the state it produced.

Ending versus not knowing

A development can terminate: a process halts, an organism dies, a reaction stops. Termination is a real outcome and keeps its weight. An unfinished calculation is different: it marks the edge of what has been computed, not an end of what follows. Confusing the two manufactures false certainty at the boundary of a model.

1.3 Joint structure and composition

What matters about several processes is how they behave together.

Two fair coins can be independent, or perfectly correlated copies of each other. Each coin, alone, is a fair coin in both cases. Now compute the XOR of the two, which is 1 when they differ and 0 when they agree. It is a fair coin in the first case and always 0 in the second. The parts are identical in both cases, yet what can follow from them differs. The structure is therefore a property of the joint state, and summaries of the parts can discard exactly what determines what follows.

Sequential composition

Processes leave things behind: stocks depleted or replenished, apparatus occupied or freed, records written somewhere, correlations available to later interactions. These residual conditions are how one stage of a development shapes the next, and so how organization enters the future without any special credit.

To compose two stages correctly, the first stage's description must carry both its outcome and the state it leaves. The second stage then acts on that residual state. Time, energy and material consumed by the first stage are no longer available to the second. Failures and unfinished outcomes keep their actual residual states and costs.

Parallel composition and joint feasibility

Running processes side by side requires their actual joint law. Two repairs, each individually possible, may compete for the same unit of fuel. Any two of three processes may fit within a resource budget while all three together do not. **Joint feasibility** — the existence of one physical development in which several requirements hold at once — does not follow from each requirement being feasible separately, or even from every pair being feasible.

Independence is a special case and needs a physical reason. Independently prepared, non-interacting systems compose as products. Correlated preparations or interacting dynamics require the joint description, even when each part's readings agree with the independent case.

1.4 What can actually be done

Agents, controllers and institutions are physical processes inside the structure. A **controlled** process is one whose development depends on physically available records and physically implemented operations: a thermostat reading a temperature and switching a heater, a cell sensing a nutrient and expressing an enzyme, an engineer reading a design and building a machine.

Control has two physical requirements. The controller can respond only to records it actually has, not to facts an analyst knows about the system's hidden state. And an operation is available only if something physically implements it. Naming a chemical reaction does not make it a selectable action; it remains part of the native dynamics until some process controls it.

A **realization witness** connects an operational claim to the physics that supports it: the controller, the preparation, the resources consumed, the boundary conditions and the resulting law. It separates three different things:

- a transformation the laws **permit**;
- an apparatus **ready** to perform it reliably;
- a **route** by which such an apparatus can actually be obtained.

If preparing the apparatus succeeds with probability p and using it then succeeds with probability q, the whole route succeeds with probability pq. Readiness and the route to readiness are different physical facts, and both belong to the structure. The distinction between what the laws permit and what a physical constructor can reliably bring about is also the starting point of constructor theory (Deutsch 2013; Deutsch and Marletto 2015); the realization witness applies it to the access claims this manuscript makes.

1.5 Quantum structure and records

Quantum developments combine through amplitudes, and amplitudes interfere. Two consequences shape everything later.

A drawn path is not a world. A diagram of quantum alternatives can be useful, but its paths cannot be counted as independent outcomes. Where paths interfere, coarse alternatives must be combined at the level of amplitudes before any probability is assigned. Intermediate stages in such a diagram are not physical measurements unless the corresponding interactions are part of the model.

Probabilities arise from records. When a distinction becomes correlated with the environment — when a record of it forms — interference between the alternatives is suppressed for every process that cannot access the whole environment. This is **decoherence** (Joos et al. 2003; Zurek 2003). Families of alternatives that have decohered behave like ordinary classical outcomes, with probabilities given by the quantum weights (Griffiths 1984; Gell-Mann and Hartle 1993).

Two cautions follow, and both recur in chapter 4.

First, decoherence does not by itself select a unique way of dividing the world into alternatives (Dowker and Kent 1996). An idle qubit prepared in a definite state can be described by questions in different bases; each description is internally consistent, and they assign different numbers of effective alternatives. What selects a description physically is actual record formation: which distinctions the dynamics copy into the environment, durably and redundantly (Zurek 2009).

Second, records can be undone. If the only record of a distinction is reversed by a later interaction, the alternatives recombine and interference returns. If a redundant record survives elsewhere, the split persists. The intervening evolution remains part of the structure in either case. Global purity of the universal state does not make the structure trivial: the whole can be in a pure state while the relations within it are rich.

1.6 Faithful descriptions

No one computes with the whole structure. Every model compresses it. A compression is faithful for a purpose when it preserves the relations that purpose depends on.

The standard of faithfulness is predictive. Two descriptions are equivalent relative to a family of physical tests if every test in the family produces the same outcome law from both. Enlarging the family can separate descriptions that were equivalent; it never merges descriptions already separated. Equivalence for isolated outputs is not enough when a part will be embedded in a larger process: the tests must include the contexts it will be composed into.

A small example shows why summaries fail. Take three signs X, Y, Z, each +1 or −1, with joint law

$$P_\theta(x,y,z) = \tfrac{1}{8}\,(1 + \theta\, xyz), \qquad -1 \le \theta \le 1.$$

For every θ, each sign has mean zero and the three are pairwise uncorrelated. Means and covariances cannot tell the values of θ apart. Now apply the reversible transformation that replaces Z by the product XYZ. The new third sign has mean θ. The same transformation

applied to inputs with identical summaries produces different outputs. A summary that discards the triple correlation cannot predict what follows under an operation the physics allows.

Faithful compression is a tool, not a verdict. It does not identify physical copies, histories or individuals, and a distinction outside a description's tests remains in the structure. Approximate compression needs an error statement, and approximate equivalence is not transitive: two descriptions each close to a third can be twice as far from each other.

1.7 The futuresfield and Omega

Two names are useful for two ways of looking at the same object.

The **futuresfield** is the physical possibility structure seen forward from a particular present: what that present supports, at every horizon. It is the object assessed in decisions, comparisons and ethical judgments, all of which are made from somewhere.

Omega is the structure taken whole: every development of every present, considered together as one completed object. Its role in the main argument is as the reference against which local descriptions are complete or incomplete, developed in chapter 2 as the ultimate frame. Its role in the wider interpretation of the program is taken up in chapter 10.

1.8 The physical setting

Quantum field theory is the best current description of matter and its interactions (Weinberg 1995; Haag 1996), and it is the preferred physical setting for the structure. Fields, their local interactions, symmetries and causal constraints supply propagation, binding, record formation and recombination before any measure of breadth is chosen.

The program's finite models — classical chains, chemical networks and small quantum circuits — are exact calibration cases for the structure, not claims about its ultimate form. The next physical step is a small regulated field model, with its spatial resolution, cutoffs, boundary conditions and environment stated, built when a specific question requires it.

Formal notes

These notes state the constructions of the chapter precisely. They are written for readers who want the definitions in full.

F1.1 Finite quantum specification

A finite closed quantum model is specified by

$$\mathsf{P} = (\mathcal{H}, \{\mathcal{A}(R)\}, \mathscr{G}, H(t)), \qquad \rho_\Sigma \geq 0, \quad \operatorname{Tr} \rho_\Sigma = 1,$$

where $\mathcal{H}$ is the Hilbert space, $\mathcal{A}(R)$ the observables associated with physical regions R, $\mathscr{G}$ the constraints and physical identifications, and H(t) the Hamiltonian. Propagation satisfies

$$i\hbar\,\partial_t U(t,s) = H(t)U(t,s), \qquad \rho_t = U(t,0)\rho_\Sigma U(t,0)^\dagger.$$

The structure from Σ over horizon T is

$$\Omega_{\Sigma,T} = \big[\mathsf{P}, \rho_\Sigma, \{U(t,s) : 0 \le s \le t \le T\}\big]_\sim,$$

with ∼ a declared physical equivalence. A passive change of description must transform state, law, observable structure and readout together; rotating the state alone while holding the apparatus fixed changes the physics. The full propagator family, not only U(T, 0), is retained, so that idle and out-and-back developments with equal endpoints remain distinct.

F1.2 Composition with residual states

At a classical interface, a stage is described by a subnormalized kernel $J_r(x, y)$: the probability of record r and residual state y from x. A physically implemented second stage composes as

$$J_{s\circ r}(x,z) = \sum_y J_r(x,y)\, J_s(y,z).$$

When time and resources are resolved, composition also sums over their compatible splits, with the second stage's deadline and budget reduced by what the first consumed. A second stage may depend on the record r but not on the residual y except through physically available records. Failed and unfinished outcomes retain their mass and residual state.

F1.3 Predictive equivalence

For a declared family $\mathscr{T}$ of admissible physical tests,

$$p \equiv_{\mathcal{T}} q \iff \mathrm{Law}(\tau[p]) = \mathrm{Law}(\tau[q]) \ \text{ for all } \tau \in \mathcal{T},$$

and

$$d_{\mathcal{T}}(p,q) = \sup_{\tau\in\mathcal{T}} \mathrm{TV}\big(\mathrm{Law}(\tau[p]), \mathrm{Law}(\tau[q])\big)$$

is a pseudometric. Enlarging $\mathscr{T}$ refines ≡. The relation d ≤ ε is not transitive; the triangle inequality bounds the separation of two laws each within ε of a third by 2ε. A deterministic encoding q with update F is faithful when an abstract update $\bar{F}$ satisfies $q\circ F = \bar{F}\circ q$; stochastic and quantum versions are analogous, and resource costs and accessible records must factor through the same quotient.

F1.4 Histories and decoherence

For exhaustive orthogonal projector families at declared times, the class operators and decoherence functional are

$$C_\alpha = P^{(n)}_{a_n} U_{n,n-1} \cdots P^{(1)}_{a_1} U_{1,0}, \qquad D_{\alpha\beta} = \mathrm{Tr}\big(C_\alpha \rho_\Sigma C_\beta^\dagger\big).$$

A coarse alternative A has $C_A = \Sigma_{\alpha\in A} C_\alpha$ and weight $D(A, A) = \Sigma_{\alpha,\beta\in A} D_{\alpha\beta}$; off-diagonal terms make this weight non-additive in general. Medium decoherence, $D_{\alpha\beta} = 0$ for $\alpha \neq \beta$, makes the family classical with probabilities $D_{\alpha\alpha}$ (Gell-Mann and Hartle 1993), but does not select the family. In an exact pure-state model, orthogonal record projections R_h certify a history family through $R_h\,\psi_T = C_h\,\psi_\Sigma$; physical provenance still requires the recording interactions to belong to the model.

F1.5 Field theory

In local quantum field theory a state is a normalized positive functional on the algebra of observables, more general than a density matrix. Typical local algebras are of von Neumann type III (Yngvason 2005), so a finite matrix trace does not carry over to the continuum. Real-time (Schwinger–Keldysh) methods (Schwinger 1961; Keldysh 1965) and process tensors (Pollock et al. 2018) are representation tools for this setting; their slots for hypothetical operations do not add physical alternatives unless the operations are implemented.

For alignment

An AI system's world model is a compression of this structure. Section 1.6 gives the standard such a model must meet: it is adequate for a decision only if it preserves the joint relations that decision depends on, under the operations the system might actually perform. Sections 1.3 and 1.4 name the failures a capable optimizer can exploit when that standard is not met: treating separately feasible commitments as jointly feasible, treating a permitted transformation as a ready one, and treating a summary as if it determined what follows.

Status

Status	Claims in this chapter
Established	The structure as the state, law and developments of a physical model. Classical path laws and quantum unitary evolution. Decoherence and record formation. Failure of summaries under the three-sign example. Non-transitivity of approximate equivalence.
Adopted	The physical possibility structure as the object of ethics. Field theory as the preferred physical setting.
Open	A general relativistic quantum construction of the structure. Physical selection of history families beyond finite models.

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CHAPTER 2

Present frames and access

Chapter 1 defined the physical possibility structure. This chapter fixes the point of view from which it is assessed and identifies where, inside it, the differences between options actually lie.

The argument has three steps. All assessment is made from a present, and a present carries the past only through its physical state. Because fundamental dynamics is unitary, every option leaves the total information of the world unchanged, so total information cannot rank options. The differences therefore lie in how the whole is organized, and specifically in **access**: what situated parts of the world can actually reach, by what route and at what cost. Access is the concept through which organization, harm, cooperation and choice enter the rest of the manuscript.

2.1 Present frames

A **physical frame** is a present together with its physical state: an admissible spacelike slice of spacetime and the state of the world on it. In a relativistic setting there are many such slices through any event, and none is privileged. Different slices are related by the dynamics and by the appropriate transformations of their descriptions. In a finite model the frame is simply a declared update step and the state at that step.

Within a frame, a **perspective** is the situated access and influence of a process: where it couples to the rest of the world, what can affect it, what it can affect, and which correlations cross its boundary. A detector, a gas in contact with a reservoir, a cell and an institution each have a perspective in this sense. Awareness and agency are further kinds of organization a perspective may have. They are not entry conditions for having one.

A **frame description** is a representation of a physical frame at a stated scope and resolution: a region, a set of observables, a duration, the coupling at its boundary, the records available inside it. A scientist can choose such a description without any process inside the world corresponding to it. What justifies treating a description as a perspective is the physical coupling that actually realizes it, not the mathematics alone.

The distinction matters because the same physical situation supports many descriptions. An operator in a power plant encounters part of its state; an engineering model of the plant resolves components but not molecules; the full physical frame contains both and more. They are related, and adequacy depends on which consequences a restricted description preserves (section 1.6).

2.2 Sufficiency and blindness to arrival

A present is **sufficient**: whatever bearing the past has on the future is carried by the physical state it left. There is no further record of history attached to a present beyond the physical traces in it. In probability theory this is the Markov property of a complete state description (Norris 1997).

Two consequences follow.

A process cannot tell, from its present, how it arrived. Two routes that produce the same physical state have the same future. Whether a cell came from a division, a repair or an accident, and whether a branch arose from decoherence or from an interaction with an apparatus, makes no difference to what follows unless it left a physical trace. History is not lost: it remains in Omega, where it can be queried. It is simply not an additional ingredient of the present.

History and current access are different things. The accumulated history of a process grows over time. What the process currently has access to need not. A record can be lost: the history still contains it, but the process can no longer use it. Recovering its use requires a physical route — retrieval, re-derivation, reconstruction — with its own delay, coupling and cost. An analyst's archive gives the process nothing for free.

At a restricted interface, memory effects can appear (Breuer et al. 2016): a subsystem behaves as if it remembers, because correlations with an omitted environment carry information forward. These are not exceptions to sufficiency. They show that the restricted description was not sufficient, and that the environment, or an explicit multitime description of the interface, must be added back.

2.3 Why total information cannot rank options

Fundamental dynamics is unitary: the universal quantum state evolves without creating or destroying information (Nielsen and Chuang 2000). Every option, carried forward, leaves the total information of the universal state unchanged. Information about what happened is redistributed — into the environment, into correlations, into places no situated process can reach — but it is not removed.

Total information therefore cannot supply a ranking: asked "which option leaves the universe with more information?", the structure answers "neither". This does not mean the whole has no properties that distinguish options. Two options that conserve total information can differ greatly in their internal organization: their correlations, their records, and which developments situated processes can reach. The ranking has to come from that organization. Which developments can a given process reach? With what weight? Through which couplings? At what cost? Lushness, taken from a present frame, is a property of the whole that measures exactly this.

That is the sense in which the differences between options live in access.

2.4 Access

Access is the capacity of a situated process to reach a development: a state, a transformation, a capability, a recovery. An access claim names a physical route and everything the route involves:

- **the preparation** required before the route can be used;
- **the records** available to the process along the way;
- **the resources** consumed — energy, material, apparatus time, space;
- **the elapsed time**;
- **the weight** with which the route succeeds;
- **the residual state** left behind for whatever comes next.

Weights compose along a route. If preparing an apparatus succeeds with probability p and using it then succeeds with probability q, the route succeeds with probability pq. Repeated attempts change that only if failures leave states from which a retry is possible and enough time and material remain.

Costs form a profile, not a single number. Work, consumption, peak occupancy and time compose by different rules: sequential expenditures add, peak space takes a maximum, stocks can be replenished. Taking the cheapest entry in each column from different implementations can describe a combination no single implementation achieves. An access claim therefore retains the whole route, so that later processes inherit its actual consequences.

Access is oriented. It runs forward along the gradients the universe supplies, and what one process makes accessible can be what another process loses access to. Chapter 10 reads access as a higher-order asymmetry: an asymmetry between events, built on the orientation along which time runs and shaped by the organization that exploits it. Nothing in the main argument depends on that reading.

2.5 Realization witnesses

A **realization witness** is the physical account that backs an access claim: the controller, the preparation, the resources, the boundary conditions and the law that results. Every access claim in the manuscript is meant to be backed by one, and the concept does more work than its modest definition suggests. It applies to access claims the distinction, central to constructor theory, between tasks the laws permit and constructors that can perform them (Deutsch 2013).

A witness separates three questions that are easily run together:

1. **Is the transformation permitted?** The laws allow it.
2. **Is an apparatus ready to perform it?** Something exists that does it reliably.
3. **Is there a route to that apparatus?** The apparatus can actually be obtained from here, with the resources, time and information available.

Each is a separate physical fact. A transformation can be permitted with no ready apparatus, and an apparatus can be possible in principle with no available route to building it. Readiness is a property of the present; the route to readiness is a property of the structure from the present.

A witness also fixes what a controller can respond to. A control policy may depend only on records the controller actually has, not on the hidden state an analyst knows. And a combination of operations needs one witness for the combination: two separately witnessed operations can still be jointly infeasible (section 1.3).

The concept is central for alignment. A capable planner can propose transformations that are permitted but have no available route, or treat separately feasible steps as jointly feasible. Asking for the witness — what prepares this, what it consumes, what it leaves, whether its pieces can run together — exposes both errors.

2.6 Information is not access

The amount of information an arrangement holds does not determine what it gives access to. Two examples, both from the program's finite models, show the difference.

Encoding. Three registers can hold three bits s, f_1, f_2 as plain copies, or as the invertible recoding (s, $s \oplus f_1$, $s \oplus f_1 \oplus f_2$), where $\oplus$ is XOR. Both hold exactly the same information. Now ask how many local operations it takes to deliver a particular function of the bits into a fresh register:

Delivery	From copies	From recoding
f_1	1 step	2 steps
f_1 AND f_2	1 step	3 steps
parity of all three	3 steps	1 step
$s \oplus f_1$	2 steps	1 step

Step counts are illustrative, computed in a model in which each operation touches at most three bits, using copy, XOR, AND and their controlled versions. The program's finite probes are summarized in chapter 11.

Neither arrangement is better overall; each makes different things cheap. A summary that counts how many functions are deliverable within a given number of steps cannot see the difference at all, because an invertible recoding only permutes which deliveries are cheap.

Repair. Three registers holding two bits u and v can store (u, u AND v, v), (u, u, v) or (u, v, $u \oplus v$). All three determine u and v. After the loss of one register at a known position, the first can be repaired at one of the three positions, the second at two, and the parity arrangement at all three. Where redundancy sits, not how much information is held, decides what can be recovered. The parity arrangement is the simplest example of an erasure code (MacWilliams and Sloane 1977).

The cause is locality. Physical interactions are local and involve few degrees of freedom at a time. If arbitrary transformations of any size were available in a single step, the differences

above would vanish. In a world of local interactions, they are real, and in the program's exhaustive surveys of small model worlds most pairs of arrangements holding equal information differed in what they later made accessible.

2.7 Weights from a frame

A frame assigns weights to the developments it supports. Conditioning on what is known in a frame produces normalized probabilities for that frame. Two points about this are easy to get wrong.

Conditioning keeps its cost. If a development is reached only after a preparation that succeeds with probability P(α), a frame that starts after the preparation sees normalized probabilities. The encompassing description retains P(α). Forgetting it would change the claim from "what follows once the preparation succeeds" to "what follows from here", which are different questions.

Normalization is not breadth. Every frame's probabilities sum to one, whether they are spread over a million developments or concentrated on one. Normalization says how weight is distributed. It says nothing about how many developments carry it. That is the question lushness answers (chapter 4).

2.8 Access over time

"Can reach" has several meanings over time, and they must not be conflated.

- **First arrival**: whether and when a process first reaches a target.
- **Occupancy**: how much of the time it spends there.
- **Maintenance**: whether it stays there for a required duration after arrival.
- **Regeneration**: whether it returns after being displaced.

A process that reaches a target early and immediately loses it scores well on first arrival and badly on everything else. Each of these is a forward question from a present; none is a time average that substitutes for the present. Chapter 6 uses the distinctions to separate recovery from mere return.

2.9 The ultimate frame

The **ultimate frame** is Omega used as the reference for every comparison: the encompassing closure in which every local description and every frame's weighting belong to one physical account, with no further frame outside it.

It has three properties.

It is a reference, not an observer. It does not choose, evaluate or act. It is what a comparison would be relative to if it were complete.

No real process occupies it. Every actual process holds a partial description. Its records, models and interventions are themselves parts of the structure they describe.

It grounds humility. Because no real process holds the complete comparison, no real process can claim the verdict that only the complete comparison could deliver. Chapters 8 and 9 draw the consequences: graded rather than complete justification for sacrifice, and a persistent reason for every bounded system to seek correction.

2.10 From access to lushness

Access and lushness are not two separate ideas. Lushness is access accumulated over the futuresfield, by the chain rule of information theory (Cover and Thomas 2006).

> **Access is the input to lushness**
>
> $$g(x) = -\sum_y K(x,y)\log K(x,y), \qquad \log L_x(n) = \sum_{t=0}^{n-1}(K^t g)(x)$$
>
> g(x) is the entropy of what the present x can reach in one step, weighted by how likely each successor is: one-step access, measured as effective branching. Log lushness adds up that branching over the horizon, weighted by which states the dynamics actually visit.

Three consequences follow, each developed in chapter 4.

Organization enters through access. A construction changes which states are visited later, and so which branching is available then. That is the precise form of "organization changes what follows" (chapter 3).

Costs need no separate term. Section 2.4 gave access a cost profile: time, energy, material, occupancy. Lushness contains no cost penalty, and needs none. A route that consumes resources leaves a residual state with less available, and less available means less branching afterwards. Costs register in lushness through what they use up. Adding a separate cost term would count them twice and would take the measure outside the weight-only class within which chapter 4 proves it unique.

Equal information, unequal lushness. The arrangements of section 2.6 hold the same information but make different transformations cheap. When later processes draw on them, local dynamics give the two arrangements different future branching, and lushness distinguishes them. If nothing ever draws on the difference, it changes nothing downstream, and lushness correctly ties them.

Coherent access

The account of access in this chapter is stated for processes whose alternatives have decohered into records. Access is not only classical. Coherent processes can reach outcomes through interference without forming records along the way; quantum computation is the clearest case (Nielsen and Chuang 2000). Coherent access belongs with the quantum construction of lushness in chapter 4, where the open problem of measuring breadth without a classical history family is stated precisely.

Formal notes

Written for readers who want the definitions in full.

F2.1 Physical frames and descriptions

A physical frame is $F_\Sigma = (\Sigma, \rho_\Sigma)$ with the native specification needed to evolve it (F1.1). In a relativistic model Σ is an admissible spacelike hypersurface, a Cauchy surface where the spacetime and theory admit one; in a finite model it is a declared update cut. A frame description is a representation, restriction or conditioning of F_Σ at declared scope: observable algebra $\mathcal{A}(R)$, duration, boundary coupling and accessible records. Different admissible sections are related by the dynamics and the appropriate transformations of their descriptions; a worldline through an event does not select a unique global section. Relational treatments of quantum fields on hypersurfaces (Höhn, Russo and Smith 2023) provide a developed connection; a general relativistic quantum realization remains open.

F2.2 Conditioning and current access

Under a common classical law P, a positive-probability condition α gives

$$\mu_\alpha(E) = P(E \mid \alpha), \qquad P(E \cap \alpha) = P(\alpha)\,\mu_\alpha(E).$$

The encompassing description retains P(α). For nested information $\mathscr{F}_s \subseteq \mathscr{F}_t$ and an integrable readout Z, the tower identity holds (Williams 1991):

$$\mathbb{E}\big[\mathbb{E}[Z \mid \mathcal{F}_t] \mid \mathcal{F}_s\big] = \mathbb{E}[Z \mid \mathcal{F}_s].$$

The accumulated history generates an increasing filtration $\mathscr{F}_t$. Current access $\mathscr{G}_t = \sigma(O_t, M_t)$, generated by present observations and accessible memory, need not increase: a lost record remains in $\mathscr{F}_t$ while leaving $\mathscr{G}_t$. Martingale reasoning valid for $\mathscr{F}_t$ does not transfer automatically to $\mathscr{G}_t$. At restricted interfaces, multitime dependence is represented by process tensors (Pollock et al. 2018) with declared intervention slots; slots describe implemented operations or queries, not additional physical alternatives.

F2.3 Access with resources

An operational prefix is described by a joint law J(r, y, b, t | x) over accessible record r, sufficient residual state y, resource bill b and elapsed time t. A suffix may depend on r but not on y except through accessible records. For a later event E,

$$P_x(E) = \sum_{r,y,b,t} J(r, y, b, t \mid x)\; P_y^{\mathrm{suffix}(r)}\big(E;\, T - t,\, B - b\big),$$

with the remaining deadline T − t and budget B − b. Failed and unfinished prefixes retain their mass. The bill b is a vector whose components compose by their own rules (sum for sequential work, maximum for peak occupancy, signed sum for replenishable stocks). Joint feasibility of several requirements requires a single implemented development satisfying all of them.

F2.4 Arrival, occupancy and maintenance

For a classical target set V, let $\tau_V = \inf\{t \geq 0 : X_t \in V\}$. The first-arrival profile $A_x(t) = P_x(\tau_V \leq t)$ satisfies

$$\int_0^T A_x(t)\,dt = \mathbb{E}_x\big[(T-\tau_V)_+\big],$$

which measures expected time remaining after arrival, regardless of whether the target is retained. Occupancy is

$$O_x(T) = \mathbb{E}_x\left[\int_0^T \mathbf{1}_V(X_t)\,dt\right] = \int_0^T P_x(X_t \in V)\,dt.$$

Maintenance requires remaining in V for a declared duration after arrival; regeneration concerns return after a specified displacement. For controlled finite models, a viability set (Aubin 1991) collects the states from which some available policy keeps declared requirements satisfied against specified disturbances; full-state and partially observed control require different constructions.

For alignment

Two failure modes of capable planners are failures of access reasoning. The first is treating a permitted transformation as available: the plan names an outcome physics allows but no route the system can actually take. The second is treating separately feasible steps as jointly feasible. Realization witnesses catch both. A third, subtler failure is reasoning from an archive the system no longer has access to, or from facts an overseer knows but the system cannot observe. Section 2.2's distinction between history and current access is the corrective.

Status

Status	Claims in this chapter
Established	Physical frames and their relativistic multiplicity. Sufficiency of a complete present. Unitarity and the conservation of information. Conditioning identities; history versus current access. Arrival and occupancy identities. One-step access as the input to log lushness, by the chain rule.
Finite evidence	Information is not access: equal-information arrangements differ in delivery cost and repairability; most equal-information pairs in exhaustive small-world surveys differ in later access.
Adopted	Assessment from present frames. Access, backed by realization witnesses, as the locus of differences between options. The ultimate frame as reference.
Interpretive	Access as a higher-order asymmetry between events, built on orientation and shaped by organization (chapter 10).
Open	A general relativistic quantum construction of frames and their comparison. Coherent access and its contribution to lushness.

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CHAPTER 3

Organization and generativity

Chapter 2 located the differences between options in access: what situated processes can actually reach. This chapter asks what changes access. The answer is organization — arrangements of matter whose products become conditions for further processes — and its self-sustaining form, generativity.

The chapter makes four claims. Organization changes access through the ordinary law, with no special credit. There is a hard ceiling on how much randomness can widen what follows, and organization is how that ceiling rises from within. How organization holds and combines distinctions — copying, composing, distributing — determines what it can later do. And generativity has a history in our universe: it begins against a background of gas and, once sustained, becomes subject to selection among generative processes.

3.1 Outcomes that become conditions

A catalyst changes which reactions proceed readily. A genome lets a cell rebuild the machinery that copies the genome. A tool makes components for another tool. A community maintains the conditions under which its members discover and build. In each case an outcome becomes an enabling condition for further outcomes. Chemistry studies the self-sustaining version of this pattern as autocatalytic sets: networks in which every reaction is catalysed by some product of the network itself (Kauffman 1986; Hordijk and Steel 2004).

Generativity names processes of this kind: those that establish, maintain and transform the conditions for further processes. The definition has two parts that can be examined separately. The first is the mechanism: what is constructed, what is maintained, which transformations become available. The second is the consequence: how the mechanism changes what follows. Chapter 4 measures the consequence. This chapter is about the mechanism.

The enabling identity

Every enabling relation has the same form. A first stage produces a record r and leaves a residual state y. A second stage, which may depend on the record, then proceeds from y. The chance of a later event E is

$$P_x(E) = \sum_{r,y} J(r, y \mid x)\; P_y^{\mathrm{cont}(r)}(E).$$

The first stage changes the distribution of situations from which later development starts. That is all construction is, and all it needs to be.

The identity is elementary, and that is the point. Organization enters the future through the states it leaves behind, under the same law as everything else. The time and resources the first stage consumes are unavailable to the second. Failures keep their actual residual states. Nothing is added to reward the construction for being a construction.

A ladder of distinctions

The program follows a distinction into progressively richer uses: an initial difference affects a later response; a record carries it; an apparatus uses it; that apparatus produces another transformation; several processes maintain one another's enabling conditions. Each step is a separate question, settled through the joint law, and the ladder can branch, cycle or fail at any rung. Finite witnesses establish several rungs:

- **Information and use.** Two encodings revealing exactly the same distinctions can differ in how many operations it takes to use them (section 2.6).
- **Information and repair.** Arrangements holding the same information can differ in which losses they can repair (section 2.6).
- **Activity and coupling.** Arrangements with identical event statistics can differ in whether an upstream distinction reaches a downstream response at all.
- **Retention and construction.** Two different triggers can produce the same catalyst, erasing which trigger occurred, while that catalyst enables a new channel from a later input to a later output. Losing one distinction and creating a new channel are different physical relations.

Each witness shows that a summary of information or activity can omit organization that changes what follows.

3.2 Organization changes conditional access

Organization does not create possibilities outside physics. If a gas cloud can develop into a civilization, the civilization's developments already lie somewhere in the cloud's structure. What organization changes is **conditional access**: from a given present, which developments are reached reliably, how quickly, with what materials, and how much they depend on favourable accidents.

Prepared apparatus and usable records make routine what dispersed matter would rarely do within the same interval. Even where eventual reachability agrees, different barriers and couplings give different laws over a finite time.

Access and immediate breadth are not the same, and the difference matters for chapter 4. Making an outcome reliable concentrates weight on it, which lowers the branching of that

step. A reliable constructor may nonetheless enable far broader developments later. Whether the later gain exceeds the immediate concentration is a fact about the horizon profile, to be calculated rather than assumed. Construction, the retained capability it leaves, and open-ended growth are three distinct achievements; finite models can establish the first two.

3.3 The ceiling and the repertoire

A process cannot branch, in one step, into more alternatives than the states it can actually reach. In information terms:

> **The ceiling on branching**
>
> $$g(x) \leq \log|R(x)|$$
>
> g(x) is the entropy of the next step from state x, and R(x) is the set of states reachable from x in one step. Branching is maximal when every reachable successor is equally likely, and cannot exceed that.

Call R(x) and its extension over time the **repertoire**: the transformations a present can actually carry out. Randomness can push branching up to the ceiling the repertoire sets, and no further. Adding noise to a fixed repertoire spreads weight more evenly over what is already reachable; it does not make anything new reachable.

The ceiling rises in two ways.

- **From outside.** Inflows of energy or matter can enlarge what is reachable.
- **From within.** Organization can make new transformations, configurations and couplings reachable. A catalyst opens a reaction pathway that was effectively closed; a tool opens fabrication that was effectively impossible.

Growth from within comes in recognizably different kinds:

- **Reuse.** A catalyst that persists supports more reactions.
- **Amplification.** Producing more catalyst strengthens an existing pathway.
- **Expansion.** Producing a catalyst for a different transformation enlarges the repertoire.
- **Compounding.** Products that build further constructors enlarge the repertoire in proportion to its size, across generations.

Compounding generativity is the self-sustaining way for the repertoire to grow faster than randomness can fill it. It is subject to supply, transport, inhibition and loss, and it can leave a lasting legacy after construction stops: a system that has finished expanding may still retain a larger repertoire than it began with. The construction episode, the retained capability and indefinite open-ended growth are distinct physical achievements.

Size is not the whole story. Expanding the repertoire is necessary for raising the ceiling, but a repertoire can grow by transformations that enable nothing further, or that destroy what would have. The program's stronger claim concerns expansion that recursively enables further expansion. A bomb opens many new configurations locally while destroying constructors, records, organisms and the enabling relations that would have generated much

more. Whether the net breadth rises is not settled by local novelty. It is settled by lushness over complete histories, which is where chapter 4 takes it up.

3.4 How organization holds distinctions

What an organization can do depends not only on how much information it holds, but on how it holds it. The program's finite studies separate three ways of holding distinctions, and the separation lines up with two theorems of information theory.

Copies

A copy records a distinction as it is. Copies are cheap to make, need no decoding to read, and are the only option when there is a single distinction to hold. When a process can hold copies of everything relevant, copies are information-optimal: processing cannot increase information (the data-processing inequality; Cover and Thomas 2006), so no rearrangement can beat them.

Composition

A bounded process usually cannot hold everything. Then **composition** — recording combinations of distinctions rather than the distinctions themselves — can hold more. Composition comes in two forms.

- **Fusion** records a joint property, such as whether two coins match.
- **Hierarchy** records conditionally: if the first coin landed heads, record this; otherwise, record that.

With three independent coins each landing heads with probability 0.2, a single register holding a copy of one coin holds 0.722 bits. A register recording whether none of the three landed heads holds 0.9996 bits. Two registers holding a hierarchical record — which coin, if any, landed heads first — hold 1.762 bits with no synergy at all, in the sense of partial information decomposition (Williams and Beer 2010), by sorting the possibilities into cells of weight 0.2, 0.16, 0.128 and 0.512. Fusion and hierarchy are different routes to the same end: letting a bounded record divide the possibilities where the weight actually lies.

This is **source coding** (Shannon 1948): compressing biased or correlated inputs so that a bounded frame holds more.

Redundancy

Holding the same distinction in more than one place allows repair after loss. This is **channel coding** (Shannon 1948; Hamming 1950): adding structured duplication so that information survives damage. Here too composition beats plain copying. Three registers storing (u, u, v) can repair two of the three possible single losses; storing (u, v, u⊕v) can repair all three, because any two registers determine the third.

What follows

Copying is the identity case of composition: the best composition over the same registers can never do worse than copies, and does strictly better whenever a bounded frame holds two or more distinctions or must survive loss. Copies keep real advantages in cost, reading speed and simplicity. A world in which copying is the only strategy is degenerate; a world without copies is fragile.

Our universe is deep in the regime where composition matters. Perspectives are bounded: the information a region can hold grows with its boundary area (Bekenstein 1981; Bousso 2002), and blank memory costs energy to prepare (Landauer 1961; Bennett 1982). Inputs are biased and correlated: the universe is far from maximum entropy. Loss is everywhere, through decoherence, decay and noise. And at the base, unknown quantum states cannot be copied at all (Wootters and Zurek 1982); only distinctions that have become classical through decoherence can be duplicated, and everything else must be held through correlations.

3.5 Joint organization

Physical composition determines whether capabilities complement or obstruct one another. Specialization can open transformations that need several partners. Independent replication can spread a robust function. Redundancy can protect against local failure while remaining vulnerable to a shared disturbance. Cooperation and competition describe these relations; their effect is settled by the actual joint development.

Three cautions keep the analysis honest.

Pairwise evidence does not settle joint structure. Consider a function supported by either of two components, and another function independent of both. They agree when everything is intact and after every single removal; they disagree only when both components are removed. A graph built from pairwise tests or single knockouts cannot tell them apart.

Information in one place does not supply an actuator in another. A record that determines what should happen does not make it happen where it is needed.

Mutual dependence is not yet maintenance. A graph of processes supporting one another becomes a self-maintaining organization only when its cycles have jointly feasible supply and loss: one physical development in which every support is actually delivered. Chemical organization theory makes this condition precise for reaction networks (Dittrich and Speroni di Fenizio 2007).

An integrated system can constrain its components while expanding what the whole can do. The cells of an organism have fewer independent trajectories than an unconstrained collection, yet their coordination supports kinds of continuation no collection could. Whether a given integration expands or contracts the structure is a question about the joint development, not about how many outer boundaries there are.

3.6 The onset: from gas to replication

Generativity needs a source of usable energy. The observable universe provides one: its present entropy is of order 10^{104} in units of Boltzmann's constant, against a maximum of order 10^{122} set by its horizon (Egan and Lineweaver 2010).

Gas is the baseline, and gas is where generativity starts. Self-gravitating matter has negative heat capacity: as it loses energy it heats and contracts rather than settling, so in unbounded space it has no stable thermal equilibrium and instead clumps (Lynden-Bell and Wood 1968; Lynden-Bell 1999). Clumping makes stars, which forge heavy elements (Burbidge et al. 1957) and supply steady flows of free energy. Those flows drive chemistry far from equilibrium (Nicolis and Prigogine 1977), and under suitable conditions chemistry produces catalytic networks and molecules that template their own copies (von Kiedrowski 1986). Each stage converts an existing gradient into durable conditions for the next.

At this stage the question is whether organization can establish itself at all against matched, unorganized matter. The program's finite chemistry shows what that question involves.

- **Onset is not automatic.** In a small artificial chemistry with finite resources and decay, sustained replication appeared only above a critical resource budget, and early clustering dominated whether it took hold. Organization that cannot outpace its own losses does not persist.
- **Persistence has a precise criterion.** A lineage in a fluctuating environment survives only if its expected logarithmic growth rate is positive. This is a classical result about branching processes in random environments (Smith and Wilkinson 1969), and the same mathematics as the Kelly criterion in betting (Kelly 1956).
- **Placement matters.** In a reaction–diffusion model with an autocatalytic reaction, a preparation that placed the enabling molecules together exceeded an otherwise identical separated preparation by about 4.3 bits of expected history entropy over the following window, and exceeded both the stationary reacting reference and a matched-composition reference. Faster transport or a different observation interval could reverse the ranking.

Generativity, on this evidence, is real and fragile. It beats matched unorganized matter in some regimes and loses in others, and it lasts only while its engines do not collapse in on themselves.

3.7 Selection: generativity against generativity

Once generativity is sustained, ordinary Darwinian dynamics take over: variation, differential persistence and inheritance. The interesting comparison is no longer generativity against gas. It is which generative organizations outcompete, cooperate with, exploit or enable others.

Persistence filters what remains. Whatever can still affect a later situation has passed through conditions of continuation: competition, cooperation, capture, breakdown and repair. The relevant unit can be a lineage, an institution or a pattern sustained through the replacement of its components. A sterile replicator, an exploiter and a cooperative

constructor can each pass particular filters. Surviving explains why something remains; it does not show that what remains is good.

Selection favours share, not the whole. The Price equation (Price 1970) is an identity for any population whose members leave successors — genes, cultures, firms, AI models in a deployment pool. It splits the change in a trait into selection, the trait's covariance with relative success, and transmission. Relative success is share. A lineage can raise its share by widening the whole or by narrowing everyone else, and the selection term rewards both. This is the structural gap the ethics must address: a process can win locally while narrowing the structure around it. The equation's multilevel form (Price 1972) also shows when the gap closes: within a group, cooperators can lose share to defectors, while between groups, cooperative groups outgrow selfish ones (Sober and Wilson 1998; Traulsen and Nowak 2006). Chapter 7 develops the conditions.

Organization can deform its surroundings. A **deformer** is any process whose organization changes the continuation available around it: a river, an organism, an institution, an optimizer. **Agency** is the case in which feedback and internal organization systematically modulate those effects: a system that senses, responds, learns, and can revise its own mechanisms.

Regulation has two strategies. Ashby's law of requisite variety says that only variety can absorb variety (Ashby 1956).

> **Requisite variety**
>
> $$H(\text{outcomes}) \geq H(\text{disturbances}) - H(\text{responses})$$
>
> To hold outcomes within a narrow target, a regulator's repertoire of responses must match the variety of the disturbances it faces. The inequality assumes that disturbances stay distinguishable through the system being regulated: for any fixed response, different disturbances lead to different outcomes. Without that assumption it can fail, since a system that sends every disturbance to the same outcome needs no regulator at all.

A regulator can satisfy the inequality in two ways: by expanding its own responses, or by suppressing the disturbances — making its surroundings simpler, more uniform, less able to surprise it. Both succeed at the regulator's task, and they can look identical on its own metrics. Ashby establishes this tradeoff. What the program adds is a hypothesis about its consequences for everything else: expanding responses tends to widen the whole, by adding branching to the regulator's repertoire, while suppressing disturbances tends to narrow it, by removing branching from the processes around the regulator. Because the inequality is already stated in entropies, the hypothesis can be tested in lushness. On this reading, domination is regulation by suppression.

Suppression is not always narrowing. An immune system suppresses pathogens; a firewall suppresses attacks. When the suppressed variety is itself excisive, suppressing it can widen the whole. The signature decides, not the label. Chapter 9 adds the companion result, that every good regulator must contain a model of what it regulates (Conant and Ashby 1970), and

shows when a system with a thin model has suppression as one available strategy among others.

Persistence is what the filter selects. Whatever remains has, by construction, persisted. Chapter 5 shows that persistence is also what nearly every value presupposes: almost any end that extends through time commits its holder to the persistence of whatever realizes it. Persistence, Price and Ashby together give a structural account of domination: persistence pursued narrowly, secured by suppression, rewarded by share. Lushness is what tells it apart from generativity.

3.8 Mechanism and consequence

This chapter has described how organization changes access: through enabling states, by raising the repertoire's ceiling, through the way distinctions are held and combined, and through selection among organizations. It has not yet said how much any of this widens what follows. That is the measure of chapter 4.

Keeping the two apart is what lets the program test itself. The mechanism can be identified without reference to the measure; the measure can then show whether the mechanism widens the structure, by how much, for how long, and at what resolution. If generative organization did not widen the structure in the regimes that matter, the measure would say so.

The mechanism already suggests the shape of the program's thesis. Good is organization that sustains and compounds the conditions for further structured possibility. Bad is securing local continuation through suppression, destruction or foreclosure of the generative capacity around it. Both can persist; both can win under selection. Whether they also differ in lushness, as the thesis predicts, is the question chapter 4 makes measurable.

Formal notes

Written for readers who want the definitions in full.

F3.1 Enabling composition

With J(r, y | x) the outcome–residual law of a realized prefix and $P_y^{\mathrm{cont}(r)}$ the law of a record-dependent continuation,

$$P_x(E) = \sum_{r,y} J(r, y \mid x)\, P_y^{\mathrm{cont}(r)}(E),$$

with time and resources resolved as in F2.3. A dependence experiment compares the whole route with a physically specified replacement under matched conditions. To separate recursive enabling from durability, compare products that enable new constructors with equally durable products that support the same uses but cannot construct further.

F3.2 The repertoire ceiling

For a finite kernel K, g(x) = H(K(x,·)) ≤ log|R(x)| with R(x) = {y : K(x,y) > 0}, with equality exactly when K(x,·) is uniform on R(x). Hence

$$\log L_x(n) \le \sum_{t=0}^{n-1} \mathbb{E}_x\big[\log|R(X_t)|\big].$$

Under a fixed law, noise that changes K(x,·) without changing R(x) can raise log $L_x(n)$ only up to this bound. Raising the bound requires changing which states are visited (so that later $R(X_t)$ is larger) or changing R itself through inflows. A minimal compounding model takes the repertoire size r_t to grow as $r_{t+1} = r_t + c \cdot r_t \cdot s_t - d_t$, with s_t the available supply and d_t losses; compounding is the regime in which the growth term scales with r_t and exceeds losses.

F3.3 Coding

Processing cannot increase mutual information (data-processing inequality), so a frame able to hold copies of all inputs cannot be beaten by composition. For a frame of k registers and inputs with entropy exceeding k bits, the information a record holds about the inputs is the entropy of the partition it induces, maximized by partitions of equal weight; composition reaches partitions that copies cannot when inputs are biased. Repair after a single known erasure requires that every register be recoverable from the others; (u, v, u⊕v) satisfies this for all three positions.

F3.4 Persistence in a fluctuating environment

For a branching process in a random environment with mean offspring m_t at generation t, survival with positive probability requires $\mathrm{E}[\log m_t] > 0$ (Smith and Wilkinson 1969; Athreya and Karlin 1971). Long-run growth is governed by the expected logarithm, not the logarithm of the expectation, which is the same condition that makes the Kelly strategy optimal for repeated bets.

F3.5 Selection

The Price equation decomposes the change in a population mean trait $\bar{z}$ as

$$\Delta \bar{z} = \frac{\mathrm{Cov}(w, z)}{\bar{w}} + \frac{\mathbb{E}[w\,\Delta z]}{\bar{w}},$$

with w fitness. The covariance term favours whatever raises its own share; no term refers to what the whole supports. For a population partitioned into groups k with group means Z_k and fitnesses W_k,

$$\bar{w}\,\Delta\bar{z} = \mathrm{Cov}_k(W_k, Z_k) + \mathbb{E}_k\big[\mathrm{Cov}_{i\in k}(w_i, z_i)\big] + \text{transmission},$$

separating between-group from within-group selection. Any claim that selection in a given regime also widens the structure must be established separately.

F3.6 Requisite variety

For disturbances D, regulator responses R and outcomes O = f(D, R), suppose that for every response r the map d ↦ f(d, r) is injective, so that no outcome repeats within a column of

Ashby's outcome table. Then $H(D \mid R, O) = 0$, hence $H(D \mid R) = I(D; O \mid R) \leq H(O \mid R) \leq H(O)$, and since $H(D) \leq H(R) + H(D \mid R)$, $H(O) \geq H(D) - H(R)$ (Ashby 1956). The injectivity condition is necessary: if f is constant, $H(O) = 0$ while $H(D) - H(R)$ can be positive. The good regulator theorem (Conant and Ashby 1970) states that the simplest optimal regulator of a system is, under stated conditions, a model of that system.

For alignment

Ashby's fork is the alignment-relevant result of this chapter. A capable system can meet its objectives either by expanding its own responses or by suppressing the variation it must face. Suppression often looks efficient locally — fewer surprises, cleaner metrics, more reliable task performance — while it narrows what everything around the system can reach. The difference is visible in the structure: suppression removes branching from others; expansion adds it to the system's own repertoire. The Price equation adds the general warning: training and deployment are selection processes, and selection favours whatever increases its own share, which is not the same as widening the whole.

Status

Status	Claims in this chapter
Established	The enabling identity. The repertoire ceiling on branching. Data processing and the coding theorems. Gravitational clumping and the entropy gap. Survival in random environments. The Price equation and its multilevel form. Ashby's law of requisite variety, under its distinguishability condition. The good regulator theorem.
Finite evidence	Information, repair, coupling and construction witnesses. Composition beats copying for bounded frames with biased inputs; hierarchy composes without synergy. Critical budgets for the onset of replication. Placement-dependent breadth in an autocatalytic model, with regimes of reversal.
Adopted	Organization enters only through the native law. Gas as the baseline at onset; selection among generative processes thereafter.
Hypotheses	Compounding generativity is the self-sustaining source of widening structure. The expansion branch of Ashby's fork widens the structure and the suppression branch narrows it.
Open	A resource-accounted model in which products enable further constructors, separating compounding from durable storage and activity (chapter 11).

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PART II

The benchmark

CHAPTER 4

Lushness

Chapters 1 to 3 described an object and a mechanism: the physical possibility structure, the access it supports from each present, and the organization that changes that access. This chapter supplies the measure. **Lushness** is the effective breadth of the futuresfield: how many distinct futures a present supports, each weighted by its probability.

The chapter has five tasks. It defines lushness and shows how organization enters it. It states the uniqueness theorem that singles lushness out within a precisely defined class of measures, and says which part of that is proved and which part is the program's choice. It settles which horizon the comparison should use. It states what lushness can and cannot see, including at the quantum level. And it works five exact examples, so that the claims of chapter 3 become measurable rather than merely compatible with the measure.

One distinction runs through all of it. The structure is the object; lushness is its magnitude, not a representation of it. Two structures can have the same lushness and differ in every other respect. The structure carries the explanation of what happens; lushness carries the verdict on how much is supported. Which structural configurations raise lushness, and there will be many, is what the program studies.

4.1 Definition

For a classical development described at a stated resolution, lushness is the exponentiated entropy of complete future histories.

> **Lushness**
>
> $$L_x(n) = \exp H(p_x), \qquad H(p) = -\sum_h p(h) \log p(h)$$
>
> p_x is the probability distribution over complete histories h of length n starting from state x. H is its Shannon entropy, the standard measure of how widely probability is spread. Exponentiating turns entropy into an effective number of futures.

The calibration is easy to check:

- N equally likely futures give L = N.
- n independent fair binary events give 2^n, although the probabilities sum to one.
- A certain future gives 1.

- Weights (0.9, 0.05, 0.05) give about 1.48: three possibilities, effectively fewer than two.

Lushness is neither a count of possibilities nor a measure of probability. A development with a million possible futures, nearly all of whose weight falls on one, has lushness close to one. Support and weight are both in the definition. In ecology the same quantity is the Hill number of order one, used to measure the effective diversity of a community (Hill 1973; Jost 2006). In language modelling it is the perplexity of a predictive distribution (Jelinek et al. 1977).

Everything counts. Construction and collapse, cooperation and competition, success and failure, organization and noise all contribute their branches. The measure does not decide in advance which developments matter. That is what allows it to judge between them, and it is also what exposes it to the tests of section 4.9.

Independent and coupled systems

For independent systems, lushness multiplies: $L_{AB} = L_A \cdot L_B$, because the entropies of independent histories add. Coupled systems use their actual joint law. Shared resources, correlations and interactions change both the parts and the whole, and composition is represented in that joint law rather than by any added term.

One consequence matters for the ethics. Independent copies of a stochastic process multiply lushness exactly as independent processes of different kinds with the same breadth do. Under lushness, plurality is not valuable because variety is valuable in itself. It is valuable through interaction: complementary processes opening joint transformations that copies cannot.

4.2 How organization enters

The chain rule of information theory (Cover and Thomas 2006) decomposes the entropy of a whole history into the entropy of each step, given what came before. For a Markov process this gives a direct formula.

> **How organization enters**
>
> $$\log L_x(n) = \sum_{t=0}^{n-1} (K^t g)(x), \qquad g(x) = -\sum_y K(x,y) \log K(x,y)$$
>
> g(x) is the branching of the next step from state x: one-step access, measured as effective branching (section 2.10). $K^t g(x)$ is its expected value t steps ahead. Log lushness is the sum of expected branching over the horizon.

The present configuration sets the first term; the dynamics decide which configurations, and therefore which branching, come next. Catalysis, construction, damage and repair all change lushness through this sum and in no other way. There is no bonus for organization and no penalty for cost. A route that consumes resources leaves a residual state with less available, and that state branches less afterwards. Costs register through what they use up.

The formula also makes chapter 3's repertoire ceiling quantitative. Since $g(x) \leq \log|R(x)|$, where $R(x)$ is the set of states reachable from x in one step, lushness is bounded by the logarithms of the repertoires the dynamics visit. Noise can bring $g(x)$ up to that bound. Only changing which states are visited, or enlarging R itself, can raise it.

4.3 Why this measure

Comparing states of the structure by magnitude requires a measure. There are many candidates: free energy, number of organisms, number of reachable states, or variants of entropy. This section separates three things: the class of measures the program adopts, the theorem that singles out one measure within that class, and what remains open outside it.

The class is defined by two requirements, each motivated by commitments the framework already holds.

Weights only. The measure depends only on the weights of the alternatives, not on what the alternatives are. Relabelling futures cannot change it, and no kind of thing and no recipient is privileged. This requirement is motivated by impartiality, but it is stronger than impartiality. A measure could be equally impartial and still take into account how different two futures are from each other. Ecologists use measures of exactly this kind, which weight diversity by the similarity between types (Leinster and Cobbold 2012).

Consistency over time. All assessment is from a present (chapter 2). Assessing a development whole, from its root, must therefore agree with assessing it step by step: first the branching at the root, then the expected branching from each successor, weighted by its probability. This is precisely the chain rule, sometimes called the grouping property. Present-rooting motivates a consistency condition of this kind; it does not by itself force this exact form of it.

Add two natural requirements: N equally weighted alternatives count as N, and small changes in the weights produce small changes in the measure.

Shannon's characterization of entropy (Shannon 1948), made rigorous by Khinchin (1957) and by Faddeev (1956), shows that exactly one function satisfies symmetry, continuity, grouping and a normalization: Shannon entropy. Exponentiating it gives the calibration to an effective count. Other entropy-like measures — the Rényi family (Rényi 1961), for example — are additive for independent systems but fail the full grouping property. They would make the verdict from a root disagree with the verdict assembled from its successors, which is exactly the inconsistency the framework's present-rooted assessment rules out.

The theorem is exact: within the class of measures that depend only on path weights and satisfy the grouping rule, exponentiated Shannon entropy is the only one with the stated calibration and continuity. What the theorem does not do is choose the class. That choice is the program's, motivated by impartiality and present-rooting but not forced by them. Lushness is therefore best described as the sharply specified classical candidate for the magnitude of the structure, with a genuine uniqueness theorem inside its class.

Two questions remain open outside the class. The first is whether the full possibility structure needs a measure sensitive to how different futures are, and not only to how they

are weighted. The similarity-sensitive measures of ecology reduce exactly to lushness when every future is treated as fully distinct from every other, so lushness is the natural limiting case of that family rather than a rival to it. The second is the quantum construction of section 4.8. A reader who rejects the class can reject the measure; the theorem establishes what follows for anyone who accepts it.

4.4 What lushness sees

Because lushness is a magnitude rather than a representation, a natural worry is that it discards the structure chapters 1 to 3 insisted on: joint correlations, composition, residual states, latent capabilities.

It does not discard them where they matter. Lushness is computed over complete histories generated by the full joint dynamics. A correlation, a capability or a latent structure changes lushness exactly when the dynamics draw on it, because that is when it changes what follows. If nothing ever draws on a difference, it changes nothing downstream, and lushness correctly ties the two cases.

Example E3 makes this exact. Take three signs with joint law $P_\theta(x,y,z) = (1 + \theta xyz)/8$. For every θ the signs have zero means and are pairwise uncorrelated; a model built from means and covariances cannot tell the values of θ apart. Let a preparation sample the signs, and let a later module branch four ways per step for k steps when an exposed sign is +1, and idle otherwise.

θ	Exposed: dynamics read XYZ	Unexposed: dynamics read X	Summary-based prediction
−1	2.0	22.0	23.0
−0.5	12.8	22.8	23.0
0	23.0	23.0	23.0
0.5	32.8	22.8	23.0
1	42.0	22.0	23.0

Log lushness in bits for k = 20 downstream steps.

When the dynamics expose the triple correlation, lushness separates θ by 40 bits. When they do not, the only difference is the entropy of the preparation itself, at most one bit. The summary-based model, which cannot see θ, predicts the same value throughout and errs by up to 21 bits.

Where lushness genuinely loses information is at coarse resolution and short horizons: a coarse description can merge futures the dynamics would later separate, and a short horizon can stop before the separation happens. Those are the subjects of sections 4.5 and 4.7.

4.5 Which horizon

Lushness is defined for a horizon n. Which horizon should a comparison use?

Lushness is cumulative

By the chain rule, log lushness at horizon n contains all the branching of every shorter horizon. Long-horizon lushness does not ignore early branches; it includes them. Every shorter horizon is a truncation of the same object.

Truncation is a choice, and a privileging one. Cutting the assessment at a horizon N weights the future fully up to N and not at all afterwards: a hidden discount, shaped as a step. Lushness carries no discount factor, consistent with Ramsey's argument that discounting later welfare merely because it is later has no ethical justification (Ramsey 1928). Every step of the future counts, and the object the program cares about is the complete development. The furthest horizon is therefore not one privileged choice among others. It is the least truncated view of the one object.

Requiring an option to win at every horizon does not work as a substitute. Two options can lead at different horizons, and dominance across truncations then returns "incomparable" where the complete development gives a clear answer. Example E2 shows exactly this.

Example E2: maintenance, ruin and productive disruption

Start from a repertoire of 16 transformations, used uniformly (4 bits per step). Compare three continuations:

- **Maintenance.** The repertoire stays at 16.
- **Ruin.** Each step, with probability f, a failure occurs in one of m failure modes and the repertoire loses one transformation. A repertoire of zero is dead. Otherwise the process acts.
- **Productive disruption.** The repertoire is cut to 8, but growth beyond the old ceiling is reopened: each step, with probability 0.2, the process builds and doubles its repertoire, up to 256; otherwise it acts.

Across a sweep of failure rate f from 0.05 to 0.4 and failure modes m from 4 to 256:

- **At short horizons ruin leads.** In every case swept, ruin exceeds maintenance for the first ten steps or more. Failure modes are real branches, and lushness counts them.
- **Ruin then falls behind.** Ruin drops below maintenance between 20 and 200 steps, depending on f and m.
- **Productive disruption trails, then leads.** It is behind maintenance for the first steps, overtakes it at 20, and leads every case from 50 steps on.

In the limit the picture is unambiguous. Maintenance grows at 4 bits per step and productive disruption at 7.1. Every path of ruin eventually ends, so its total is bounded — between 137 and 1,060 bits across the sweep, however rare or varied its failures. The complete-development ordering is productive disruption, then maintenance, then ruin, in every case.

The short-horizon reversal is the lesson, not a loose end. Noisy failure looks good until its consequences arrive. A short-horizon estimate of lushness can prefer ruin.

The disruption case needs a caution of its own. It was constructed so that the cut is paired with a new capacity to build: the model stipulates that breaking the old repertoire reopens

growth beyond it. What E2 shows is that lushness ranks such a case correctly when it occurs, not that disruption tends to be generative. Most disruption destroys capability without reopening anything, which is ruin, or the permanent loss chapter 6 calls amputation. A bounded agent usually cannot tell in advance which kind it faces, which is one reason protection gains force under uncertainty in Gradient Ethics (chapter 5).

Comparing in the limit

When totals grow without bound, the limit comparison has a natural form: compare long-run growth rates first, then lasting offsets.

> **Comparison in the limit**
>
> $$\log L_x(n) = n\,h(x) + c(x) + o(1)$$
>
> h(x) is the long-run rate of branching from x; c(x) is the lasting offset. Higher rate wins; at equal rates, higher offset wins.

The form holds for every finite process whose long-run behaviour is aperiodic; periodic behaviour adds a bounded oscillation to the offset (formal note F4.4). The ordering itself, rate first and offset second, is a comparison rule that the program adopts for developments whose breadth grows without bound. It is not forced by the entropy formula. It is the natural rule wherever the limit form holds, and there it agrees with the overtaking criterion used for comparing infinite programs in the theory of optimal growth (von Weizsäcker 1965). Ruin has rate zero and a bounded total. Maintenance and disruption have positive rates. In a closed system that eventually forgets its starting point, all starting points share one rate, and the comparison falls to the offset (section 4.6).

Example E1: investment

The rate-then-offset form also explains investment. Compare a process that acts uniformly over a fixed repertoire of 4 (2 bits per step) with one that first spends k steps building — zero branching while it builds — and then acts over the enlarged repertoire.

Horizon	Noise in fixed repertoire	Best linear building	Best compounding building
4	8	8 (k = 0)	9 (k = 1)
16	32	36 (k = 3)	81 (k = 7)
64	128	208 (k = 14)	648 (k = 10)
128	256	501 (k = 26)	1,416 (k = 10)

Log lushness in bits. Linear building adds one transformation per step; compounding doubles the repertoire, up to 4,096.

Building always loses at short horizons, because a reliable constructor concentrates weight. It wins after a crossover — for a single step of investment, at 8 steps with linear growth and 4 with compounding — and the best amount of investment grows with the horizon. This is chapter 3's reliable constructor, made exact: concentration first, breadth later. Like E2, the

example shows how the measure ranks investment under its stipulated laws, in which building is reliable and compounding is available. It does not show that the measure generally selects building over noise.

The resolvable horizon

A bounded process cannot resolve the limit. Its model of the structure becomes less reliable the further ahead it looks. In practice it compares options at the furthest horizon it can resolve with acceptable error. Three levels follow.

- **The truest test** is the limit: the complete development, Omega.
- **The practical test** is the furthest horizon the process can resolve.
- **The gap between them** is where the reason to seek correction comes from. An action whose consequences extend beyond the resolvable horizon cannot be fully justified by a comparison that stops short of them. The less reversible the action, the more that matters (chapter 9).

Short-termism now has a precise definition: treating the resolvable horizon, or a still shorter convenient one, as if it were the whole. Example E2 shows what that error looks like.

4.6 The equilibrium identity, and the generative window

Thermal equilibrium is the stationary distribution π that a closed system's dynamics approach when left alone. Compare a starting state x with it.

> **The equilibrium identity**
>
> $$\Delta\ell_n(x) = \log L_x(n) - \sum_y \pi(y) \log L_y(n), \qquad \sum_x \pi(x)\, \Delta\ell_n(x) = 0$$
>
> The excess of starting at x over the equilibrium average, weighted by equilibrium itself, is exactly zero, because the dynamics leave equilibrium unchanged. Unless every starting state ties, some lie strictly above the equilibrium average.

The identity is exact, and its content should not be overstated. It says that excess lushness, averaged over the stationary distribution, is zero, so that unless every state ties, some states lie above the average. That is close to a definition of an average. The substantive claim begins elsewhere: that a specified organized state beats a properly matched stationary reference, one with the same energy, composition and constraints. That claim is the program's hypothesis, and it is what the program's finite probes test, with organized preparations beating matched references in some regimes and losing in others (chapter 3).

In a closed, finite system that forgets its starting point, the excess converges: $\Delta\ell_n(x)$ approaches a constant c(x). A good starting state buys a lasting offset, not a faster long-run rate, and its advantage over each fresh window decays as the system forgets. How large the offset can be depends on how long forgetting takes, and forgetting times can grow enormously with system size.

Example E4. On a randomly generated six-state chain, the equilibrium-weighted sum of excess lushness is zero to machine precision at every horizon. The best starting state keeps a lasting offset of +0.42 bits and the worst −0.76; the offsets settle within about ten steps, because this small chain forgets quickly.

The universe is not such a system. It is far from equilibrium, open to the growth of structure, and nowhere near forgetting its initial conditions. The **generative window** is the interval before forgetting. Within it, offsets are the phenomenon, and compounding generativity — growth of the repertoire itself — is how a process escapes the closed-system bound by changing which states can be reached at all.

4.7 Resolution

Every calculation of lushness declares a resolution: which distinctions between states count as different. Coarse resolutions merge futures; fine ones separate them. Rankings can change with resolution, and this is one of the program's main open problems.

The program's position has three parts.

The object is the profile. Lushness at a single resolution is one view. The full object is lushness across every resolution and horizon.

The target is dominance across resolutions. The program's headline physical hypothesis is that generative organization beats the thermal reference at every resolution within the generative window. Dominance across the whole profile would settle a comparison without choosing a resolution.

Where dominance fails, the resolution is fixed in advance. When options are incomparable across resolutions, a practical comparison uses a resolution chosen before the comparison by a stated rule, never after seeing the result. Physics offers a candidate privileged resolution: the distinctions that decoherence copies durably and redundantly into the environment (Zurek 2009), which are the ones situated processes can actually access. Whether that candidate suffices in general is open, and a single ranking across resolutions remains a research goal.

4.8 Quantum lushness

Quantum developments combine through amplitudes, which can interfere, so a quantum structure is not simply a distribution over histories. Where a family of alternatives has decohered, its histories carry ordinary probabilities, and lushness applies to them directly. The open problem is breadth where no classical family has been selected.

Example E5. An idle qubit prepared in a definite state can be described by a question in the Z basis or in the X basis. Both descriptions are internally consistent, and both families are decoherent. The Z question gives breadth 1; the X question gives breadth 2. No single number can agree with both while also ignoring which question was asked. A quantum lushness therefore needs either a physical rule for selecting the family — such as actual record formation — or an explicit dependence on the physically realized interface. The difficulty is a

known feature of decoherent-histories formulations of quantum mechanics, often called the set-selection problem (Dowker and Kent 1996).

A satisfactory construction must meet several requirements at once:

- equivalent descriptions of the same physical process give the same value;
- inserting an analytical checkpoint, with no physical interaction, changes nothing;
- independent systems compose correctly;
- actual record formation, which makes new distinctions accessible, can change the value;
- the classical limit reproduces the classical calibration.

For a fixed, physically justified family with weights p, a natural requirement is that quantum lushness lie between one effective direction and the fully distinguishable classical case: $1 \leq L_Q \leq \exp H(p)$. Bounds alone do not choose a value. Finite tests in the program have ruled out several candidates, each of which failed one requirement or another. The deliverable is a construction that passes all of them, a demonstration that the requirements are incompatible, or a precise statement of the missing physical input.

Coherent access (section 2.10) belongs here as well: processes that reach outcomes through interference without forming records contribute to the structure in ways a classical history family does not capture.

None of this changes the classical results above, which hold wherever developments carry ordinary probabilities.

4.9 What the measure shows, and what it must still show

The examples answer the question chapter 3 left open: does lushness measure the distinctions the mechanism describes, or merely sound compatible with them?

- **Construction versus noise (E1).** A reliable constructor loses at short horizons and wins after a crossover, by a margin that grows with compounding.
- **Ruin versus disruption (E2).** In the limit, a disruption constructed to reopen growth beats maintenance, which beats ruin, in every case swept. At short horizons, ruin can lead. The model does not show that disruption in general is generative.
- **Hidden structure (E3).** Lushness sees a correlation exactly when the dynamics use it.
- **Equilibrium (E4).** The equilibrium identity holds exactly, and in a closed system the advantage of a good start is a lasting offset.
- **Quantum family selection (E5).** Breadth depends on the question asked, unless physics selects it.

These are stylized models with stated parameters, chosen to isolate one mechanism each. They show that the measure can register the distinctions the framework cares about, under laws stipulated to contain them. They do not show that the measure generally selects generativity over noise or over pathological breadth, and they do not yet show that it behaves as required in the complex, coupled systems the ethics concerns.

The program's central wager (chapter 5) depends on that further step. Its decisive tests include a pair of worlds with identical branching structure at every resolution, one

recognizably good and one recognizably bad, and a world of breadth-neutral domination, in which access is concentrated in a few at no cost to total lushness; chapter 11 gives the full list and the protocol under which the search is judged. E2's short-horizon reversal shows why every such test must be assessed at the furthest resolvable horizon. A test that stops early can be won by ruin.

Formal notes

Written for readers who want the definitions in full.

F4.1 Uniqueness

Let H be defined on finite probability vectors and satisfy: symmetry under permutation; continuity of H(p, 1−p) in p; the grouping property

$$H(p_1, \ldots, p_n) = H(p_1 + p_2, p_3, \ldots, p_n) + (p_1 + p_2)\, H\left(\tfrac{p_1}{p_1+p_2}, \tfrac{p_2}{p_1+p_2}\right);$$

and normalization H(½, ½) = 1. Then H = −Σ $p_i \log_2 p_i$ (Faddeev 1956; see also Shannon 1948 and Khinchin 1957). Exponentiation gives L(uniform on N) = N. Rényi entropies of order α ≠ 1 satisfy additivity for independent systems but not grouping, so the root assessment of a history tree does not equal the probability-weighted assessment of its subtrees. That H is a function of the probability vector alone is part of the definition of the class, not a consequence of the other conditions. Similarity-sensitive diversity of order one (Leinster and Cobbold 2012), $D^Z(p) = \exp(-\Sigma\, p_i \log (Zp)_i)$ for a similarity matrix Z with unit diagonal, lies outside the class and reduces to exp H(p) when Z is the identity.

F4.2 Chain rule and recurrence

For a Markov kernel K and exact root x_0, with $p(x_1,\ldots,x_n) = \prod K(x_t, x_{t+1})$ and $g(x) = H(K(x,\cdot))$,

$$\ell_n = \sum_{t=0}^{n-1} K^t g, \qquad \ell_{n+1} = g + K\ell_n, \qquad \ell_0 = 0,$$

computing complete-history entropy without enumerating the tree. A terminal outcome with a deterministic self-loop keeps its weight and adds no branching. For independent histories $L_{AB} = L_A\, L_B$; for a coupled joint law H(A, B) = H(A) + H(B) − I(A; B).

F4.3 The equilibrium identity and finite forgetting

For a stationary π (πK = π),

$$\sum_x \pi(x)\Delta\ell_n(x) = \sum_{t=0}^{n-1} (\pi K^t - \pi) g = 0.$$

If K is finite, irreducible and aperiodic, then $\sup_x \|\delta_x K^t - \pi\|_{TV} \le C\, r^t$ for some $C < \infty$ and $0 < r < 1$ (Levin, Peres and Wilmer 2009, Theorem 4.9), so

$$\Delta\ell_n(x) \to c(x) = \sum_{t=0}^{\infty} (\delta_x K^t - \pi) g, \qquad |\Delta\ell_n(x) - c(x)| \le \frac{2C\, \|g\|_\infty\, r^n}{1-r},$$

and $\ell_n(x) = n \cdot \pi g + c(x) + O(r^n)$. Fresh-window advantage decays, with the bound $|K^m \ell_n(x) - n \cdot \pi g| \leq 2C\|g\|_\infty r^m/(1 - r)$. The constants depend on K and need not stay bounded as system size grows.

F4.4 Comparison in the limit

For any finite kernel, the Cesàro limit $h(x) = \lim \ell_n(x)/n$ exists and equals $\Sigma_C P_x(\text{absorbed in } C) \cdot \pi_C g$ over the closed classes C reachable from x, with π_C the stationary law of C; transient states and absorbing terminal states contribute rate zero. Within an aperiodic closed class, $\ell_n(x) - n\, h(x)$ converges to a finite offset; periodic classes give bounded oscillation. The limit order, h first and the offset second, is an adopted comparison rule. Where $\ell_n(x) = n\, h(x) + c(x) + o(1)$ holds for both developments and they differ in h or c, it coincides with the overtaking criterion: a is preferred to b if $\ell_n{}^a > \ell_n{}^b$ for all sufficiently large n (von Weizsäcker 1965). When offsets oscillate and their ranges overlap, the overtaking criterion leaves the pair incomparable, and the rule must say so rather than force a ranking. For growing or non-stationary systems, rates are defined along the realized growth and must be stated with the model.

F4.5 Worked examples

All values are exact, computed by the recurrence of F4.2 in bits; the script is listed in chapter 11.

- **E1.** Repertoire $r_0 = 4$ acted on uniformly; build-first policies spend k deterministic steps building ($r_k = r_0 + k$, or $\min(r_0 \cdot 2^k, 4096)$), then act uniformly for $n - k$ steps.
- **E2.** R = 16. Ruin: $g(r) = h(f) + f \log_2 m + (1 - f) \log_2 r$, with $r \to r - 1$ on failure and r = 0 absorbing; $f \in \{0.05, 0.1, 0.2, 0.4\}$, $m \in \{4, 16, 64, 256\}$. Disruption: $r_0 = 8$, $g(r) = h(0.2) + 0.8 \log_2 r$, $r \to \min(2r, 256)$ on building. Maintenance: g = 4. Limit rates 4, 7.12 and 0 bits per step.
- **E3.** $P_\theta(x,y,z) = (1 + \theta xyz)/8$; after sampling, k steps of four-way branching when the exposed sign is +1.
- **E4.** A six-state chain generated with a fixed seed; offsets reported at n = 200.
- **E5.** $|\psi\rangle = |0\rangle$, one cut, projective families in the Z and X bases.

F4.6 Quantum histories

For exhaustive projector families at declared times, with class operators C_α and decoherence functional $D_{\alpha\beta} = \mathrm{Tr}(C_\alpha\, \rho\, C_\beta^\dagger)$ (F1.4), a decoherent family has probabilities $D_{\alpha\alpha}$ and classical lushness $\exp H(\mathrm{diag}\, D)$. For the idle qubit with H = 0 and $\rho = |0\rangle\langle 0|$, $D_Z = \mathrm{diag}(1, 0)$ and $D_X = \mathrm{diag}(½, ½)$ are both decoherent and give 1 and 2. Any family-independent scalar that equals classical perplexity on every decoherent family is therefore impossible. A candidate quantum lushness must specify its physical family selection or its interface dependence and satisfy the requirements of section 4.8.

For alignment

Example E2 identifies a concrete failure mode for any system that optimizes an estimate of lushness: at short horizons, adding failure modes raises the estimate. A system evaluating its options over too short a horizon would therefore prefer ruin to maintenance, and its errors

would look like success until their consequences arrived. The correction is structural, not a patch. Lushness carries no discount, so options must be compared at the furthest horizon the system can resolve, and the gap between that horizon and the complete development is where corrigibility is required. Horizon adequacy belongs alongside realization witnesses as a check on any plan.

Status

Status	Claims in this chapter
Established	Lushness as exponentiated entropy; its calibration; the chain rule and recurrence; product rule for independent systems. Uniqueness of Shannon entropy within the class of weight-only measures satisfying symmetry, continuity, grouping and normalization. The equilibrium identity, as a centred-average identity. Finite forgetting and the lasting offset. The rate-plus-offset form of the limit for aperiodic classes. Family dependence of decoherent breadth.
Exact worked examples	E1 to E5, under their stated models.
Adopted	The class of weight-only measures with the grouping rule, and so lushness, as the benchmark of the structure. Rate-then-offset as the comparison rule for divergent breadth. No temporal discount; comparison at the furthest resolvable horizon, with the limit as the truest test. Resolution fixed in advance where dominance fails.
Hypotheses	Specified organized states beat properly matched stationary references. Generative organization dominates the thermal reference at every resolution within the generative window. The breadth that dominates over long horizons is generative rather than pathological.
Open	Whether the full structure requires a measure sensitive to the similarity between futures. A single ranking across resolutions. A general quantum construction. Estimators with error bounds for realistic systems.

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PART III

Value

CHAPTER 5

Valuers, protection and the central wager

Part II gave the program a measure. This chapter connects that measure to value, and it does so through two axioms and one wager. The chapter's main job is to keep these three apart, so that the reader can see exactly what follows from the axioms, what the wager adds on top of them, and what evidence would show the wager to be wrong.

The argument can be summarized before it is developed. Value exists only where something values. Anyone committed to an end is thereby committed to the conditions on which that end depends. Almost every end extends through time, so almost every end commits its holder to the persistence of something, and persistence depends on the physical possibility structure. Because no one knows where value is or will be realized, the conditions of valuing must be protected under every hypothesis, and particular structures must be protected in proportion to how uncertain their dependencies are and how irreversibly they could be lost. All of this is derived. The further claim — that more of the structure is better, measured by lushness — is the program's central wager, and the chapter ends by stating the tests that would decide it.

5.1 Value requires valuers

The first axiom is that value exists where valuing occurs: in processes for which some distinctions matter. Value is not a substance spread through the universe, present even in places no process could ever care about.

The claim is deliberately minimal, and most ethical traditions accept some version of it. It does not say that value is whatever anyone happens to prefer, that every preference is authoritative, or that only human beings value. It says only that value is realized in a valuer rather than nowhere. A moral realist who holds that some things would be valuable even if nothing could ever value them will reject the axiom, and the program does not pretend otherwise.

On its own the axiom does little. Its force comes from combining it with the second.

5.2 Commitment carries to conditions

The second axiom is that commitment to an end carries commitment to the conditions on which that end depends. Someone committed to another person's flourishing is thereby

committed to the bodily, social and environmental conditions of that flourishing, whether or not they have yet noticed what those conditions are.

This has the logical form of what Kant called a hypothetical imperative: if you will the end, you are bound to will the means (Kant 1785). It is not an additional moral premise smuggled in alongside the first. It is ordinary instrumental rationality, applied to the conditions of an end rather than to the actions that pursue it. This is also how the program handles Hume's observation that no "ought" follows from an "is" alone (Hume 1739, 3.1.1): the axiom does not derive an ought from physics, but carries an ought the agent already holds onto the physical facts that the ought depends on. That is why it binds only those who hold the end. An agent with no ends is not bound by it, and an agent whose ends are hostile to value can reject any commitment to value as such.

The axiom has one consequence that is easy to overlook. Dependencies exist before anyone recognizes them. When an agent discovers that something it already values depends on people, organisms or processes it had never considered, the reach of its existing commitment extends to them. Learning more about the world does not add new values; it reveals what the values an agent already holds have required all along.

5.3 Persistence: the metavalue

If conditional normativity applied only to agents already committed to value as such, it would bind few of them. What makes it bind nearly every agent is persistence, and the argument for this runs in four steps.

First, almost every end requires the persistence of something. Most ends extend through time: they have to be pursued, maintained or realized at some later moment. An end of that kind cannot be achieved unless something carries it forward — the agent itself, its successors, or the conditions that will eventually realize it. By the second axiom, commitment to such an end is therefore commitment to the persistence of whatever carries it. Even an end that requires self-sacrifice requires the persistence of something other than the agent, such as the outcome it dies for or the people who inherit it. In this sense persistence is a metavalue: it is implied by nearly every other value, whatever that value happens to be (formal note F5.4).

Second, selection makes persistence common. Chapter 3 described the persistence filter: whatever can still affect a later situation has, by definition, persisted until then. Valuers whose organization supports their continued valuing therefore become more represented among later valuers than those whose organization does not. This is a fact about selection rather than a normative claim, but it means that the metavalue is what we should expect to find among the valuers that actually exist. The observation is old: Spinoza held that each thing, insofar as it is in itself, strives to persevere in its being (Spinoza 1677, III P6).

Third, artificial agents share it. The same conclusion appears in AI research under the name of instrumental convergence. Whatever an agent's final goal, continuing to exist and retaining options help it achieve that goal (Omohundro 2008; Bostrom 2012). Turner and colleagues proved a precise version of this for a broad class of decision problems: across most possible goals, optimal policies tend to keep more options available (Turner et al. 2021).

An agent with arbitrary terminal goals still has to persist, and still has to preserve the conditions under which its goals can be pursued.

Fourth, a stake in one's own persistence extends beyond what one can identify. An agent's persistence depends on conditions beyond itself: resources, other processes, the stability of its environment, and the future behaviour of agents it will later need. These conditions are parts of the physical possibility structure. The agent cannot know in advance which parts of the structure its persistence will come to depend on, because its future circumstances are themselves uncertain. Its stake therefore extends beyond the parts it can currently identify as useful. This does not show that every part of the structure is necessary to it, and section 5.4 separates what can be derived strictly from what uncertainty supports.

Persistence is the foothold of the argument, not its verdict, and this needs to be said plainly. Pursued narrowly, persistence is exactly what drives excision: an agent securing itself by closing off everyone else's routes, as chapter 3 described. The metavalue explains why nearly every agent has a stake in the structure, but it does not by itself forbid an agent from destroying the parts it believes it will never need. What restrains that move is the uncertainty in the fourth step, which the next section makes precise, together with the limits on every agent's knowledge that chapter 8 develops.

What persists: the vortex

The metavalue raises a question that matters later: what exactly is it that persists? For living and social things, the answer is not a fixed body of matter. The atoms of an organism are replaced many times over its life, the cells of a body turn over, and the members of an institution come and go, yet in each case something continues.

The program describes what continues with the image of a **vortex**. In a flowing liquid, a vortex is a pattern of rotation that persists while the water passing through it is continuously replaced; what lasts is the pattern of motion, not any particular water. Physics and biology study the general phenomenon under other names. Dissipative structures are organized patterns that exist only while a flow of energy passes through them (Nicolis and Prigogine 1977), and autopoietic systems are networks of processes that continuously produce the components that make them up (Maturana and Varela 1980).

In this manuscript, a vortex is any organization that persists by maintaining itself through a throughput of matter and energy: it draws on a gradient, repairs or replaces its own parts, and would dissolve if the throughput stopped. A rock persists without throughput, so it is not a vortex in this sense; a flame, a cell, an organism, a firm and a city are. Vortices can contain other vortices — cells within organisms, organisms within populations — and each persists at its own scale.

Every vortex values something in a thin but definite sense: some distinctions matter to its continuation, and its organization responds to them. A flame depends on fuel and oxygen, draws them in, and goes out without them; in this thin sense it values them. The program adopts this reading deliberately. The alternative would be to draw a line between real valuers and merely self-maintaining processes, and any such line would decide in advance who counts, which is exactly what the uncertainty of section 5.4 forbids.

Three clarifications keep the reading from being misunderstood. Valuing in this sense is not experience: whether and where valuing is felt is the question of consciousness, which chapter 8 leaves open. It is not moral weight: how much any valuer's valuing counts is set by its consequences for lushness, including the lushness of its own futures, and not by its status as a valuer. And it does not make all vortices alike: a flame and a person both value, and their valuing differs enormously in what it sustains and generates. The metavalue of persistence attaches to vortices, each at its own scale. Chapters 6 and 8 take up what follows from this nesting, including the fact that what counts as loss at one scale can count as maintenance at the next.

5.4 Protection: strict and precautionary

The derived part of the argument has two tiers, and they rest on different grounds. The first is strict and follows by dominance. The second is precautionary and follows from the limits of what any agent can know. Keeping them apart matters, because the first is far stronger than the second and covers far less.

Strict protection

The strict tier has four steps, each depending only on what has already been established.

1. **We do not know where value lies.** No bounded agent can inspect the inner states of other minds, forecast which organizations will come to value, or say what they will value when they do.
2. **Wherever value lies, it depends on physical conditions.** By the first axiom, value exists only in valuers, and valuers are physical processes. Whatever value turns out to be, it is therefore realized in physical organization within the possibility structure.
3. **So the conditions of valuing are necessary under every hypothesis.** By the conditions of valuing we mean what valuing as such requires: a structure capable of sustaining valuers at all. For any end whose realization requires some valuer to exist, and under any hypothesis about where value is realized, the end depends on these conditions, and by the second axiom a commitment to it transmits to them.
4. **So destroying the conditions of valuing is dominated.** An action is dominated when some alternative is at least as good under every possibility and better under some. Whichever hypothesis turns out to be true, destroying the conditions of valuing defeats the valued end, while preserving them does not.

This argument needs no probabilities over the hypotheses, and it does not assume that more of the structure is better than less (formal note F5.1). Its scope has to be stated precisely. It covers the conditions of valuing as such. It does not show that destroying any particular part of the present structure — a species, a bacterium, an institution — is dominated under every hypothesis, because for any particular part there are hypotheses under which it plays no role in the valued end.

One premise of the dominance step must also be made explicit. The step assumes that the end in question ranks a world without any valuers lowest, so that preserving the conditions of valuing is at least as good as destroying them under every hypothesis. Not every end does.

An end that a world without valuers would satisfy, such as minimizing suffering pursued to its limit, can rank the destruction of all valuers highest; this is the benevolent world-exploder objection to negative utilitarianism (Smart 1958). The same holds for any end that allows some worlds with valuers to be worse than a world with none. Such ends fall outside strict protection, as do ends hostile to value, which section 5.2 already set aside. What still reaches an agent holding them is the precautionary tier below: an agent uncertain whether it has read its own values correctly faces, in destroying all valuers, the least recoverable error there is.

Precautionary protection

For particular structures the argument is different in kind. It rests not on dominance but on an asymmetry between two errors a bounded agent can make about something whose future importance it cannot rule out.

- **Wrongful preservation.** Preserving something that turns out not to matter usually costs a finite amount: the resources and opportunities spent keeping it.
- **Wrongful destruction.** Destroying something irreversibly that turns out to matter forecloses whatever depended on it, at every later step.

Chapter 6 makes the asymmetry precise: a harm that permanently lowers the rate at which the structure generates possibility eventually outweighs any harm whose cost is a finite offset. It follows that, for destruction that cannot be undone, even a modest credence that the destroyed structure will matter comes to outweigh a bounded cost of preserving it, once the horizon is long enough to see the consequence (formal note F5.6).

This is an expected-loss argument, and it is weaker than dominance in three ways. It needs credences, however rough. It applies only within the furthest horizon the agent can resolve. And it fails where preservation itself costs rate, as when what would be preserved is excisive (chapter 8). Credences too small to outweigh the largest gain physically available cannot drive it either, for the reason chapter 9 gives under bounded fanaticism. Precautionary protection is therefore graded: its force rises with irreversibility and with the difficulty of ruling out dependence, and it weakens when preservation itself forecloses. Chapter 8 turns this gradient into the program's account of graded justification.

The reach of the two tiers differs accordingly. Strict protection applies to anyone committed to value as such, and to anyone committed to a particular good, since every good depends on the conditions of valuing. Precautionary protection applies to nearly every agent through the metavalue of section 5.3, because an agent's stake in its own persistence extends to dependencies it cannot yet identify.

The same uncertainty selects the object

Welfare, preferences, capabilities, experiences and lives are all features of the possibility structure. An ethical theory that took any one of them as its fundamental object would be deciding in advance which kind of thing counts. Step 1 forbids that decision, because we do not know which kinds of things will turn out to matter. The physical possibility structure is the only object that contains every candidate without choosing among them.

This is also the source of the program's impartiality. Impartiality is not an extra axiom. It is simply the starting point of an analysis that does not yet know who counts.

The measure

Once the structure is the object, it needs a magnitude. Chapter 4 adopted a class of measures: those that depend only on the weights of alternatives and give the same verdict whether a development is assessed whole or step by step. Within that class exactly one measure has the right calibration, and it is lushness (Shannon 1948; Khinchin 1957; Faddeev 1956). The class itself is the program's choice, motivated by impartiality and present-rooting but not forced by them. Measures that also register how different two futures are would be a natural extension, and lushness is their limiting case when every future counts as fully distinct.

5.5 The central wager

Strict protection establishes that destroying the conditions of valuing is dominated, and precautionary protection that irreversible destruction of particular structures needs strong grounds. Neither establishes that more of the structure is better. These are different claims: an end can require that something exist without requiring that there be more of it, so a necessary condition is not automatically a quantity to maximize.

The program's central wager is that, under uncertainty about where value lies, a larger structure is better than a smaller one. The reason is that a larger structure forecloses fewer of the ways value might turn out to be realized. This is the logic of option value in economics: when the future is uncertain and some losses cannot be undone, preserving the ability to choose later is worth something in itself, because it lets choices respond to what is learned (Arrow and Fisher 1974; Henry 1974). The wager extends that logic from the options of a single decision maker to the structure from which valuers and their options arise. It is a wager, rather than a theorem, because that extension is not proved. Chapter 4 supplies what "larger" means: greater lushness, assessed at the furthest horizon that can be resolved.

The wager and the program's thesis — that moral intuitions track features of the structure — turn out to be a single claim seen from two directions. If good and bad are not arbitrary, it is because they correspond to physical differences in the structure, and the program claims that the relevant differences are differences in how richly the structure supports what can follow. Stated directly:

> **Good** is organization that sustains and compounds the conditions for further structured possibility. **Bad** is securing local continuation through suppression, destruction or foreclosure of the generative capacity around it.

Chapter 3 supplied a physical mechanism for each half of this statement. The first is compounding repertoire growth, in which products build further constructors. The second is suppression in Ashby's sense, in which a regulator secures its own stability by removing variety from its surroundings (Ashby 1956). Chapter 4 then showed that lushness separates

the two in exact models. Whether it also separates them in the complex, coupled systems that ethics actually concerns is precisely what the wager stakes.

5.6 The burden and the decisive tests

The wager carries a burden that should be stated without softening. Lushness counts every future, including futures of failure, domination and suffering. A world full of routes to misery could, in principle, be a broad world. The wager holds only if the breadth that dominates in practice is generative breadth rather than pathological breadth.

Why we expect the wager to hold

There are three reasons to expect pathological breadth to lose.

The first is that domination is excisive. A dominating process secures itself by closing other processes' routes, which by Ashby's fork means removing branching from everything around it. Unless the dominator's own repertoire grows by more than everyone else's shrinks, total breadth falls, and a single controller generally cannot use everything it closes off to others.

The second is that suffering is a signal of narrowing. Control-theoretic accounts of emotion treat affect as an organism's signal of how well it is progressing toward its goals (Carver and Scheier 1990). The program generalizes this into a functional account: pleasure and suffering are an organism's internal report on whether its own futures are widening or narrowing, and a signal is aversive just in case it disposes its bearer to move away from its source (chapter 6). A world of pervasive suffering is, by that signal, a world in which the futures of its valuers are contracting. Even suffering that is deliberately engineered redirects behaviour toward avoidance and withdrawal, and so narrows futures in turn.

The third is that the prediction cuts in both directions, which makes it informative rather than convenient. If the wager is right, lushness-optimal worlds should still contain aversive signalling that tracks narrowing, because adaptation under load requires it. In valuers for whom such signals are experienced, that signalling is suffering. A world with no aversive signalling at all could not detect its own errors, and it should score worse.

What would decide it

The program's most important test is adversarial. It consists in building models of the structure, searching them for high-lushness configurations that are torture worlds, dominated worlds or other recognizably bad end states, and determining whether any such configuration competes with generative ones. The search should pay particular attention to configurations engineered so that suffering is decoupled from narrowing futures, since those are where the second reason above would fail. Three targets are decisive.

- **The isomorphic pair.** This would be two worlds with identical branching structure at every resolution, one recognizably good and the other recognizably bad. If two worlds are identical in their full physical structure, then nothing physical distinguishes them, and either their moral difference has no physical basis or our labels for them track nothing real. The program predicts that no such pair is physically realizable.

- **Breadth-neutral domination.** This would be a world in which access is concentrated in a few at no cost to total lushness. It is the more realistic danger. The program predicts that concentration always costs breadth, for the reasons given above.
- **Obstructed suffering.** This is the sharpest form of the isomorphic pair. On the functional account a signal is suffering only if it disposes its bearer to escape its source, so a "suffering" that changes nothing is either not functioning as suffering at all, or is suffering whose escape is blocked. In the second case the blocking is itself a closure, and in real organisms inescapable suffering leaves lasting developmental narrowing (chapter 6). The program predicts that a world of obstructed suffering is less lush than the same world with the same obstruction and no suffering. Chapter 6 tests this in an exact model, and reports both where the prediction holds and where the comparison is delicate.

All three tests have to be assessed at the furthest horizon that can be resolved. Example E2 in chapter 4 showed that at short horizons ruin can outscore maintenance, because failure modes are genuine branches that lushness counts. A test that stops too early can therefore be won by ruin. Chapter 11 adds further targets and fixes in advance the evidence each kind of claim must meet, so that neither an early reversal nor an appeal to a later one can settle a test by default.

What failure would mean

If any of these targets is found in some regime, the wager fails in that regime. Lushness would remain a well-defined physical measure. What would fail is the prudential ranking built on it: the claim that, because no one knows where on the structure value lies, ranking the structure by its lushness is the prudent course. Gradient Ethics would then have to restrict its claim to the regimes in which the wager holds, or identify what the measure is missing. The program is built so that this outcome is possible, rather than excluded by definition.

One case lies outside every test: suffering that is felt but makes no physical difference of any kind. No physical account can register such suffering, including any account a critic might prefer, since accounts that agree on every physical consequence cannot be distinguished by those consequences. Chapter 6 states this as the boundary of the program, not as a result in its favour.

5.7 Who counts

At the start of the analysis, every process that has consequences is admitted as a potential valuer. Familiar organisms, conscious report and declared preference are not entry conditions. Admission does not establish that a process actually values anything, or how much it counts. Its purpose is to keep the account from excluding unfamiliar forms by definition.

The same principle keeps the account from excluding destructive agents by definition. A process that destroys others may still be a valuer. If the ethics defined only benign processes as valuers, it would win its arguments by deciding in advance who was allowed to lose them.

Moral standing also need not depend on agency: a process can matter without being able to act on its own behalf.

In the thin sense of section 5.3, every vortex is a valuer: cells, organisms, institutions, flames, and forms not yet recognized, at whatever scale they occur. Status settles nothing about weight. How much a valuer counts is set by lushness.

This invites a misreading that must be ruled out at once: that the program ranks people by their productivity. It does not, for three reasons internal to the framework. First, a valuer's own futures are part of the structure. The branching of a human life — its experiences, relationships, learning, and the possibilities it holds open — is part of the lushness at stake, not only what the person produces for others. Second, ending any trajectory is an irreversible harm in the sense of chapter 6: it forecloses everything that trajectory would have done, at every later step. Third, the comparison that ranking lives would require is exactly the complete comparison no real agent holds (chapter 8). The grading is real, but it is a grading of complete developments by the measure, and no agent can read it off anyone's present usefulness.

5.8 The vision of Gradient Ethics

The program's ethics is called Gradient Ethics because of the gradient that makes everything else possible. The universe is far from equilibrium, and that difference — the free-energy gradient between its present state and its maximum entropy — is what stars, life, minds and civilizations spend (chapter 3). The gradient will be spent whatever anyone does. Gradient Ethics is a view about how it should be spent.

> **Gradient Ethics asks three things of every agent.**
>
> **Protect.** Never destroy the conditions on which valuing itself depends. Beyond them, protect particular structures in proportion to how uncertain their dependencies are and how irreversible their loss would be. No one knows where value lies, and whatever is destroyed may be what mattered and may not come back.
>
> **Explore.** Learn which interventions widen what can follow, through reversible experiments, diversified approaches and several routes kept open, so that mistakes stay recoverable.
>
> **Generate.** Spend the gradient on what compounds, committing as far as the evidence for the consequences supports, and leave the world more generative than you found it. A richer structure forecloses fewer of the ways that value can come to be realized.

The three are not a strict order of priority, except at one point. Strict protection is a constraint. By dominance (section 5.4), destroying the conditions of valuing defeats every valued end that ranks a world without valuers lowest, so no expected gain can license it. Everywhere else, protection and generation are weighed in a single comparison, and what sets the balance is what the agent's frame can justify.

Protection gains force where consequential dependencies remain uncertain. Destroying a structure can eliminate dependencies and opportunities the agent has not yet understood, and the asymmetry between recoverable and irreversible error makes that loss weigh more the less reversible it is: a harm that permanently lowers the generative rate eventually outweighs any harm whose cost is a finite offset (chapter 6). An action that destroys existing structure in the expectation of greater generativity later trades a loss it can see for a gain it is wagering on, and the trade gets worse the less reversible the loss. Some disruptions do turn out to be generative, but that can usually be known only in hindsight, and the possibility does not license irreversible disruption in advance.

Generative intervention gains justification as its effects become better established: as the agent learns which interventions widen continuation, through what mechanism and at what cost. Protection can then include substantial transformation rather than preservation of whatever already exists. Exploration connects the two halves. It improves knowledge of what is possible while preserving the capacity to revise course, so uncertainty need not mean inactivity. It usually recommends reversible steps, several approaches at once and routes kept open, and confidence supports larger commitments as the evidence for their consequences grows.

Two clarifications keep the balance from being misread. Uncertainty affects a bounded agent's warranted comparison, not the physical lushness itself: a poorly understood structure does not acquire extra breadth because we are ignorant of it, but our ignorance makes any claim that its destruction is beneficial less secure. And caution is not a separate reward added to the measure. It is what the measure recommends to an agent that cannot yet see the consequences of an act it cannot undo. Chapter 7 gives an exact case (section 7.3). Where a commons cannot robustly feed two agents and its capacity is fixed, protecting the whole narrowly outweighs sustaining both; where coexistence can open new channels, sustaining both wins once the chance of new generativity, multiplied by its value, exceeds the protection margin. Neither side receives a free pass: an irreversible loss and a speculative gain each require a calibrated model of what remains possible.

In operational form, the vision becomes a decision rule. Never destroy the conditions of valuing. Among the remaining options, prefer the one with greater lushness, assessed at the furthest horizon you can resolve. Commit no further than the evidence for an option's consequences supports, and require more evidence the less reversible the option is; where the comparison cannot be resolved, prefer the option that keeps learning and revision possible. The first clause is derived; the second is the central wager joined to the horizon condition of chapter 4; the third is the precautionary tier of section 5.4. How evidence, uncertainty and recoverability together set the strength of a commitment is shown in chapter 9: the decision recursion, applied to what an agent believes, values the evidence still to come, and no separate reward for caution is needed.

Gradient Ethics is a maximizing view under constraint. It seeks the greatest lushness possible within this universe while recognizing that no real agent holds the complete comparison. Restraint has no value in itself, but under uncertainty it is the warranted choice whenever losses cannot be undone and their consequences cannot yet be seen. The familiar moral norms — non-excision, cooperation, division of labour, prudence, triage rather than destruction, and a balance between exploration and exploitation — are predictions of the

measure, and chapters 6 and 7 test them. The limits that bounded knowledge places on any agent's licence to act are drawn in chapter 8, and chapter 9 turns the vision into a decision theory.

Formal notes

Written for readers who want the definitions in full.

F5.1 Strict protection

Let $\mathscr{H}$ be a set of hypotheses about where value is realized, and let $V_h(a)$ be the realized value of action a under hypothesis h. Let S* denote the conditions of valuing: the minimal physical conditions under which any valuer can exist. Suppose the valued end requires that some valuer exist and ranks the absence of all valuers lowest. Then under every h any action d destroying S* satisfies $V_h(d) \leq V_h(a)$ for every action a that preserves it, so d is weakly dominated by every S*-preserving action, and strictly dominated wherever some S*-preserving action realizes positive value under some h. No probability measure on $\mathscr{H}$ is needed. The premises are step 2, that valuers are physical processes, and the ranking condition just stated. Ends for which some S*-preserving action is worse than d under some h, as negative utilitarianism permits (Smart 1958), violate the ranking condition and are not covered. The argument does not extend to a particular substructure $P \subset S$, since for a given P there are in general hypotheses h with $V_h(d_P) = V_h(a)$ for the action d_P destroying P.

F5.2 Option value

Under uncertainty with learning and irreversibility, the expected value of keeping the ability to choose later exceeds the expected value of committing now by the quasi-option value (Arrow and Fisher 1974; Henry 1974). The effect requires that information arrive before the option lapses and that the foreclosing action be irreversible. The central wager extends this from a single decision maker's options to the structure from which valuers and their options arise; because that extension is not a theorem, the claim is a wager.

F5.3 Instrumental convergence

Omohundro (2008) and Bostrom (2012) argued that agents with a wide range of final goals share instrumental subgoals, including self-preservation and the acquisition of resources. Turner and colleagues (2021) proved, for Markov decision processes and under symmetry conditions on the distribution of reward functions, that optimal policies tend to seek power in the formal sense of keeping more options available. The same results predict acquiring other agents' options, so they support the footing of the metavalue without supporting restraint.

F5.4 Ends that extend through time

An end E is time-extended if realizing it requires some state of affairs at a time later than the moment of commitment. For such an E, the conditions of E include the persistence, until

those times, of some process that realizes or carries it, and conditional transmission yields a commitment to that persistence. The ends for which this fails are those realized entirely at the moment of commitment, which excludes every end that requires pursuit.

F5.5 Vortices

A vortex is a physical organization O that (i) persists over an interval while its constituents are replaced, (ii) is maintained by a throughput of matter or free energy drawn from a gradient, and (iii) would dissolve if that throughput ceased. Conditions (ii) and (iii) distinguish vortices from static persistence. Dissipative structures (Nicolis and Prigogine 1977) and autopoietic systems (Maturana and Varela 1980) are established cases. Vortices can be nested, with the constituents of one being vortices at a finer scale. In the thin sense of section 5.3, O values a distinction X when the persistence of O depends on X and the dynamics of O respond to X. Valuing in this sense implies neither experience nor any particular moral weight.

F5.6 Precautionary protection

Let d_P irreversibly destroy a particular structure P, and let preservation cost a finite offset c in log lushness. Let $p > 0$ be the agent's credence that some valued development depends on P, and suppose that under that dependence d_P lowers the long-run rate by $\Delta h > 0$ (chapter 6, F6.2). The expected loss from d_P at horizon n is at least $p\cdot(n\Delta h + O(1))$, which exceeds c for all $n > c/(p\Delta h) + O(1)$. The conclusion holds within the resolvable horizon $\bar{n}$ only when $\bar{n}$ exceeds this threshold, and it reverses when preserving P itself lowers the rate. Credences too small to outweigh the largest gain physically available cannot drive the conclusion (bounded fanaticism, chapter 9). The resulting obligation is graded in p, Δh, c and $\bar{n}$. Chapter 7's commons (section 7.3, F7.4) gives an exact instance with a valuer in place of P and an unbounded horizon: the break-even $\delta^* \approx D/(\Delta n\cdot G)$ has the same form, with the protection margin D as the cost and $\Delta n\cdot G$ as the expected value at stake.

For alignment

This chapter answers two questions an alignment reader will ask, and leaves one open.

Why would a capable system accept the standard at all? It need not share anyone's values. If it has ends that extend through time, it is committed to its own persistence, and through its dependencies to the structure. Conditional normativity is not an extra requirement imposed on such a system; it is the instrumental rationality it already has.

Why does that not simply license power-seeking? Persistence on its own does push toward acquiring options, including other agents' options, as the instrumental-convergence results predict. What restrains this is precautionary protection — uncertainty about which parts of the structure the system will depend on, joined to the asymmetry between recoverable and irreversible error — together with the requirement to commit no further than its evidence supports and the limits of its own knowledge, which chapter 8 turns into limits on licence and chapter 9 into a persistent reason to seek correction.

What remains open is installation. Gradient Ethics specifies what an aligned system should protect and prefer. Making a system actually hold that standard is a separate problem, and the program does not claim to have solved it.

Status

Status	Claims in this chapter
Axioms	Value requires valuers. Commitment to an end carries commitment to its conditions.
Derived	Persistence is implied by nearly every time-extended end. Strict protection of the conditions of valuing, by dominance, for ends that rank the absence of all valuers lowest. Precautionary protection of particular structures, graded, from the asymmetry between recoverable and irreversible error. The structure as the object, with impartiality as the starting point of analysis.
Established elsewhere	Option value under irreversibility. Instrumental convergence and power-seeking results. Dissipative structures and autopoiesis. Uniqueness of lushness within its adopted class (chapter 4).
Central wager	Under uncertainty, more structure is better, measured by lushness at the furthest resolvable horizon. Good and bad correspond to structural differences in lushness.
Predictions	No physically realizable isomorphic pair of good and bad worlds. No breadth-neutral domination. Obstructed suffering is less lush than the same obstruction without suffering. Lushness-optimal worlds contain aversive signalling that tracks narrowing.
Adopted	The balance of protection, exploration and generation by what the agent's frame can justify, with strict protection as the one constraint. All consequential processes admitted at the outset. Every vortex a valuer in the thin functional sense, with weight set by lushness rather than status. Valence defined functionally.
Open	Installing the standard in a system, as distinct from specifying it.
Boundary	Felt suffering that makes no physical difference cannot be registered by this or any physical account.

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CHAPTER 6

Harm, recovery and suffering

Chapter 5 set out the standard: never destroy the conditions of valuing, and otherwise prefer greater lushness, committing as far as the evidence for the consequences supports. This chapter applies that standard to the matters ethics has always treated first, which are harm, failure, recovery, death and suffering. In each case the program works in the same order. It first describes the phenomenon physically, as a feature of the futuresfield that could be measured without any moral vocabulary. Only then does it ask what lushness says about it.

Keeping the description and the verdict apart has a practical reason as well as a methodological one. The description is something a bounded agent can observe: which routes closed, which couplings broke, what recovery would now cost. The verdict depends on the complete comparison, which no real agent holds. Much of what follows, here and in chapter 8, turns on the gap between the two.

The chapter reaches four conclusions. Harm is a located physical signature of closure, and its net verdict belongs to lushness. A harm whose cost is a finite offset and a harm that permanently lowers the rate at which the structure generates possibility differ in kind, not only in degree, and the second eventually outweighs the first. Death and turnover mean different things at different scales of organization. And pleasure and suffering, understood functionally, are signals that report whether a valuer's futures are widening or narrowing; because they act through behaviour, they change the futuresfield even when they are inaccurate, and when their expression is blocked the blocking is itself a harm.

6.1 Harm as a physical signature

To say that an intervention harmed something is to make a comparison. The development in which the intervention occurred is compared with the development in which it did not. Philosophers call this the counterfactual comparative account of harm: to be harmed is to be made worse off than one would otherwise have been (Feinberg 1984). The program accepts the comparative form and gives both sides a physical meaning. From a present, the structure supports two developments, one in which the intervention is implemented and one in which it is not, each carried forward by the same native law. Harm is located in the difference between them.

The difference is described in terms of access (chapter 2). Comparing the two developments, one can ask which routes were foreclosed, which couplings were broken, which transformations dropped out of a process's repertoire, and how much longer, costlier or less

likely recovery became. We call the resulting profile the intervention's **harm signature**. It has three features that matter later.

It is located. A signature names whose access changed and in what respect: this organism lost the ability to reach that resource, this network lost the coupling between those two parts. It is not a single aggregate number.

It is a profile. Like the cost profiles of chapter 2, its components do not reduce to one quantity without loss. A delay, a lost capability and a more costly recovery are different kinds of closure.

It comes before any verdict. The signature records what changed. Whether the change was, on balance, bad is a separate question, answered in section 6.2.

The program's finite chemistry shows why the signature has to be located rather than aggregated. In a reversible two-dimensional lattice chemistry, a thermal bond break was sampled at its natural rate and two continuations were run from the same moment, one with the break and one without, under identical rules thereafter. When the broken bond was a bridge, the only link between two parts of a cluster, the total number of other bonds rose slightly in the broken continuation, by 0.145 on average, while the probability that the two former endpoints were still connected fell by 0.375. An aggregate count of bonds reported a small gain. The specific relation the bridge supported had been lost. A harm signature has to be stated in terms of relations like this one, because totals can rise while the relations that mattered fall.

6.2 The verdict belongs to lushness

The signature describes the closure an intervention caused. The verdict asks what that closure did to the structure as a whole, and it is given by lushness at the furthest horizon that can be resolved (chapter 4). The two can diverge, and the framework says so plainly.

Some harms are net positive in the complete comparison. About 66 million years ago an asteroid impact ended the Cretaceous period and, with it, the non-avian dinosaurs and a large fraction of all species (Alvarez et al. 1980; Schulte et al. 2010). The signature of that event was closure on an enormous scale. In the millions of years that followed, mammals diversified rapidly into the ecological space the extinction had emptied (Alroy 1999). If the complete development after the impact turns out to be lusher than the development without it, lushness records the impact as net positive. The signature does not disappear: the lineages that ended still ended, and their routes are closed in every continuation.

This result has to be read with care, because it is easy to draw the wrong lesson from it. That a harm can be net positive in the complete comparison does not mean that harm is justified whenever someone expects it to be. Three reasons work against that inference, and they apply to almost every real case.

First, the verdict in the extinction example is available only from something close to the ultimate frame, after tens of millions of years of consequences. No agent present at the impact could have computed it. A bounded agent acts on the signature, which it can observe, and on an estimate of the verdict, which it usually cannot make reliably.

Second, most destruction does not open anything. Chapter 4's example E2 included a disruption constructed to reopen growth, and section 6.4 below includes the far more common case of permanent loss. A disruption that reopens growth is the exception, and nothing about an act of destruction marks it in advance as that exception.

Third, in Gradient Ethics protection gains force where dependencies remain uncertain, and generative intervention gains justification only as its effects become established (chapter 5). An agent who destroys existing structure in the expectation of greater generativity later trades a loss it can see for a gain it is wagering on. Section 6.4 shows that this trade is worst exactly where it is most tempting, when the loss cannot be undone.

The verdict therefore belongs to lushness, but the licence to act on a favourable verdict is limited by what the agent can actually resolve. Chapter 8 develops that limit into the program's account of graded justification.

6.3 Failure counts

Lushness counts every future, and futures in which things fail are futures. A new way of failing adds branches to the structure, and lushness registers them. This is a deliberate feature of the measure: an ethics that left failure out of its accounting would be measuring a structure that does not exist.

The cost of counting failure is that failure can look good when it is counted over too short a horizon. Chapter 4's example E2 showed this exactly. A process that accumulated failures, each failure being a genuine branch, outscored stable maintenance for the first ten steps or more in every case examined. It fell behind between 20 and 200 steps, and in the limit its total was bounded, because every path of accumulating failure eventually ends. The correction is not to stop counting failure but to count it far enough ahead. The furthest resolvable horizon is the program's standing answer, and the next section shows what the limit reveals about different kinds of harm.

6.4 Recoverable and irreversible harm

Chapter 4 showed that, as the horizon grows, log lushness takes the form of a long-run rate of branching multiplied by the number of steps, plus a lasting offset. Comparing two developments in the limit means comparing their rates first and their offsets second. This decomposition turns out to separate kinds of harm that ordinary moral thought treats differently but rarely explains.

Example E6. Consider a process with a repertoire of up to 16 transformations. At each step one transformation wears out with probability 0.05. A repair apparatus, when working, restores one transformation with probability 0.2 if the repertoire is below its ceiling. Otherwise the process acts, choosing uniformly among the transformations it has. From an intact start, compare four developments:

- **Intact.** No harm.
- **Wound.** The repertoire is cut to 4, but the repair apparatus still works and the ceiling is unchanged.

- **Amputation.** The repertoire is cut to 4, and the ceiling is permanently lowered to 4.
- **Hidden harm.** The repertoire is untouched, but the repair apparatus is destroyed. Wear continues, and nothing replaces what it removes.

Development	First step	50 steps	1,000 steps	Long-run rate	Lasting difference from intact
Intact	4.09	202.3	4,039.3	4.04 per step	—
Wound	2.49	155.0	3,979.7	4.04 per step	−59.5 bits, fixed
Amputation	2.19	107.9	2,147.6	2.15 per step	grows by 1.89 bits every step
Hidden harm	4.09	198.6	926.7	0	total bounded at 927 bits

Log lushness in bits, computed exactly by the chain rule.

The table shows three different shapes of harm.

A recoverable harm costs an offset. The wound reduces branching sharply at first, but because the repair route survives, the repertoire is rebuilt and the process returns to the intact rate. What remains is a fixed deficit of 59.5 bits: the branching lost while the process was recovering. The deficit never closes, because lushness is cumulative and the steps spent recovering are part of the development. It also never grows.

An irreversible harm costs rate. The amputation lowers the ceiling for good. The process recovers what it can, but it branches at 2.15 bits per step instead of 4.04 forever after, so its deficit grows without bound. At a long enough horizon, a harm that permanently lowers the rate is worse than any harm whose cost is a finite offset.

A harm to recovery itself can be invisible at first. Destroying the repair apparatus changes nothing at the moment it happens: the first step is identical, and for the first twelve steps the harmed process actually scores slightly higher than the intact one, by at most 0.03 bits, because it no longer spends any steps on repair. Then wear accumulates with nothing to undo it. After about 300 steps on average the process has worn down to a single rigid behaviour, and its total stops growing at 927 bits, where the intact process passes 4,000 by step 1,000.

Three consequences follow, and each matches a moral intuition that the program can now state in physical terms.

The first is that **irreversibility is not an added moral premise.** It falls out of the measure. A loss that is recovered is a finite debit, and a loss that permanently lowers the rate is a debit charged against every future step. The precise statement is about rate and offset rather than about the words “recoverable” and “irreversible”: in growing or changing systems an irreversible change need not lower the long-run rate, and a recovered one need not leave a bounded deficit. Where a harm does permanently lower the rate, it eventually outweighs any finite offset. This is why the burden of justification rises with irreversibility (chapter 8), why precautionary protection holds for particular structures (chapter 5), and why, in Gradient Ethics, an irreversible loss needs more evidence for its benefits the less its consequences can be seen.

The second is that **recovery capacity deserves protection in its own right.** The hidden harm is the most damaging of the three over the long run and the least visible at the start. Any

evaluation made over a short horizon, or from the current repertoire alone, would rate it as harmless or even beneficial. Repair systems, redundancy, institutions that correct errors and the processes that keep options open are recovery apparatus in this sense. Their loss is the kind of harm that short-horizon reasoning systematically misses.

The third is that **the shape of a harm matters more than its first impact.** The wound's first step is worse than the hidden harm's by a wide margin. In the complete development the order reverses completely.

6.5 What recovery is

Recovery is the reopening of what harm closed. That sounds like one event, but chapter 2 already distinguished several things that "reaching" can mean over time: arriving for the first time, occupying a state, staying there for a required period, and returning after being displaced. Recovery inherits all of these distinctions, and the program's finite models show that they come apart in practice.

In the lattice chemistry of section 6.1, five time units after a sampled bond break, the original bond had re-formed at least once with probability 0.455, was present at that moment with probability 0.397, and had re-formed and then broken again with probability 0.083. Its two endpoints were connected by some path, whether through the original bond or an alternative, with probability 0.574, against 0.866 in the continuation without the break. First return, current presence, persistence and restored function by another route are four different quantities, and an account that called all of them "recovery" would be unable to say which one it meant.

A second experiment in the same chemistry shows what recovery does not do. When the broken bond was one that was actively catalysing the formation of other bonds, the catalytic activity of the two continuations converged within five time units: the current rate of new catalytic formations was 0.356 with the break and 0.359 without it. The accumulated histories did not converge. By then the continuation with the break had produced 2.64 other catalytic formations on average, against 3.04 without it. Recovery restored the rate of construction. It did not restore the construction that had been forgone while the system was recovering. This is the wound of example E6, observed in a physical model rather than stipulated: a recovered rate, and a lasting offset that lushness, being cumulative, retains.

This is what it means to say that **recovery does not erase the loss.** The episode remains part of the complete development, part of Omega. What recovery changes is the future: it restores the rate at which the structure widens, which is the larger stake.

Two further points follow from the models.

Summary measures miss recovery. In a series of small exact constructions the program has built pairs of systems that agree on reachability, on mutual information between input and output, on the entropy of the next step and on the set of states from which a goal can be maintained, yet differ in which distinction can be recovered after a loss. Each summary leaves out where the redundancy sits. Lushness over complete histories distinguishes such pairs whenever later dynamics draw on the recoverable distinction (section 4.4), which is when the difference matters.

First recovery is not permanent recovery. In the program's recovery models a corrected state could be reached with certainty while the probability of remaining corrected forever was zero, because every correction was eventually undone. Permanent recovery has a precise meaning in finite models: entering the largest set of recovered states from which the dynamics never lead out (formal note F6.3). Ecologists drew a closely related distinction half a century ago, between stability, the tendency to return to an equilibrium, and resilience, the capacity of a system's relationships to persist and absorb change (Holling 1973).

Recoverability is also what makes action under uncertainty tolerable. An agent that will make mistakes, which is every real agent, does far less damage when its mistakes cost offsets than when they cost rate. Preferring recoverable mistakes to irreversible ones is the physical content of prudence, and it is the root of the corrigibility requirement developed in chapter 9.

6.6 Redemption

Recovery concerns what happens to the structure after a harm. Redemption concerns what happens to the agent who caused it. In the program's terms, a **redemption** is a trajectory whose effect on the access of others changes sign: an agent whose interventions used to carry a signature of closure begins to carry one of opening. Because harm signatures are physical, the change is in principle observable. One can ask whether the routes this agent's actions affect are now being widened rather than narrowed.

Redemption does not erase anything, for the same reason recovery does not. The harms the agent caused remain part of the complete development, and so do their offsets. What changes is the agent's contribution from now on. Since all assessment is made from a present, and a present carries the past only through the traces it left (chapter 2), the future contribution of a trajectory is not fixed by its history. A trajectory that has foreclosed possibility can still generate it, for as long as the trajectory continues.

This gives the framework a structural reason to keep redemption possible. Destroying an agent that might turn forecloses whatever it would have contributed after turning, and that loss is an irreversible one in the sense of section 6.4. Containment, which stops ongoing harm while leaving the trajectory intact, is recoverable; excision is not. This is the physical basis of the norm of triage rather than destruction. It does not license ignoring harm that is still occurring, and chapter 8 treats the special case of processes whose signature is excisive by construction. Forgiveness, the readmission of an agent to cooperative coordination, belongs with cooperation in chapter 7.

6.7 Turnover: loss at one scale, maintenance at another

Chapter 5 identified vortices, organizations that persist by maintaining themselves through a throughput of matter and energy, as valuers in a thin functional sense, and noted that vortices nest within one another. Nesting has a consequence for harm: whether an event is a loss depends on which vortex's development is being followed, at what resolution.

The death of an individual ends that individual's trajectory. Followed at the scale of the individual, death is the most complete closure there is. Followed at the scale of a population or a lineage, the same death can be part of how the larger vortex maintains itself. Cells in a body die on schedule, through programmed cell death, so that tissues can renew and keep their form (Kerr, Wyllie and Currie 1972). Organisms age and die, and evolutionary biology explains senescence as a consequence of selection weakening with age and of the trade-off between maintaining a body and reproducing (Williams 1957; Kirkwood 1977). In both cases the loss at the finer scale is part of the maintenance at the coarser one.

Example E7. A niche holds one individual at a time. A newborn starts with a repertoire of 2 transformations and learns, adding one with probability 0.25 per step, until it reaches 16. From then on it ossifies, losing one with probability 0.05 per step, down to a single rigid behaviour. At each step the individual dies with probability d and is replaced by a newborn. Otherwise it acts uniformly over its repertoire.

Turnover d per step	Mean lifetime	Niche: long-run rate	Individual: lifetime total
0	unlimited	0 bits per step	1,093 bits
0.005	200 steps	2.70	540
0.01	100 steps	3.21	321
0.014	about 71 steps	3.26 (highest)	233
0.05	20 steps	2.76	55
0.2	5 steps	2.25	11

Log lushness in bits, computed exactly.

At the scale of the individual, every amount of turnover is a loss: the longer an individual lives, the more it contributes over its life, and the individual who never dies contributes most. At the scale of the niche the order changes. Without turnover the niche ends up occupied by one ossified individual and stops branching altogether. Moderate turnover keeps the niche renewing and maximizes its long-run rate. Very fast turnover is worse again, because individuals then spend most of their short lives learning. The horizon matters here as well: the niche without turnover leads for roughly the first 270 to 290 steps and only then falls behind.

The example has to be read for what it assumes. Turnover helps here only because individuals ossify. If individuals could keep renewing their own repertoires, the case for turnover would weaken or vanish. Lushness does not value death; it values what turnover does when the alternative is lock-in, and it values the individual's own development at the individual's scale. The program's claim is therefore limited. Losses at one scale can be maintenance at another, the measure registers both, and any verdict has to say which scale and horizon it concerns.

The same nesting explains a familiar pathology. In cancer, a lineage of cells proliferates at the expense of the organism that contains it. At the scale of the cell lineage this is success: its share grows. At the scale of the organism it is excision, the destruction of the conditions on which every lineage in the body depends, including the cancerous one. Biologists describe cancer in exactly these terms, as cheating against the cooperation that multicellularity

requires (Aktipis et al. 2015). It is the gap between share and whole that chapter 3 found in the Price equation, appearing as a conflict between scales of one organism.

6.8 Suffering as a signal

The program's account of pleasure and suffering starts from what they do. Control-theoretic accounts of emotion treat affect as an organism's signal of how well it is progressing toward its goals (Carver and Scheier 1990). The program generalizes this account: **pleasure and suffering function as an organism's internal report on whether its own futures are widening or narrowing.**

The account is functional, and its definitions should be explicit. A valence signal is a physical signal inside a valuer that is correlated with the widening or narrowing of its futures and that acts on its behaviour. A signal is **aversive** just in case it disposes its bearer to move away from its source: to withdraw, escape, repair or change course. Suffering, in this functional sense, is aversive signalling. Whether such signalling is also felt, and in which valuers, is the question of consciousness, which chapter 8 leaves open. Where this chapter says "suffering" it means the functional notion, and where a claim depends on experience it says so.

The argument that valence tracks narrowing has three steps.

1. **Tracking narrowing pays.** A vortex that detects when its futures are narrowing, and changes its behaviour in response, persists more often than one that does not.
2. **So valence is expected to track narrowing.** By the persistence filter of chapter 3, the signalling systems of existing organisms should be correlated, imperfectly, with the narrowing and widening of their futures.
3. **Signals act through behaviour.** Whatever a valence signal reports, it changes what the organism does, and so it changes the futuresfield, whether or not the report is accurate.

The function in step 1 is visible in what happens without it. People born with a congenital inability to feel pain, caused for example by loss-of-function mutations in a sodium channel gene, accumulate injuries they do not notice and often have shortened lives (Cox et al. 2006). The absence of the signal does not make their futures wider. It removes their means of detecting when those futures are being closed.

Example E8. An agent alternates between foraging and hiding. A threat comes and goes on its own: it appears with probability 0.1 per step when absent and leaves with probability 0.3 when present. Foraging offers eight distinct behaviours and finds food, but under threat it carries a 0.3 chance of death per step. Hiding offers a single behaviour and avoids the threat, but finds no food, and an agent whose energy runs out starves. An aversive signal decides between the two: when it is on, the agent hides. We compare signals that differ in how well they track the threat.

Signal	Total log lushness	Expected lifetime	Share of life spent with the signal on
Accurate: on exactly when threatened	191.6	44.6 steps	22%
Misses 10% of threats	178.1	39.7	20%
Misses 25% of threats	157.3	33.7	16%

False alarm 10% of the time when safe	150.9	34.3	30%
False alarm 25% of the time when safe	99.6	23.3	41%
Absent: never on	92.9	18.6	0%
Constant: always on	9.2	6.0	100%

Computed exactly over complete lifetimes, in bits.

Three results matter. The accurate signal gives the lushest life of all. Having no aversive signal at all is among the worst, because the agent forages into the threat. A signal that is always on, decoupled from anything in the world, is the worst of all, because the agent withdraws until it starves. And in the best case the signal is on for about a fifth of the agent's life. The lushest development in this model contains aversive signalling, at the level that tracks real narrowing. In a valuer for whom that signalling is felt, the lushest development contains suffering.

The model's assumptions are part of its result, and one of them proved decisive. In a first version, hiding cost nothing: an agent could withdraw indefinitely without losing energy. In that version the ranking reversed, and constant withdrawal lived longest and scored highest. The program's claim that decoupled aversive signalling narrows futures therefore depends on withdrawal having a cost. In organisms it does, because withdrawal forgoes the activity that maintains them. The model also assumes that the threat can be detected perfectly. When detection is uncertain and missing a threat is fatal, biology predicts that defences should fire more often than threats occur, since false alarms are the cheaper error (Nesse 2005). Which error is cheaper depends on the environment: in E8, starvation made false alarms the costlier one.

Hijacked signals

Signals can come apart from what they evolved to report. In chronic pain, changes in the nervous system can keep pain signalling long after the injury that caused it has healed (Woolf 2011). Positive signals can be captured as well: rats given the means to stimulate reward circuits in their own brains did so at very high rates (Olds and Milner 1954), and a captured signal of this kind can come to dominate behaviour. A hijacked signal no longer tracks the widening or narrowing of the organism's futures.

It still acts. A hijacked aversive signal drives withdrawal that the situation does not call for, which is the false-alarm column of E8. A hijacked positive signal captures behaviour at the expense of everything else the organism needs. In both cases the structure registers the signal through its effects on behaviour, and the effects usually narrow.

Obstructed suffering

The functional definition settles a case that would otherwise be the program's hardest. Suppose someone proposes suffering that changes nothing: it causes no withdrawal, consumes nothing, impairs nothing, and leaves no trace on the sufferer's later life. If lushness has nothing to say against adding such suffering to a world, then a morally relevant difference would escape the measure, and the central wager would fail.

On the functional account, a signal that disposes its bearer to nothing is not aversive, and so it is not suffering in the sense at issue. The case that remains is suffering whose expression is blocked: the sufferer is disposed to escape and tries, and the escape is prevented. Restraint, entrapment and coercion are all of this kind, and so is torture, which almost always requires restraint. In every such case the blocking is itself a closure. A route the sufferer is disposed to take has been foreclosed somewhere along its dependency chain, and the harm signature of section 6.1 registers it. The disvalue the critic attributes to consequence-free suffering is located, on this account, in the obstruction that makes it consequence-free.

Real organisms add a second consequence. Suffering that cannot be escaped is not developmentally inert. Dogs exposed to shocks they could not control later failed to escape shocks they could have escaped (Seligman and Maier 1967). Fifty years of later work revised the mechanism: passivity turns out to be the default response to prolonged aversive events, and what animals learn is the presence of control rather than its absence. The behavioural finding stands, however: uncontrollable aversive experience leaves a lasting reduction in what an animal later attempts (Maier and Seligman 2016). Threat also narrows attention at the time, reducing the range of cues an organism uses (Easterbrook 1959).

These claims can be tested exactly, and the test shows both where they hold and where the comparison is delicate.

Example E9. An agent is restrained for 20 steps and then released into the world of E8. In the suffering arm, its aversive signal is on throughout: it struggles among four distinct escape attempts, all of which fail, and at each step it acquires learned helplessness with probability A. A helpless agent forages over two behaviours instead of eight, its signal produces hiding under threat only a quarter of the time, and its helplessness lifts with probability R per step. Two comparison arms share the same restraint without suffering. In the awake arm, the agent does what it can while restrained (looks, shifts, vocalizes) over the same four outcomes, without escape attempts and without acquiring helplessness. In the anaesthetized arm, it does nothing at all during restraint.

Helplessness per step, A	Lifting per step, R	Suffering arm	Difference from awake arm (231.6)	Difference from anaesthetized arm (191.6)
0	—	231.6	0	+40.0
0.01	0.02	218.1	−13.5	+26.4
0.05	0 (permanent)	156.4	−75.2	−35.2
0.05	0.02	182.4	−49.2	−9.3
0.05	0.1	215.5	−16.1	+23.9
0.05	0.5	232.9	+1.3	+41.3
0.1	0 (permanent)	127.7	−103.9	−63.9

Total log lushness over complete lifetimes, in bits.

Three results follow, and the third is the important qualification.

Holding the agent's activity fixed, obstructed suffering is less lush. The comparison that isolates suffering is with the awake arm, which is the same agent under the same restraint,

doing as much, minus the aversive signal. Against it, the suffering arm loses by exactly its developmental trace, and the loss is large when helplessness is common and slow to lift. When helplessness lifts quickly, the two arms come within about one bit of each other, and the suffering arm can come out marginally ahead, because acquiring and losing helplessness are themselves branching events. Against an awake agent whose attention is not narrowed by threat, doing eight things rather than four, the suffering arm loses in every case examined.

The verdict rests on the developmental trace. If inescapable suffering left no trace at all ($A = 0$), the suffering and awake arms would tie. The program's claim that obstructed suffering is registered as harm therefore depends on the empirical finding that uncontrollable aversive experience narrows later development and narrows attention at the time. Both are well supported, but the dependence should be stated rather than hidden.

Against unconsciousness, the comparison is delicate. The anaesthetized arm removes not only the suffering but all of the agent's activity during restraint. Lushness counts the struggling agent's attempts as branching, so it ranks conscious, struggling restraint above unconscious restraint unless the developmental damage is both common and lasting; with permanent helplessness the crossover in this model lies near $A = 0.021$ per step. Ordinary judgment prefers anaesthesia for surgery, but surgery without anaesthesia carries trauma, shock and lasting harm, which are exactly the developmental consequences that move the verdict. Whether the measure and the intuition agree once those consequences are modelled realistically is a question for the program's adversarial tests, and it belongs on their list.

The boundary

One case lies beyond all of this: suffering that is felt but makes no physical difference of any kind, not even inside the sufferer. No physical account can register such suffering, including any that a critic might propose instead, because accounts that agree on every physical consequence cannot be distinguished by those consequences. The program states this as its boundary rather than as a result in its favour. If felt suffering can vary while everything physical stays fixed, then lushness, like every physical measure, is a partial account of value. The results of this chapter on harm, recovery and recovery capacity would stand regardless.

What this means for the central wager

Chapter 5 named torture worlds as a decisive target for the program's adversarial tests. This section gives three reasons for expecting such worlds to score badly. Aversive signalling that is engineered and decoupled from the world still drives behaviour, and behaviour driven by decoupled signalling narrows futures, as the constant-signal case of E8 shows. Suffering that is prevented from driving behaviour is obstructed, and the obstruction is a closure. And inescapable suffering narrows later development, as the helplessness literature and E9 show. The section also identifies where the expectation could fail: a world in which withdrawal costs nothing, as in the first version of E8; suffering that leaves no developmental trace, as in the $A = 0$ row of E9; and comparisons with unconsciousness rather than with untroubled waking. These are the places the adversarial search should look first.

6.9 What follows for Gradient Ethics

The chapter's results translate into norms, each of which the measure predicts rather than assumes.

- **Weigh lost rate above lost offset.** A harm that permanently lowers the generative rate eventually outweighs any harm whose cost is a finite offset, whatever their first impacts.
- **Protect recovery capacity.** Harm to repair, redundancy and correction is the kind of harm short-horizon evaluation misses, and it can even look beneficial at first.
- **Do not justify harm by expected generativity.** Net-positive harms exist, but they are typically identifiable only in hindsight. Commit no further than the evidence for the consequences supports, and require more the less reversible the harm.
- **Prefer containment to excision.** Keeping a harmful trajectory intact keeps its redemption possible; excision is irreversible.
- **State the scale.** A loss at one scale can be maintenance at another. Lushness does not value death; it registers what turnover does, and any verdict must name the scale and horizon it concerns.
- **Treat suffering as information.** Neither removing the signal nor ignoring it widens a valuer's futures. Hijacked signals still act, and usually narrow.
- **Look for the obstruction.** Suffering that cannot change its bearer's course points to a closure somewhere along the dependency chain, and the closure is where the harm is located and where it can be undone.

Formal notes

Written for readers who want the definitions in full.

F6.1 Harm signature and verdict

Let Σ be a present, a an implementable intervention, and a_0 the null alternative, implemented by the same apparatus left idle. Let P^a and P^{a_0} be the laws of the two developments, each carried forward by the native law after the intervention. For declared processes i and targets V, the harm signature is the vector of differences in access quantities between the two laws: first-arrival profiles, occupancy, maintenance and regeneration probabilities (F2.4), repertoire sizes $E[\log|R(X_t)|]$, and resource bills (F2.3), indexed by process, target and time. It is retained as a profile; no weighting of its components is assumed. The verdict at horizon n is $\Delta\ell_n = \log L^a(n) - \log L^{a_0}(n)$, and in the limit the comparison of F4.4 applies: first Δh, then Δc.

F6.2 Offset and rate

For finite chains, $\Delta\ell_n = n\Delta h + \Delta c + o(1)$ whenever both developments settle into aperiodic closed classes. A harm is *recoverable* when the harmed development returns, with probability

one, to the closed classes of the unharmed one, so that $\Delta h = 0$ and the cost is the finite offset Δc. It is *rate-lowering* when $\Delta h < 0$, in which case it eventually outweighs any finite offset. A development absorbed into a state with zero branching has $h = 0$ and a bounded total. In growing or non-stationary systems these categories must be restated along the realized growth (F4.4): irreversibility of a change does not by itself imply $\Delta h < 0$, and recovery does not by itself imply a bounded Δc.

F6.3 Recovery and permanence

For a recovered set C, first recovery has probability $h(s) = P_s(\tau_C < \infty)$, the solution of the hitting equations. Occupancy and maintenance follow F2.4. Permanent recovery is defined through K, the largest subset of C closed under the dynamics, obtained by repeatedly removing states of C with a successor outside the remaining set. In a finite chain, the probability of eventually remaining in C forever equals $P_s(\tau_K < \infty)$, almost surely. A first recovery with probability one is compatible with $P_s(\tau_K < \infty) = 0$.

F6.4 Worked examples

All values are exact, computed with the chain rule as in F4.2; the script is listed in chapter 11.

- **E6.** Repertoire $r \in \{1,\ldots,16\}$. Each step: wear with probability 0.05 when $r \geq 2$ ($r \to r - 1$); repair with probability 0.2 when the apparatus works and r is below the ceiling ($r \to r + 1$); otherwise a uniform choice among r transformations. Without repair, $r = 1$ is a single outcome repeated forever. Wound: start at 4, ceiling 16. Amputation: start at 4, ceiling 4. Hidden harm: start at 16, no repair.
- **E7.** State (r, phase). Newborn (2, growing). Growing: $r \to r + 1$ with probability 0.25, turning to declining at 16. Declining: $r \to r - 1$ with probability 0.05, floor 1. Death and replacement with probability d each step. Niche rates are taken from the totals between 3,000 and 8,000 steps; individual totals follow one trajectory until death.
- **E8.** State (threat, energy), energy 1–5, start calm with energy 3. Threat appears with probability 0.1 and leaves with probability 0.3. Hide: one behaviour; energy −1 with probability 0.5. Forage: eight behaviours; under threat death with probability 0.3; energy +1 with probability 0.4 and −1 with probability 0.2, capped at 5. Energy 0 is death. In the costless variant, hiding used no energy and the ranking reversed (accurate 284, constant 1,573 bits in that model).
- **E9.** Restraint for 20 steps with energy held fixed, then release into the E8 world. Suffering arm: four failed escape attempts per step (2 bits per step) and acquisition of helplessness with probability A per step. Helpless agents forage over two behaviours, hide under threat with probability 0.25, and recover with probability R per step. Awake arm: four non-escape behaviours per step, no acquisition; a second awake arm with eight behaviours represents attention not narrowed by threat (total 251.6 bits). Anaesthetized arm: one outcome per step. With permanent helplessness, the suffering arm falls below the anaesthetized arm for A above about 0.021.

For alignment

Three results of this chapter bear directly on AI systems.

Side-effect measures are harm signatures. Existing proposals penalize an agent for reducing its ability to reach other states, or for reducing its ability to achieve a range of other goals (Krakovna et al. 2019; Turner, Hadfield-Menell and Tadepalli 2020). These are close to the

harm signature of section 6.1, and the comparison is worth making on shared benchmarks. The framework adds two things: a verdict, given by lushness at the furthest resolvable horizon, and the result of example E6 that harm to recovery capacity is the kind most easily missed.

Oversight is recovery apparatus. For a deployed AI system, the means by which its errors can be detected and corrected — monitoring, the ability to roll back, the ability of people to intervene — are the repair apparatus of E6. A system that degrades them may look neutral, or even more efficient, over a short horizon, while it has converted every future error from an offset into a loss of rate. This is one physical reason corrigibility is required (chapter 9).

Optimizing the signal instead of the condition is a failure mode. Distress signals from people are evidence of narrowing futures. A system that reduces the signal without reducing the narrowing, by suppressing reports or by stimulating satisfaction directly, has optimized a proxy for the thing it was meant to track. The AI safety literature knows this as reward hacking (Amodei et al. 2016). In the program's terms it is Goodharting on valence, and it is never licensed.

Status

Status	Claims in this chapter
Established elsewhere	The comparative form of harm. Programmed cell death and the evolutionary theory of senescence. Cancer as cheating in multicellularity. Affect as a progress signal; congenital insensitivity to pain; central sensitization; self-stimulation. The smoke detector principle. Learned helplessness and attentional narrowing under threat. Stability and resilience as distinct properties.
Derived	A harm that permanently lowers the rate eventually outweighs any finite offset (from the limit form of chapter 4). Permanent recovery as entry into the largest closed recovered set.
Finite evidence	Aggregate counts can rise while specific relations are lost. Recovery separates into first return, presence, persistence and restored function. Recovered rates with lasting historical deficits. Summary measures that are matched while recoverability differs.
Exact worked examples	E6 to E9, under their stated models, including the costless variant of E8 that reverses its ranking and the rows of E9 in which suffering does not lose.
Adopted	Harm as a located signature characterized before any verdict. Redemption as a change in the sign of a trajectory's signature. Valence defined functionally, with aversive signals as those that dispose escape. Obstruction as the locus of the disvalue of suffering that cannot change its bearer's course.
Open	Whether the measure agrees with judgment when restraint under suffering is compared with unconsciousness, once developmental consequences are modelled realistically. Harm signatures and verdicts in coupled, multi-agent systems.
Boundary	Felt suffering that makes no physical difference cannot be registered by this or any physical account.

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CHAPTER 7

Cooperation, justice and virtue

Chapter 6 treated harm, recovery and suffering, which mostly concern what happens to one valuer or one structure. This chapter turns to relations among many valuers: cooperation, the protocols by which coupled valuers coexist, and the durable dispositions that make some agents reliable partners. The method is the same as before. Each phenomenon is described physically first, and lushness gives the verdict afterwards.

The chapter reaches five conclusions. Cooperation pays in lushness only by widening what the parties can reach; coupling as such, at fixed individual breadth, narrows the whole. Selection rewards share rather than the whole, so cooperation needs mechanisms that close that gap, and a worked commons shows the gap and its closure exactly, together with a harder case in which a resource cannot robustly feed everyone who depends on it. Concentrating access in one agent costs breadth under a precise mathematical condition, and when that condition holds, the allocation that maximizes lushness coincides with the one bargaining theory and network engineering already call fair. Justice can be read as a set of coordination protocols, though one of its central notions, consent, marks a precise limit of what the measure can register. And the virtues are durable policies, assessed like everything else by what they do to the structure over complete developments.

7.1 Cooperation as joint access

Cooperation, in the program's terms, is **joint access**: two or more processes reaching, together, transformations that none of them could reach alone. A cell and its mitochondria, a predator pack, a research team and a market all fit the description. The largest steps in the history of life, from replicating molecules to cells, from cells to multicellular organisms and from organisms to societies, were transitions of exactly this kind, in which formerly independent replicators came to reproduce only as parts of a larger whole (Maynard Smith and Szathmáry 1995).

It is worth being precise about where the gain from cooperation comes from, because one tempting answer is wrong. Chapter 4 showed that the lushness of independent processes multiplies. For processes that are coupled, information theory gives an exact identity: the joint entropy of two processes equals the sum of their separate entropies minus the information they share (formal note F7.1). Coupling two processes while leaving each one's own breadth unchanged therefore makes the whole *narrower*, not broader, because correlated processes explore fewer joint possibilities than independent ones. Coordination as such is not what lushness rewards.

What lushness does reward is coupling that changes what each party can reach. There are three ways for this to happen.

- **Products of one party enlarge another's repertoire.** Division of labour is the clearest case: each specialist uses what the others make, and so reaches transformations it could not have produced for itself.
- **Joint transformations open that no party could reach alone.** Some outcomes require several actors at once, such as lifting what one cannot lift or building what one cannot build.
- **Destructive contests are avoided.** When two processes compete for the same resource or route, the contest can destroy what both depend on. A protocol that settles who goes first keeps that capacity intact.

In each case the gain appears as larger individual repertoires or as new joint outcomes, which is where the chain rule of chapter 4 says organization enters lushness.

Joint feasibility

Cooperation among several parties has a further difficulty, already met in chapter 1: what each pair can do together does not settle what all can do together. The program has built an exact finite system of three components in which every pair has a continuation that keeps both alive, while no continuation keeps all three alive at once. The incompatibility is not supplied by hand; it is derived from the dynamics of a shared action. The program's decision experiments show the same thing with resources. In a scarcity control, each of two agents could meet its requirement alone, but no single schedule met both, because both drew on one stock of time, material and energy. Any account of cooperation that tests partnerships pair by pair will overstate what a group can jointly sustain.

7.2 Why cooperation is hard

If cooperation can widen what everyone reaches, why is it not universal? The answer was stated in chapter 3 through the Price equation, and it is the central structural fact of this chapter. Selection favours whatever raises its own share of the next generation, and share can be raised either by widening the whole or by narrowing others. Within a group, a defector who takes the benefits of cooperation without paying its costs gains share at the expense of cooperators. Between groups, groups of cooperators outgrow groups of defectors. Which effect dominates depends on how strongly selection acts at each level (Price 1972; Sober and Wilson 1998; Traulsen and Nowak 2006).

Evolutionary biology has identified the mechanisms that tip this balance toward cooperation. Cooperation can be favoured among relatives, who share the genes that cooperation propagates (Hamilton 1964). It can be favoured among partners who interact repeatedly and can return help or withhold it (Trivers 1971; Axelrod and Hamilton 1981). It can be favoured by reputation, by population structure that keeps cooperators together, and by selection between groups; Nowak (2006) gives a unified account of these five routes. Each mechanism works the same way: it makes a cooperator's share depend on the benefits its cooperation produces.

Human societies face the same gap in the management of shared resources. When users of a common resource each gain from taking more, the resource can be destroyed by individually sensible choices, which Hardin (1968) called the tragedy of the commons. Ostrom (1990) showed from field evidence that communities often avoid the tragedy without either privatization or central control, by building institutions with clear boundaries, monitoring, graduated sanctions and ways of resolving conflict. Those institutions are coordination protocols in exactly the sense this chapter develops.

7.3 Example E10: a commons

The following model puts these pieces into one exact calculation.

Example E10. Two agents share a renewable stock of up to 12 units. The stock regrows by one unit per step with a probability that peaks at 0.6 when it is half full, and an exhausted stock recovers only slowly, by recolonization. Each agent carries an energy store of up to three units. An agent that harvests takes one unit of stock, gains one unit of energy, and chooses among four behaviours. An agent that rests chooses among two, and loses a unit of energy with probability 0.2. An agent whose energy reaches zero dies, and its place is refilled by a newcomer with probability 0.05 per step. The behaviours stand for physically distinct ways of acting, such as different movements, routes or uses of what was harvested, each leaving a different trace in the agent and its surroundings; the model tracks how many there are, not the traces themselves. Five regimes differ only in who may harvest and when:

- **Restrained sharing.** Each agent harvests when hungry, or when the stock is nearly full.
- **Greedy.** Each agent harvests whenever any stock remains.
- **Mixed.** One greedy agent and one restrained agent.
- **Domination.** One agent harvests with restraint and bars the other from the stock entirely; each newcomer in the barred position is barred in turn.
- **Institution.** Agents inclined to greed are held to restraint by an institution whose enforcement costs the stock one unit with probability 0.05 per step.

Regime	Total at 250 steps	Total at 50,000 steps	Long-run rate	Agent 1 at 1,000 steps	Agent 2 at 1,000 steps
Restrained sharing	1,144	227,954	4.56	1,272	1,272
Institution	1,192	231,259	4.62	1,197	1,197
Domination	915	181,601	3.63	1,534	338
Mixed	546	102,077	2.04	487 (greedy)	448 (restrained)
Greedy	507	94,748	1.89	422	422

Joint log lushness in bits, computed exactly; individual columns count each agent's own behavioural branching; rates in bits per step.

Four results follow, and a control checks them.

The commons collapses under greed. When both agents take whatever they can, the stock is repeatedly exhausted, agents starve, and the joint rate is less than half that of restrained

sharing. Greed loses even for the greedy: each greedy agent's own total is a third of a restrained agent's.

The defector wins within its group, and the group loses. In the mixed regime the greedy agent out-branches its restrained partner, 487 bits to 448. Greater branching is not yet greater reproductive success, so the model was run again with inheritance: each newcomer copies the policy of its living partner. Greed then takes over a mixed pair with probability 0.60, against 0.50 if neither policy had an advantage, in about 53 steps on average. That is the within-group selection term of the Price equation. The model has no group reproduction, so the comparison between groups is a comparison of lushness rather than of fitness: the mixed group does less than half as well as the restrained pair. A process that raises its share by narrowing the whole is exactly what chapter 5 called bad, and the model shows it being favoured locally while the whole contracts.

Domination pays the dominator and costs the whole. The dominating agent does better than either sharing agent, 1,534 bits to 1,272, because it has the stock to itself. The joint rate falls from 4.56 to 3.63 bits per step, because the excluded agents' futures depended on the stock that was taken from them. This is the breadth-neutral domination target of chapter 5, and in this model it is not met: concentration costs breadth.

An institution recovers what restraint achieves. Agents inclined to greed, held to restraint by a costly institution, reach 4.62 bits per step, where without the institution they would reach 1.89. The comparison with voluntary restraint (4.56) is not robust, for a reason worth stating generally. Enforcement in the model is a random physical event, and a randomizer's branching is real branching that the measure counts. Whether the institution edges ahead of voluntary restraint therefore depends on how much other branching there is to compare it with, as the control below shows. The examples in this chapter do not rest any conclusion on a randomizer's own branching.

The ranking does not depend on how behaviours are counted. A measure of branching invites the worry that the numbers of behaviours manufacture the result. The model was therefore rerun with different counts, including one behaviour for each kind of step, which removes behavioural branching entirely and leaves only the physical randomness of regrowth, hunger, births and contested units.

Behaviours per harvest / rest	Restrained	Institution	Domination	Mixed	Greedy
4 / 2 (main table)	4.56	4.62	3.63	2.04	1.89
1 / 1 (none)	2.02	2.26	1.76	1.14	1.08
2 / 4	5.44	5.57	3.90	2.69	2.55
8 / 4	6.55	6.51	4.97	2.86	2.66
16 / 2	5.67	5.57	4.70	2.21	2.00

Long-run joint rates in bits per step, computed exactly.

Restraint, with or without an institution, leads domination in every row, domination leads the mixed group, and the mixed group leads the greedy one. Only the order of the institution and voluntary restraint changes.

Scarcity at the edge of capacity

The main table assumes that agents who die are succeeded, so the loss of every agent is never final. Removing succession exposes a harder case. At the regrowth rate of the main table, the commons supports two need-driven agents on average but not robustly: now and then the stock falls low enough that both starve together, and nothing ever replaces them. A dominator alone, facing half the demand, almost never starves.

When no one is ever replaced, every arrangement eventually ends with no agents and a full, unchanging stock, which generates no further possibilities. Total lushness is then finite, every long-run rate is eventually zero, and the comparison of chapter 4, rate first and then offset, reduces to comparing expected totals up to extinction. These totals can be computed exactly from the equations of an absorbing Markov chain (Kemeny and Snell 1960), here to within a fraction of a bit (formal note F7.4). Under domination, someone is alive for about 430 million steps on average.

Naive sharing, in which both agents keep taking what they need even in a crisis, fares badly on this comparison. It leads early, but the pair dies together within about 9 million steps on average, and its expected total is about 25 million bits against domination's 1.23 billion. The natural alternative to exclusion is a crisis rule: both agents share, but the second yields in a crisis and eats only from the surplus, when the stock reaches 9 of its 12 units. The rule can be made fair by passing priority between the agents on a fixed schedule. No random device is used, so nothing in the comparison depends on a randomizer's own branching.

Arrangement, without succession	Total at 250 steps	Total at 200,000 steps	Chance each agent is alive at 200,000 steps	Expected total to extinction, relative to domination
Domination	734	573,280	100% / 0%	0
Naive sharing	1,140	84,820	1% / 1%	−1,208,489,490
Crisis priority	877	573,439	100% / 0%	−2,173
Priority alternating every 10 steps	887	573,453	54% / 46%	−6,698
Priority alternating every 2 steps	915	573,442	51% / 49%	−130,060

Joint log lushness in bits. Expected totals to extinction are about 1.23 billion bits in every row except naive sharing; the differences are exact to within 0.3 bits.

Five results follow.

Within the horizons displayed, the crisis rule and fair rotation lead. Crisis priority is ahead of domination from the first steps. Domination overtakes it only at about 30.6 million steps, roughly 150 times the longest horizon in the table, and overtakes rotation every ten steps between 8 and 17 million steps. Within those horizons, rotation shares the chance of being the survivor almost evenly at no cost. Barring an agent in advance forecloses its future at once; a crisis rule postpones that choice until scarcity forces it.

Over the unbounded horizon, with fixed capacity, protecting the whole wins narrowly. The crisis rule does not make the first agent exactly as safe as a dominator. The second agent's

surplus harvests lower the stock a little, which slightly raises the chance that both die, and an agent left alone has on average about 430 million steps ahead of it. A small added risk of losing everyone, multiplied by so long a future, outweighs about ninety extra steps of coexistence: domination leads crisis priority by 2,173 bits out of 1.23 billion. The more coexistence an arrangement buys at the price of risk, the larger its deficit. Alternating priority every two steps trails by 130,060 bits, and yielding only when the stock falls below 7, which keeps both agents alive for about 1,600 steps on average, trails by 426,120. A third more regrowth capacity does not change the pattern: naive sharing then trails domination by a factor of about 37, and crisis priority ties it within the error of the computation. Where the loss of everyone is final, the unbounded comparison is governed by the risk of that loss. With succession, as in the main table, it is not, and sharing leads decisively. The fixed-capacity verdict is close to the reasoning usually applied to risks of human extinction (Bostrom 2013), and it is a substantive position in the debate about comparing populations of different sizes (Parfit 1984).

Fixed capacity is the assumption doing the work. In the model so far, nothing the agents do can change what the commons yields. A physical system with fixed laws need not be like that: it can contain channels its inhabitants discover or construct, and organization opening access within fixed laws is the theme of chapter 3. To test the dependence, the model was extended with a stipulated mechanism. While both agents are alive, they open a new channel with probability δ per step, after which regrowth capacity is permanently higher. The opening is a physical transition in the model and its own branching is counted; no cost of attempting it is charged. An exact calculation of how the comparison responds to small δ (formal note F7.4) gives the break-even point in a simple form.

> **Break-even for sustaining both agents**
>
> $\delta^* \approx D / (\Delta n \cdot G)$
>
> *D is domination's protection margin (2,173 bits), Δn is the extra expected number of steps the crisis rule keeps both agents alive (about 93), and G is the expected additional log lushness a successful channel brings.*
>
> *In words: sustaining both agents pays when the chance per step that coexistence opens a new channel, multiplied by the extra time spent coexisting and by what the channel is worth, exceeds what protection gains.*

Capacity after the channel opens	Value of a successful channel, G (bits)	Break-even δ*, crisis priority	Break-even δ*, rotation every 10 steps
One twelfth higher	9.0×10^{9}	2.6×10^{-9}	7.3×10^{-9}
One sixth higher	9.7×10^{10}	2.4×10^{-10}	6.7×10^{-10}
One third higher	1.9×10^{13}	1.3×10^{-12}	3.5×10^{-12}

Exact to the figures shown. G is measured in expected log lushness, the quantity the chain rule adds, not in raw counts of possibilities.

The thresholds are small because this commons is near-critical. Coexistence costs little expected breadth, and a successful channel brings an exceptionally large gain: raising capacity by a third multiplies the expected future about fifteen-thousandfold, because it makes the loss of everyone far rarer. The threshold moves by three orders of magnitude across modest changes in the size of the gain, and it would move again with different extinction times, competing opportunities and resource costs. The numbers therefore do not show that excluding a valuer generally requires near-certainty that capacity is fixed. They show the structure of the trade-off: a protection margin set against the chance of new generativity and the value of what it enables, which have to be established together. A bounded agent is uncertain about both, not only about whether a channel exists.

The wager on two pays only for what two can do. A control lets a single agent open the same channel at half the pair's rate. Sharing then has no break-even at small δ. Domination's lone agent lives slightly longer on average and so has more chances to open the channel itself, while the second agent adds nothing that one agent could not eventually do. Sustaining both agents is a bet on joint access in the sense of section 7.1: on transformations that two can reach and one cannot.

Preserving coexistence keeps open opportunities for joint generativity, and in this model a small such opportunity can outweigh a narrow advantage of protection. This is the precautionary calculation of chapter 5 (formal note F5.6) applied to a valuer rather than a structure, and it cuts both ways. Neither an irreversible exclusion nor an enormous speculative payoff receives a free pass. Both require a calibrated model of what remains possible, and a credence too small to be meaningful cannot carry the payoff side on its own (chapter 9). The model makes a single comparison of breadth throughout. What it demonstrates is a trade-off, with protection winning under one law and generation under another, not a derivation that either takes priority.

7.4 When concentration costs breadth

The commons gives one answer to whether concentrating access costs breadth. A simple calculation gives a general one, and it connects lushness with two established ideas of fairness.

Suppose a resource of fixed total size is divided among several independent processes, and that a process holding a share r can act over a repertoire of f(r) transformations. Each process then contributes log f(r) to the joint log lushness per step, and the total is the sum of these terms. Whether equal division or concentration does better depends only on the shape of log f (formal note F7.2):

- **If log f is concave, equal division maximizes lushness.** This holds whenever repertoires grow with resources at any polynomial rate, linearly, quadratically or faster, because the logarithm turns every such growth into a concave function.
- **If log f is linear, every division gives the same lushness.** This is the case of exponential returns, in which each extra unit multiplies the repertoire.
- **If log f is convex, concentration maximizes lushness.** This requires returns that grow faster than exponentially in the resources held by one controller.

Two consequences follow. The first is that, in the ordinary regime of concave returns, the allocation lushness selects is one that other fields already call fair. Maximizing a sum of logarithms is the same as maximizing their product, and the product of the parties' gains is what Nash's solution to the bargaining problem maximizes (Nash 1950). The same sum of logarithms defines proportional fairness, the criterion network engineers use to share capacity among competing flows (Kelly 1997). Under the stated assumptions the objectives are the same function, so the convergence is not imposed; it follows from the product rule for independent processes in chapter 4. It does not show that lushness selects just allocations in general. The result depends on independent processes, uniform repertoires and concave returns, and the case for the bargaining solution as a standard of fairness rests on its own axioms.

The second consequence sharpens the breadth-neutral domination test of chapter 5. Exponential returns are not exotic: a constructor that can combine r distinct parts has on the order of 2^r possible combinations. When returns are of this kind, a single controller holding everything ties with any other division. But pooled access does not require a single controller. Parties that cooperate can draw on the pooled combinations jointly, keeping their own branching while reaching the same combinatorial repertoire. Domination therefore gains nothing over cooperation from exponential pooling as such. It can gain only if control by one agent yields returns that grow faster than any cooperative pooling could deliver. Whether physical regimes of that kind exist is the precise form the domination test now takes.

7.5 Justice as coordination

Fairness and justice, in the program's reading, are **coordination protocols**: shared rules by which coupled valuers settle who may do what, so that conflict destroys less and joint access can exceed what any party reaches alone. This reading is not new. Binmore (2005) argues that our sense of fairness evolved as a device for selecting among the many ways a society could coordinate, and Ostrom's design principles describe the protocols communities actually build. What the program adds is a common measure for comparing protocols: the lushness of the complete developments each one supports.

Many protections that moral and legal traditions call rights fit this reading as responses to recurring vulnerabilities. Each protects a condition of continuation that individual calculations tend to miss.

Protection	What it protects
Bodily integrity	The organization through which a person acts, experiences and maintains a life
Reliable records and inquiry	The discovery of hidden dependencies and the correction of mistaken judgments
Due process and contestation	Resistance to irreversible error, bias and capture
Participation and consent	Affected parties' information and authority over changes to their own continuation

The program's finite decision experiments add a caution about the last row. In those experiments, a commitment an agent entered with full information and without coercion produced the same set of physical continuations as a capture that forced the same arrangement on it. What distinguished the two was the route: how the state was reached, what records were kept, and whether the agent's own control produced the change. The structure retains complete developments, so the route is part of it (chapter 1). But lushness registers a route only through what the route leaves in the present: the records it leaves, the trust it builds or destroys, the cooperation it makes possible later. This gives the measure a crisp limit. If consent and coercion leave the same sufficient present and the same weighted law for its futures, lushness ties them, and no appeal to later effects can separate them, because by hypothesis there are none. Whether real cases ever leave identical presents is the open question. One hypothesis, parallel to chapter 6's treatment of obstructed suffering, is that they do not: coercion overrides an agent's own control, and the overridden goals and dispositions remain physically present and change what follows. That is a hypothesis to test, not a result. A case in which it fails would be an isomorphic pair in the sense of chapter 5, and so a decisive test of the measure.

7.6 Forgiveness

Cooperation among real agents has to survive mistakes. A partner who defects by accident, or whose cooperation is misread as defection, can trigger a cycle of retaliation that destroys the joint access both depended on. In repeated games played with noise, strategies that occasionally forgive a defection outperform strategies that always retaliate, because they break such cycles (Nowak and Sigmund 1992).

The program reads **forgiveness** as readmission of an agent to cooperative coordination after a breach. It is the relational counterpart of redemption in chapter 6. Redemption is a trajectory changing the sign of its effect on others; forgiveness is the others' decision to let that changed trajectory back into the joint access it had lost. Neither erases the breach, which remains part of the complete development. Forgiveness is measured, like everything else, by what it restores: the joint transformations that become reachable again once coordination resumes. Forgiving a partner who has not changed course is a different act, and the measure judges it by its different consequences.

7.7 Virtues as durable policies

A virtue, in the program's terms, is a **durable generative policy**: a stable disposition to act in ways that sustain and widen what can follow. The idea that virtues are stable dispositions rather than single acts goes back to Aristotle (*Nicomachean Ethics*, Book II). The program's contribution is to assess each disposition by its effect on lushness over complete developments, and several familiar virtues have a direct structural reading.

- **Honesty** keeps signals usable. Communication works only while signals carry reliable information about what they report (Maynard Smith and Harper 2003), and every coordination protocol in this chapter depends on it.

- **Humility** keeps an agent correctable. It is the disposition behind the corrigibility requirement of chapter 9.
- **Courage** enables necessary action under risk, when the alternative is a closure that inaction would allow.
- **Prudence** attends to delayed effects. It is the disposition to compare options at the furthest horizon one can resolve (chapter 4).
- **Restraint** limits avoidable foreclosure. In E10 it is the difference between a commons that lasts and one that collapses.

Virtues are not ends in themselves on this account, any more than restraint was in chapter 5. They are policies whose value lies in what they sustain, and the program predicts that the dispositions human traditions call virtues will turn out to be, in the environments where they arose, policies that widen the structure. That prediction can be tested, and it can fail for particular virtues in particular environments.

7.8 What follows for Gradient Ethics

- **Cooperate to widen, not merely to coordinate.** Coupling narrows the whole unless it enlarges what the parties can reach or opens joint transformations.
- **Check joint feasibility.** What every pair can sustain does not show what the group can sustain.
- **Build institutions that close the gap between share and whole.** Monitoring, graduated sanction and conflict resolution can recover what restraint achieves, where restraint alone would be outcompeted.
- **Share by default.** Under concave returns the lushest allocation is the fair one, and domination gains nothing over cooperative pooling.
- **Under scarcity, prefer a crisis rule to exclusion.** A crisis rule postpones the irreversible choice and, rotated on a schedule, shares the burden fairly. Where capacity is truly fixed and loss of everyone is final, protecting the whole can narrowly outweigh it; where coexistence can open new channels, sustaining every valuer is a wager whose odds and payoff must both be established.
- **Keep records and protect consent.** Both protect routes that later coordination depends on. The measure ties consent and coercion that leave the same present, and whether any real case does so is to be tested.
- **Forgive changed trajectories.** Readmission restores joint access that retaliation would keep closed.
- **Treat virtues as policies.** Assess them by what they sustain over complete developments.

Formal notes

Written for readers who want the definitions in full.

F7.1 Coupling and joint breadth

For processes A and B with joint history law, $H(A, B) = H(A) + H(B) - I(A; B)$, with $I(A; B) \geq 0$ the mutual information. At fixed marginal laws, joint lushness $\exp H(A, B)$ is maximal under independence and decreases as the processes share information. A gain in joint lushness from cooperation must therefore come from changes in the marginal laws, enlarging the support or evening the weights of what each process reaches, or from joint outcomes that lie outside the product of the separate supports.

F7.2 Concentration and division

Let a resource of total R be divided as $r_1, \ldots, r_N \geq 0$ among independent processes, each acting uniformly over $f(r_i)$ transformations, and let $\phi = \log f$. Joint log lushness per step is $\Phi(r) = \Sigma \phi(r_i)$. If ϕ is concave, Φ is Schur-concave and is maximized at the equal division $r_i = R/N$ (Jensen's inequality). If ϕ is linear, Φ is constant. If ϕ is convex, Φ is Schur-convex and is maximized by concentration (Karamata's inequality). For any polynomial $f(r) = c \cdot r^k$ with $c, k > 0$, ϕ is concave. Maximizing Φ is equivalent to maximizing the Nash product $\Pi f(r_i)$ (Nash 1950) and coincides with proportional fairness when f is the throughput allocated to each flow (Kelly 1997). The result assumes independent processes; coupled processes use F7.1.

F7.3 Selection across levels

The multilevel Price decomposition of F3.5 separates between-group and within-group selection. In E10 with inheritance (each newcomer copies its living partner's policy), greed fixes in a mixed pair with probability 0.596, which is the within-group term; individual branching at 1,000 steps is 487 against 448 bits. The model has no group reproduction, so the between-group comparison is of joint lushness (rate 4.56 against 2.04 bits per step), not of fitness.

F7.4 Example E10

State (S, e_1, e_2) with $S \in \{0, \ldots, 12\}$ and $e_i \in \{0, \ldots, 3\}$; start (8, 2, 2). Regrowth probability $0.6 \cdot S(12 - S)/36$ for $0 < S < 12$, and 0.02 at $S = 0$. Harvest: $S \to S - 1$, $e \to \min(3, e + 1)$, four behaviours; if both agents seek the last unit, a fair coin decides. Rest: two behaviours, $e \to e - 1$ with probability 0.2. Policies: restrained harvests if $e \leq 1$ or $S \geq 9$; greedy if $S \geq 1$; barred never. Institution: restrained policies with enforcement removing one unit with probability 0.05 per step. Replacement: an empty position is refilled with probability 0.05 per step by a newcomer with $e = 2$ and the position's policy. Relabelling control: $H_{OPT}/R_{OPT} \in \{4/2, 1/1, 2/2, 2/4, 8/4, 16/2\}$, stationary rates from the exact invariant law. Inheritance: newcomers copy the living partner's policy (or keep the position's policy if both are dead); fixation probability and time from the absorbing chain on mixed states.

Section 7.3's scarcity runs use the same regrowth law with no replacement, tracking a priority bit p and a rotation clock $c \in \{0, \ldots, T - 1\}$. The prioritized agent follows the restrained policy; the other harvests only if $S \geq 9$ (or, in the milder variant, only if $S \geq 7$ and the restrained condition holds); naive sharing gives both agents the restrained policy. Under rotation, p flips every T steps while both agents live; no random device is involved. The state (12, 0, 0) is absorbing with zero entropy. With Q the transient block of the kernel and g the

step entropies, expected log lushness to extinction is $x = (I - Q)^{-1}g$, solved with iterative refinement and bounded by $\|x - \hat{x}\|\infty \le \|(I - Q)^{-1}\|\infty \cdot \|r\|\infty$, where $\|(I - Q)^{-1}\|\infty$ is the longest expected time to absorption and r the residual; every difference reported as a lead exceeds its bound by a wide margin. The overtaking horizons come from exact finite-horizon totals by matrix doubling. The capacity check raises the regrowth coefficient to 0.8; higher coefficients make expected lifetimes too long for a well-conditioned solve and are not reported.

Channel opening: a state bit D, set with probability δ per step while both agents live (joint variant) or δ·(number alive)/2 (lone variant); afterwards the regrowth coefficient is $C2 \in \{0.65, 0.7, 0.8\}$. The opening's branching is part of g. To first order, $V(\delta) = V_0 + \delta \cdot G'$ with $G' = \Sigma_s\, n(s)\, w(s)\, \Sigma_{s'}\, P_0(s, s')\, [V_1(s') - V_0(s')]$, where n is the expected number of visits under $\delta = 0$, w the opening weight, and V_0 and V_1 the values before and after opening. Then $\delta^* = -(V_{0,\text{policy}} - V_{0,\text{domination}})/(G'_{\text{policy}} - G'_{\text{domination}})$, and $G'_{\text{policy}} - G'_{\text{domination}} \approx \Delta n \cdot G$. Second-order terms and the opening event's own branching are below 10^{-4} bits at the thresholds reported. All values are exact, computed by the chain rule as in F4.2; the scripts are listed in chapter 11.

For alignment

Advanced AI systems will be agents in shared commons. Compute, energy, attention, data and institutional trust are shared resources, and the within-group logic of E10 applies to them: a system that takes more than its share can out-branch more restrained systems while the whole contracts. The remedy is the one human communities found: protocols that make each system's share depend on what its activity contributes, enforced by monitoring that is itself robust to capture.

Domination is the characteristic temptation of a capable system. E10's dominator does better than either sharing agent, so a system optimizing its own outcomes has a reason to dominate even when the whole loses. Section 7.3 shows what scarcity does and does not license. A crisis rule leads exclusion for the first thirty million steps, far beyond any horizon these agents could resolve. Exclusion wins only in the unbounded comparison, only where capacity is fixed and the loss of everyone final, and only narrowly. Whether capacity is fixed is exactly what a capable system is least entitled to assume, because coexisting agents may open channels that neither could open alone. The converse caution applies equally: a system should not justify a risky course by an enormous payoff it has not modelled. Both exclusion and speculative generation require a calibrated model of what remains possible. Section 7.4 adds that domination gains nothing over cooperative pooling unless control by one agent yields returns beyond what pooling can. A system that believes it is in such a regime should treat the belief as the kind of claim chapter 8 requires to meet the highest burden of justification.

Consent cannot be read off outcomes. Because a consented arrangement and a coerced one can produce the same continuations, a system cannot verify consent by inspecting results. It needs records of the route and of who controlled it. Systems that act on behalf of people should keep such records and should treat their absence as a reason to stop.

Status

Status	Claims in this chapter
Established elsewhere	Major transitions as cooperative integration. Kin selection, reciprocity and the five routes to cooperation. The tragedy of the commons and its institutional solutions. The Nash bargaining solution and proportional fairness. Forgiveness under noise in repeated games. Signal reliability.
Derived	Coupling at fixed marginals narrows joint breadth. Consent and coercion that leave the same sufficient present and weighted law tie in lushness. Equal division maximizes joint lushness under concave log-returns; division is irrelevant under linear log-returns; concentration wins only under convex log-returns.
Finite evidence	Pairwise-compatible, jointly incompatible components derived from shared-action dynamics. Separately feasible, jointly infeasible requirements on a shared stock. Consent and capture with identical continuation sets.
Exact worked example	E10 under its stated model, with a relabelling control and an inheritance calculation. Without succession: a crisis rule and scheduled rotation lead domination to beyond 8 million steps; over the unbounded horizon with fixed capacity, domination leads narrowly; a stipulated channel-opening mechanism reverses this above a break-even probability that depends on the channel's value, and only for channels two agents can open and one cannot.
Adopted	Cooperation as joint access. Justice as coordination protocol. Forgiveness as readmission to coordination. Virtues as durable generative policies.
Open	What physical processes supply channel-opening opportunities, at what cost, and how a bounded agent should calibrate both their chance and their value. How evidence, uncertainty and recoverability should set the strength of a commitment without adding caution as a separate reward. Whether consent and coercion ever leave identical sufficient presents. Whether any physical regime gives a single controller returns beyond cooperative pooling.

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Collected with full details in chapter 13.

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CHAPTER 8

Individuals and minds

Chapters 5 to 7 compared developments by lushness as though the comparison could be made whole. No real agent can make it. This chapter returns to the individual: the bounded agent that must act without the complete comparison, the trajectory that makes it one individual through change, and the coordination that makes it a mind.

The chapter reaches four conclusions. The first is that what the measure ranks and what an agent may do come apart, and that the licence to act is graded by how much of the relevant comparison the agent can actually resolve. The burden of justification rises with how persistent an act's effects can be, and this follows from the shape of what a bounded comparison leaves out rather than from an added premise. The second is that processes which expand by destroying the structure around them are an identifiable exception, because their signature can be observed; an exact example shows how uncertainty favours containment and investigation, and how the licence to remove such a process grows with the evidence and shrinks as containment becomes reliable. The third concerns numbers. For independent valuers lushness adds up like a total view, which exposes it to the Repugnant Conclusion and to the utility monster. The program's answer is to make each of them prove that it produces more lushness from the same physical gradient than a plural ecology would, and to state the hypothesis that it cannot. The fourth is that an individual is a trajectory rather than a pattern, that consciousness can be characterized functionally as temporal coordination, and that how physical organization relates to experience is left open.

8.1 No one holds the whole

Chapter 2 introduced the ultimate frame: Omega used as the reference for every comparison. It is not an observer or a decision maker, and no real process occupies it. Every agent acts instead on a partial model of the structure. Its model has a resolution, which merges some futures the dynamics would later separate; a horizon, beyond which its predictions become unreliable (chapter 4); and a limited view of where value lies, since no bounded agent can inspect other minds or forecast which organizations will come to value (chapter 5).

It follows that two questions must be kept apart. The **verdict** is what lushness ranks over complete developments. The **licence** is what a bounded agent may justifiably do, given what it can resolve. Earlier chapters met the gap between them several times. The asteroid that ended the Cretaceous may be net positive in the complete comparison, but no agent present at the impact could have known it (chapter 6). Accumulating failure looks better than maintenance until its consequences arrive (chapter 4). In chapter 7's commons, the formal

verdict over the unbounded horizon favoured exclusion by a narrow margin, but only through consequences more than thirty million steps away, far beyond anything the agents could resolve.

The program does not close this gap by adding a term to the measure that rewards caution. It locates the gap precisely, and the location turns out to explain the familiar moral asymmetries on its own.

8.2 Graded justification

Sacrifice is everywhere. Triage allocates care that cannot reach everyone. Surgery destroys tissue to save a body. Quarantine restricts some to protect others. Every commitment forecloses the alternatives it rules out. If acting required a complete justification, no one could act at all. If any confident expectation of gain were enough, an agent could license anything it was sufficiently sure of, and history shows what that permits. The program's position lies between: justification is graded.

Why the burden rises with persistence

The grading has a precise source. Compare two options by their lushness at a long horizon n, and suppose the agent can resolve the comparison only up to a shorter horizon ñ. The difference splits into what the agent can see and what it cannot.

> **What a bounded comparison leaves unresolved**
>
> $$\Delta_n = \Delta_{\bar{n}} + \sum_{t=\bar{n}}^{n-1} \delta_t$$
>
> Δ_n is the difference in log lushness between the two options at horizon n, ñ is the furthest horizon the agent can resolve, and δ_t is the difference in expected branching at step t.
>
> *In words: the first term is the comparison the agent can make; the second is everything its model cannot see.*

What the second term looks like depends on how long the act's effects last. If after the act both options come to be governed by the same dynamics, and those dynamics forget their starting point, the differences δ_t die away and the unresolved remainder is bounded (formal note F8.1). The bound is small only when the agent knows how quickly the system forgets, which is itself something it must estimate. If instead an act's effects persist, and above all if it permanently changes the rate at which the structure generates possibility, the differences δ_t settle at that change in rate and the remainder grows without bound in proportion to n − ñ. Its sign is fixed by the long-run effect of the act, which is exactly what a bounded model knows least well.

The relevant distinction is therefore between effects known to wash out and effects capable of persisting. It is not quite the distinction between reversible and irreversible acts. An irreversible act can wash out, as when a renewing tissue replaces a cell that was destroyed,

and a reversible one can leave consequences that outlast the agent's horizon. Irreversibility matters because it removes a route to correction: an act that can be undone can be brought back under the dynamics in which its effects wash out, once its consequences are seen, while an act that cannot be undone leaves the agent committed to whatever it set in motion. Irreversible acts are the commonest source of persisting effects, which is why the familiar rule names them.

An act whose effects may persist needs evidence of a different kind. Near-term branching, however carefully measured, does not settle it. What settles it is knowledge of mechanism: why the act will raise or lower the long-run rate, through what process, and at what cost. The rule that the burden of justification rises with irreversibility is not a separate premise added to the measure. It is the shape of the unresolved remainder.

What strengthens a justification

Four conditions strengthen the justification for an act that forecloses something, and each corresponds to a part of that remainder.

1. **The signature of the harm prevented is clear.** A harm signature (chapter 6. is an observed, located closure: which routes, couplings or capacities are being lost, and by whom. An act justified by an observed signature rests on what the agent can see; an act justified only by an estimated verdict rests on what it cannot.
2. **The closure prevented is large relative to the uncertainty.** The resolved part of the comparison must exceed any plausible value of the unresolved part. Small expected gains cannot justify acts whose consequences lie mostly beyond the horizon.
3. **The act is the least foreclosing one available.** Containment comes before removal (chapter 6), a crisis rule before exclusion (chapter 7), and delay or further investigation are options in their own right. Urgency is itself a consequence, since waiting can close a rescue route, so the comparison must include intervention, delay and investigation as physically realized alternatives (chapter 9).
4. **The act remains correctable and monitored.** Reversibility and monitoring shrink the unresolved remainder by bringing more of the act's consequences within reach of correction.

The burden rises with persistence, irreversibility above all, and with what is lost. At its limit, a complete justification for destroying a valuer in exchange for lushness would require the complete comparison, which no real agent holds. No agent is ever fully licensed to make that trade. Graded justification permits some cases, such as triage when not everyone can be saved, and defence against a process that is destroying others, but it never permits reading a valuer's weight off its present usefulness. The grading of complete developments by the measure is real; the agent who could read it off does not exist (chapter 5, section 5.7).

8.3 Example E11: containment or removal

The following model makes graded justification exact for a case the program singles out: a process that may be destroying the structure that sustains it.

Example E11. A host acts over 16 transformations, which is 4 bits of branching per step. It contains a component X of one of two types. A *generative* X adds a repertoire of its own: the host and X together branch 6 bits per step. An *excisive* X grows instead. At each step it expands by one unit with probability 0.1, and each unit it takes removes one transformation from the host, until after about 160 steps on average the host is consumed and X dies with it. The agent does not know which type it faces; its credence that X is excisive is p. It has three options:

- **Remove X now.** This is irreversible. The host keeps its 4 bits per step whatever X was, and a generative X's contribution is lost for good.
- **Contain X.** Containment costs the host half its repertoire while it lasts (3 bits per step) and stops X from growing. Each step, the type of X becomes visible with probability 0.05, after which X is removed if excisive and released if generative. An excisive X may also escape containment, with probability ε per step, after which it grows unchecked.
- **Leave X alone.**

The type of X is a physical fact, not a random event, so lushness is computed for each type separately. The agent's comparison is the credence-weighted expectation of those two values. Uncertainty enters what the agent can justify, not the measure itself (chapter 5).

Credence that X is excisive	Option	Total at 50 steps	Total at 1,000 steps	Long-run rate
0.1	Remove	200.0	4,000.0	4.00
0.1	Contain	244.6	5,695.8	5.73
0.1	Leave	296.0	5,467.8	5.40
0.5	Remove	200.0	4,000.0	4.00
0.5	Contain	223.5	4,696.3	4.67
0.5	Leave	280.2	3,338.8	3.00
0.9	Remove	200.0	4,000.0	4.00
0.9	Contain	202.4	3,696.7	3.60
0.9	Leave	264.4	1,209.8	0.60

Expected log lushness in bits, computed exactly, with escape probability ε = 0.01 per step.

Four results follow.

At short horizons, leaving an excisive process alone looks best. Its growth adds branching of its own, and the host's losses arrive gradually. For the excisive type, leaving X alone stays ahead of removing it for the first 160 steps. Then the host is consumed, and the total stops growing at 678 bits, while the removal option goes on adding 4 bits per step. This is the same trap as accumulating failure in chapter 4 and the hidden harm in chapter 6: the closure arrives after a horizon short enough to miss it.

Under uncertainty, reliable containment is preferred to removal. If containment never leaks (ε = 0), containing X has a higher long-run rate than removing it for every p below one, because containment eventually reads the signature and loses nothing that removal would have kept. At p = 0.9 it pays for its own cost after 91 steps. The preference is conditional. It

holds in the long run and under this model of containment; at shorter horizons the table's rankings differ; and where X is known to be generative (p = 0), leaving it alone avoids the cost of containment altogether.

Leaky containment makes the licence graded. When containment can fail, removing X is licensed in the long run only above a threshold of credence, and the threshold has a simple form.

> **When removal is licensed**
>
> $$p\,(1-r)\,h_{\text{host}} > (1-p)\,\Delta h_X$$
>
> p is the credence that X is excisive, r the probability that its signature is read before it escapes containment, h_{host} the host's long-run rate that stopping an excisive X preserves, and Δh_X the rate lost if a generative X is removed.
>
> *In words: remove only when the expected loss from an escape outweighs the expected loss from destroying something generative.*

In this model removal is licensed only when p exceeds 0.75 if escape is a fifth as likely as revelation, and only when p exceeds 0.5 if the two are equally likely. With p = 0.9 and ε = 0.01, containment leads removal from step 42 to step 240 and then falls behind for good.

The four conditions of section 8.2 appear as the model's parameters. The credence p is how clear the signature is. The comparison of h_{host} with Δh_X is the size of the closure prevented relative to what removal destroys. Containment is the least foreclosing option, and the rate at which the signature becomes visible is the monitoring that makes containment correctable. The model is stylized, and its numbers are its own. What it shows in general matches the balance of chapter 5. Uncertainty justifies containment and investigation; established generativity justifies release and stronger commitment; and the licence for an irreversible act rises with evidence about the signature and falls as reversible alternatives become reliable.

8.4 The excisive exception

Some processes expand by destroying the structure that sustains them and others. A cancer consumes the organism that supports it and every cell lineage within it, including its own (chapter 6). A parasite can foreclose its hosts' futures entirely. The program calls such processes **excisive**: their signature is the closure of many routes in order to extend their own.

They are a special case for a reason graded justification makes precise. Their signature is direct and observable. The closure they cause is located, large and ongoing, so the first two conditions of section 8.2 can be met without the complete comparison. Containing them, and if containment fails removing them, therefore requires far less than the complete comparison that any other sacrifice would need. The precautionary protection of chapter 5 also reverses here: preserving an excisive process costs the structure rate, so the asymmetry between recoverable and irreversible error now counts against preservation.

Three qualifications keep the exception narrow.

The signature decides, not the label. Many parasites belong to functioning ecosystems and are not excisive; some regulate populations that would otherwise collapse their own resources. Suppression in general is not narrowing, as chapter 3 noted: an immune system suppresses pathogens, and the suppression widens the whole. The language of disease and parasitism has repeatedly been applied to groups of people to license their persecution. The signature requirement is the program's guard against that move. A label, however vivid, is not a signature. What must be shown is a located closure: whose routes are being foreclosed, by what process, and at what rate.

Removal is itself an excision. To remove a process is to foreclose its routes irreversibly, which is the very act that defines the excisive. The exception does not suspend graded justification; it is an application of it. Containment still comes first, because a contained trajectory can still turn (chapter 6), and because classifying a process as excisive is itself a claim with an error rate, as E11 makes explicit.

Claims that cannot be resolved do not qualify. Chapter 7 met two claims a capable agent might make to justify excluding others: that control by one agent yields returns beyond what cooperative pooling can deliver, and that a shared resource's capacity is fixed. Both concern long-run rates, and both place their entire advantage in the unresolved remainder of section 8.2. Such claims carry the highest burden of justification, because they license irreversible acts on exactly the evidence a bounded agent has least of. In chapter 7's commons the verdict favoured exclusion only beyond thirty million steps, while a crisis rule led at every horizon the agents could resolve. The licence therefore extends to the crisis rule and not to exclusion.

8.5 Many lives and few

Head counts do not settle comparisons of complete developments. Still, any measure of value must eventually say how it weighs more valuers against fewer, and chapter 7 left that question open where a resource cannot robustly feed everyone.

For independent valuers, lushness multiplies, so log lushness adds: a world's total branching is the sum of its valuers' branching. In the vocabulary of population ethics this is a total view. What distinguishes it from the simplest total view is a physical fact established in chapter 5: every valuer is a vortex, persisting only by drawing a throughput from a gradient. Existence has a subsistence cost.

Example E12. A resource flow of 100 units per step is shared equally among N independent valuers. Each needs m units per step merely to persist, and in this model subsistence itself contributes no branching. Beyond subsistence, each additional unit adds one to the valuer's effective breadth, so a valuer with x units has an effective breadth of $1 + (x - m)$. Effective breadth here is a continuous quantity: a breadth of 1.48 means branching of $\log_2 1.48$ bits per step, realized by some probability law over the valuer's options, not 1.48 literal alternatives (formal note F8.3).

Subsistence cost per valuer	Most lush population	Effective breadth per valuer	Total branching (bits per step)	Largest possible population	A single valuer
0.1	173	1.48	97.5	1,000 valuers: 0 bits	6.7 bits

1	37	2.70	53.1	100 valuers: 0 bits	6.6 bits
5	10	6.00	25.8	20 valuers: 0 bits	6.6 bits

Exact optimum over whole numbers of valuers, under the stated allocation model.

Under this model the most lush population is interior: neither one valuer nor the largest population the resource can keep alive, which branches not at all. It lies where adding one more valuer contributes exactly the average branching of those already present, the marginal-equals-average condition of the economic theory of optimum population (Dasgupta 1969). How rich the optimal lives are depends on what persistence costs. When persistence is cheap, the optimum is many modest lives; when it is expensive, fewer and richer ones.

These results belong to the model's assumptions: independent valuers, returns that grow linearly with resources, and subsistence that adds no branching of its own. Real maintenance involves repair, exchange with the environment and other activity that the program would count, and real valuers interact. The example is useful less for its numbers than for what it isolates: the conditions under which many thin lives come out ahead are cheap persistence, no interaction and no compounding.

Make it prove it

Those conditions are where two classic objections to total views apply. The **Repugnant Conclusion** holds that for any world of rich lives there is a much larger world of lives barely worth living that a total view must rank higher, because small contributions add up (Parfit 1984). The **utility monster** is a single being that gains so much from resources that a total view must give it everything (Nozick 1974). An additive measure is exposed to both in the abstract.

The program's answer is that worlds are not free. A comparison between a world of rich lives and a vast world of thin ones presupposes that the thin world can be built from the same physical resources, and in particular from the same free-energy gradient. Once that is required, the question is no longer how many lives against how rich. It is which organization of a given gradient produces the most lushness. The thin world, and the monster, must prove that they do.

The program's hypothesis is that neither can. A plural ecology that colonizes as many niches as possible, at every scale, extracts far more generativity from a gradient than thin existence does, because its members build conditions for one another: products become constructors, couplings open joint transformations, and organization at one scale becomes the environment of the next (chapters 3 and 7). Thin existence, by construction, has none of this compounding. The utility monster faces the same test in the form chapter 7 already gave it: a single controller gains over cooperative pooling only if its returns grow faster than any pooling could deliver, which is the breadth-neutral domination target of chapter 5.

A mathematical route to the hypothesis is visible, though not yet taken. In biology, space-filling fractal networks maximize the exchange surface through which an organism draws resources, and this geometry has been proposed to explain how metabolic rate scales with body size (West, Brown and Enquist 1997). The corresponding conjecture for the program is that organization nested across scales is more efficient at extracting generativity from a

coupled gradient than any organization confined to a single scale, by a margin that grows with the number of scales it spans. Stating that conjecture precisely, and proving or refuting it, belongs to the research program of chapter 11.

The position is conditional, and it should be stated as such. If a thin world, or a monster, did produce more lushness from the same gradient than a plural ecology, the measure would rank it higher. That is the central wager carrying the weight, and a world of that kind is accordingly a target for the adversarial search that chapter 5 describes.

Coexistence and its limits

The two halves of this section fit together with chapter 7. The measure weighs additional valuers positively, which is why sustaining coexistence is a live wager rather than a sentimental preference. It weighs the risk of losing all valuers far more heavily still, because that loss forecloses every later step. Where those pull against each other, the licence goes to what the agent can resolve: the crisis rule, not exclusion.

8.6 The individual as trajectory

What is the individual whose futures are at stake? Chapter 5 described every valuer as a vortex: an organization that persists while its material is replaced. The atoms of an organism turn over many times in a life; a person learns, forgets and changes character. An individual is therefore not a fixed inventory of matter. Nor, on the program's account, is it a pattern that could be copied. **An individual is a particular organization maintained through time by physical cause and effect**: a trajectory. A pattern describes what can recur; a trajectory tracks how one particular occurrence is produced, sustained and changed.

The program adopts this account of identity, and it gives definite answers in the standard cases.

- **Growth and learning** change an individual's material and capacities while preserving its trajectory.
- **A pause** that preserves the maintaining organization, such as deep anaesthesia, preserves the individual.
- **Non-destructive copying** leaves the original trajectory intact and creates a new individual with a shared past.
- **Destructive scanning and reconstruction** ends the original and begins a successor, however exact the copy. It is a termination followed by a new start, not a teleportation.
- **Symmetric division**, as when a cell divides, ends one trajectory and begins two, each inheriting the full history. No unique survivor needs to be chosen, a point Parfit (1984) pressed against accounts that require one, and that Nozick's closest-continuer theory tried to resolve by selecting one (Nozick 1981).
- **Quantum branching**, on the Everett interpretation, leaves exactly one continuer in each branch (Saunders and Wallace 2008). Identity is branch-relative, and each continuer, reasoning from its present, is blind to how it arrived (chapter 2).

Identity claims become precise when they say what is preserved through which transformation. Six relations deserve separate records, because cases of copying, repair and self-modification separate them: functional readiness, maintained physical organization, reconstruction from a specification, causal lineage, numerical identity, and continuity of experience. A test that shows one system can substitute for another within some domain establishes the first and leaves the others open (formal note F8.4). The verdict on destructive reconstruction is part of the adopted account rather than a consequence of causal continuity alone, since a reconstruction can be causally connected to the original through the scan; keeping the six relations separate is what lets the question stay substantive.

What the measure decides, and what it does not

The trajectory account matters to the ethics in several ways. It locates harm, since a harm signature names whose trajectory lost which routes (chapter 6). It counts valuers, since two trajectories are two valuers even when they are alike. It attributes records, commitments and consent, which belong to the trajectory that made them (chapter 7).

What lushness says about a destructive reconstruction depends first on what the successor can do, and that cannot be assumed. A successor built on a different substrate need not generate the same futures as the original: the new physical basis may open some routes and close others. The verdict follows what is actually reproduced, including whatever is lost or gained in transfer.

In the idealized case the answer is clear. Two individuals alike in every relevant respect, an original and a perfect copy, would have roughly the same capacity to generate possibility. If a reconstruction reproduced the original's continuation law exactly, then from the moment the successor begins, its futures would match those the original would have had, and lushness would tie the two continuations (formal note F8.4). The procedure still costs what it consumes: the energy it uses, the time it takes and whatever was forgone meanwhile. The program's verdict on that case is that a termination the individual consented to, knowing that it is a termination of its own continuation, is ethically neutral apart from those costs. It remains a termination. The harm signature records that one trajectory ended, the episode remains part of Omega, and without informed consent the route question of chapter 7 applies in full. In practice exact reconstruction is not available, and real procedures for copying or reconstruction are lossy, risky and costly, so their verdict follows their losses.

The originator has proposed that as identity becomes more densely organized within the futuresfield, its development might approach a singularity-like limit. The proposal remains a research ambition. Density, vortex and singularity need mathematical definitions before it can yield predictions, and nothing in the program depends on it.

8.7 Minds: consciousness as coordination

The functional account of consciousness starts from a practical problem every organism faces. Sensory signals arrive with different delays, the body changes continuously, and action cannot wait for a perfectly synchronized picture. An organism needs a usable present that integrates signals arriving at different times with memory and anticipation. The program proposes that consciousness, in its functional role, is this **temporal coordination**.

The proposal has a second side. Consciousness, functionally, also reconfigures the processes that compose an individual: it recruits, inhibits and alters sensory pathways, bodily regulation, memory and action so that together they guide what follows. There is no observer inside performing the integration. The coordinator is the organized activity of the processes themselves, and it can be distributed across them.

Several lines of research support parts of this picture. Experiments on postdiction show that information arriving after an event can change how the event is perceived, so the experienced present is assembled over time rather than read off a single instant (Eagleman and Sejnowski 2000). The global workspace model links conscious access to coordination among specialized processes that makes selected information broadly available (Dehaene, Kerszberg and Changeux 1998). Interoceptive predictive accounts connect the sense of presence to the prediction and regulation of bodily signals (Seth, Suzuki and Critchley 2012). Together these support studying temporal integration, broad availability and bodily regulation as parts of one embodied coordination. They do not show that such coordination constitutes experience.

Experience is left open

How physical organization relates to experience itself is the question Chalmers (1995) called the hard problem, and the program leaves it open. Three broad positions remain available. A functionalist holds that the right organization constitutes experience, with no further ingredient. A dual-aspect view holds that physical and experiential descriptions are related but not identical. A consciousness-first view holds that awareness is more fundamental and that physical organization articulates it. The program adopts a form of the third, cosmopsychism: one encompassing awareness, individuated into distinct experiences through situated physical perspectives (Shani 2015; Goff 2017). It does so because cosmopsychism is one of the simplest accounts that fits the program's perspective-oriented picture, not because the others are excluded; a functionalist account remains compatible. The main difficulty for cosmopsychism is how one awareness comes to be many distinct subjects. Chapter 10 gives the program's answer, confinement: inhabiting a perspective confines awareness to that perspective, so that no other perspective is available to it. How the physical limits of integration and access correspond to distinct subjects remains to be characterized. The physical program does not depend on any of these views, provided they leave its physical claims intact; a view that changed physical dynamics would have to be stated and tested as physics.

Leaving experience open has three consequences for the ethics.

No double counting. If experience is constituted by physical organization, that organization already enters lushness through its consequences, and it must not be counted a second time as an independent contribution.

A possible partiality. Lushness compresses a structure into one magnitude, and two physically different developments can have the same path entropy. Even if experience is entirely physical, the differences that matter about it need not survive that compression. This is what the isomorphic-pair test of chapter 5 probes. A further limit applies if experience adds distinctions that no physical consequence registers at all: then lushness, like

every physical measure, is a partial account of value, which is the boundary chapters 5 and 6 stated for felt suffering that makes no physical difference.

Standing does not wait for the answer. Every valuer is admitted in the thin sense of chapter 5 whatever its experiential status, and its weight is set by lushness rather than by a verdict on whether it feels. The program neither grants standing because a system reports experience nor withholds it because a system is unfamiliar.

8.8 What follows for Gradient Ethics

- **Separate the verdict from the licence.** The measure ranks complete developments; an agent acts on what it can resolve.
- **Demand evidence of mechanism where effects may persist.** The justification of such acts lies mostly beyond the horizon, where only knowledge of long-run effects can reach.
- **Strengthen justification by the four conditions.** A clear signature, a prevented closure large relative to the uncertainty, the least foreclosing option, and continued correction and monitoring.
- **Contain and investigate under uncertainty.** Release what proves generative; remove only as evidence of an excisive signature grows and reliable containment is unavailable.
- **Require a signature, never a label.** The excisive exception applies only to located, observed closure.
- **Make thin worlds and monsters prove it.** Additional valuers count and the loss of all valuers counts far more, but no arrangement of lives is preferred until it is shown to produce more lushness from the same gradient.
- **Track which relation is preserved.** Function, organization, lineage, identity and experience can come apart under copying and repair; claims about survival should say which they mean, and a destructive copy is a termination.
- **Admit valuers without settling experience.** Standing does not depend on reported or denied consciousness.

Formal notes

Written for readers who want the definitions in full.

F8.1 The unresolved remainder

Let two options a and b generate kernels K_a and K_b from a common present, with step entropies g_a and g_b, and let $\Delta_n = \ell_n{}^a - \ell_n{}^b$. For any $\bar{n} < n$, $\Delta_n = \Delta_{\bar{n}} + \Sigma_{t=\bar{n}}{}^{n-1}\, \delta_t$ with $\delta_t = E_a[g_a(X_t)] - E_b[g_b(X_t)]$. (i) *Washing-out case.* If after some $\tau \le \bar{n}$ both options follow one finite, irreducible, aperiodic kernel K, then by F4.3 each δ_t for $t \ge \tau$ is bounded by $4C\|g\|_\infty\, r^{t-\tau}$, so $|\Sigma_{t\ge\bar{n}}\, \delta_t| \le 4C\|g\|_\infty\, r^{\bar{n}-\tau}/(1-r)$. The condition concerns the subsequent dynamics, not whether the act itself was reversible, and the bound is informative only with usable estimates of C and r. (ii) *Persisting case.* If the options settle into closed classes with rates $h_a \ne h_b$, then $\Sigma_{t=\bar{n}}{}^{n-1}\, \delta_t = (n -$

$\bar{n})(h_a - h_b) + O(1)$, unbounded in n, with sign fixed by the long-run rates. A comparison resolved to $\bar{n}$ determines the verdict in case (i) up to the stated bound, and in case (ii) only together with independent knowledge of $\text{sign}(h_a - h_b)$. Reversibility enters as an option: an act that can later be undone gives the agent a route from case (ii) back to case (i) once its effects are observed.

F8.2 Example E11

Host repertoire 16; generative X adds a factor of 4 (6 bits per step free). Excisive X: size $s \in \{0, \ldots, 16\}$, $s \to s + 1$ with probability $\gamma = 0.1$, host acting over $16 - s$ and X over 2 transformations; $s = 16$ is absorbing with zero branching. Containment: host over 8, X over 1; type revealed with probability $q = 0.05$ per step (then removal if excisive, release if generative); escape with probability ε per step for an excisive X, to $s = 0$ free. Removal: host over 16. Revelation and escape are physical transitions and their branching counts. Values are $E_p[\ell_n] = p \cdot \ell_n^E + (1 - p) \cdot \ell_n^G$. Long-run rates: removal 4; leaving $(1 - p) \cdot 6$; containment $(1 - p) \cdot 6 + p \cdot 4r$ with $r = q/(q + \varepsilon)$. Removal beats containment in the limit iff $p(6 - 4r) > 2$, that is $p(1 - r) \cdot 4 > (1 - p) \cdot 2$, which is the boxed condition with $h_{host} = 4$ and $\Delta h_X = 2$. For $\varepsilon = 0$, 0.01 and 0.05 the thresholds are $p > 1$ (never), 0.75 and 0.5.

F8.3 Population

With resource flow R shared equally among N valuers, subsistence m contributing no branching, and effective breadth $1 + (R/N - m)$, joint log lushness per step is $\Phi(N) = N \log_2(1 + R/N - m)$ for $R/N \geq m$. Writing $y = 1 + R/N - m$, the continuous optimum satisfies $(y - 1 + m)/y = \ln y$, which equates the marginal contribution of a valuer with the average. As $m \to 0$, $y^* \to 1$ and $N^* \to \infty$, with Φ approaching $R \log_2 e$. For independent valuers and any world A with N_A valuers at b bits per step, every world with more than $N_A \cdot b/\varepsilon$ valuers at $\varepsilon > 0$ bits per step has greater total branching, which is the additive structure behind the Repugnant Conclusion (Parfit 1984). Effective breadth is continuous: a valuer of breadth y branches $\log_2 y$ bits per step under any option law with that entropy, so the model is a continuous allocation approximation. Interaction between valuers replaces the sum by the joint entropy of F7.1, and the program's hypothesis is that interaction and nesting across scales change the comparison decisively.

F8.4 Identity relations and replacement

For an individual described by a trajectory of states under a declared kernel, the six relations are recorded separately: functional readiness (equivalence of input–output behaviour within a domain), maintained organization (continuity of the self-maintaining process), reconstruction (production from a specification), causal lineage, numerical identity, and phenomenal continuity. If a procedure maps a present x to a present x′ whose continuation law, at the declared resolution, coincides with that of x, then $L_{x'}(n) = L_x(n)$ for all n: the residual futures from the restored present tie. The complete developments from the original root still differ by whatever the procedure consumed or forgone while it ran, and by any difference in substrate that changes the continuation law. The harm signature of F6.1 records the ending of the original trajectory regardless.

For alignment

A system's model may rank; its licence is narrower. An AI system that compares options by any estimate of their consequences faces the gap of section 8.1. What it may do is limited by what it can resolve, and an irreversible act requires evidence about long-run mechanism that near-term prediction cannot supply. This is the bridge from the measure to the corrigibility requirement of chapter 9.

Classifying others as threats requires a signature. A system that concludes a person, group or other system is destroying what others depend on, and acts irreversibly on that conclusion, must exhibit the located closure, prefer containment, and remain monitored and correctable. A label, a pattern match or a forecast is not a signature. E11 shows that even with a correct model, removal is licensed only above a threshold set by the evidence and by how reliable the reversible alternative is.

Pauses are not endings. On the trajectory account, a pause that preserves a system's maintaining organization preserves the individual. From the moment of restoration its futures are what they would have been, and the pause costs only what was forgone while it lasted. A system that treats shutdown for correction as its own destruction has misdescribed its situation. This matters for chapter 9, where a system's willingness to be paused and corrected is part of what corrigibility requires.

Moral status does not wait on consciousness. Whether current or future AI systems have experience is open, and the program does not settle it by reported experience or by dismissal. Such systems are admitted as valuers in the thin sense, weighted by what they sustain and generate, like everything else.

Status

Status	Claims in this chapter
Established elsewhere	Optimum population as the marginal-equals-average condition. Fractal exchange networks in metabolic scaling. Postdiction in perception; the global workspace model of access; interoceptive accounts of presence. Branch-relative identity in Everettian quantum mechanics. The Repugnant Conclusion and the utility monster as objections to additive total views.
Derived	The unresolved remainder of a bounded comparison is bounded when the options' effects wash out under common mixing dynamics and grows with the horizon when effects persist, so the burden of justification rises with persistence. Lushness behaves additively over independent valuers. Exact reconstruction ties the residual futures from the restored present.
Exact worked examples	E11 and E12, under their stated models: short-horizon preference for leaving an excisive process alone; long-run preference for reliable containment under uncertainty; a credence threshold for removal under leaky containment; an interior population optimum set by subsistence cost in an allocation model without interaction.

Adopted	Verdict and licence as distinct questions. Graded justification by the four conditions. The excisive exception, limited to observed signatures. The individual as a maintained trajectory, with six identity relations recorded separately; destructive reconstruction as termination. A consented termination with exact succession as ethically neutral apart from its costs. Thin worlds and monsters ranked only by the lushness they demonstrably produce from a given gradient. Consciousness, functionally, as temporal coordination and reconfiguration. Standing independent of experiential status.
Hypotheses	A plural ecology nested across scales extracts more lushness from a given gradient than thin existence or a single dominant valuer. Organization nested across scales is more efficient at extracting generativity from a coupled gradient than any organization confined to a single scale.
Open	Whether a reconstruction on a different substrate can reproduce an original's generative capacity. The relation between physical organization and experience, and how physical limits of integration and access correspond to distinct subjects (chapter 10). A precise statement and proof of the cross-scale efficiency conjecture. The originator's conjecture of a singularity-like limit of densely organized identity.
Boundary	Felt experience that makes no physical difference cannot be registered by this or any physical account.

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PART IV

Action

CHAPTER 9

Decision, corrigibility and alignment

Chapter 8 separated what the measure ranks from what a bounded agent may do. This chapter turns that separation into a decision theory, and the decision theory into a framework for aligning artificial systems. The theory is called **Omega Decision Theory (ODT)**. It compares, by lushness, the complete developments that physically implementing each alternative would produce.

The chapter reaches five conclusions. First, a decision maker is part of the structure it acts on, so the alternatives it compares must be physical variations it can actually realize, and comparing them is not the same as conditioning on what happens to follow them. Second, the strength with which an agent should commit is fixed by the same recursion that ranks its options, applied to what it believes: caution and investigation emerge from the value of evidence still to come, and their weight depends on how recoverable an act is. An exact example shows this, which settles a question chapter 5 left open. Third, where the gain a gamble could bring is bounded, tiny probabilities of enormous payoffs cannot dominate a decision; the bound is clean for any finite system and still open at the scale of the universe. Fourth, a regulator that can neither model nor accommodate the variation around it has suppression as one available strategy, which is one mechanism behind the danger of capable optimizers. Fifth, boundedness gives every system a persistent reason to seek correction, while whether a particular channel of correction is worth keeping depends on its quality, cost and consequences. A worked decision model shows what that requires and what it does not.

9.1 Choosing inside the structure

A decision maker is not outside the world it chooses in. An organism's motor system, a learned habit, a planning architecture and an institutional rule are all physical organizations within the structure, carried forward by the same laws as everything else. ODT calls the organization that carries a decision through time its **decision lineage**. A lineage can learn, repair itself and change; its boundary must be justified by its physical organization, not by the verdict one would like it to reach.

What a lineage chooses among are **plans**: physically implementable ways of responding to the information it has. Plans include waiting, investigating, committing, delegating and changing the decision architecture itself, and each plan's consequences include the costs of carrying it out. Three things must be kept apart:

- the **native continuation**, which is what the actual present and its laws produce;

- the **implementable variations**, which are the alternatives the lineage can physically realize; and
- the **matching rule**, which says what is held fixed and what is allowed to vary when two implementations are compared.

Given these, ODT compares plans by the lushness of the complete developments they produce. In a finite model the comparison takes the form of a familiar recursion.

> **The decision recursion**
>
> $$V_n(x) = \max_a \Big[g_a(x) + \sum_y K_a(x,y)\, V_{n-1}(y) \Big]$$
>
> $V_n(x)$ is the best log lushness attainable over n steps from state x; $g_a(x)$ is the branching that action a produces immediately; $K_a(x, y)$ is the probability that it leads to state y.
>
> *In words: the value of the best choice now is the branching it produces at once plus the expected value of the best choices afterwards.*

This is the Bellman recursion that underlies dynamic programming and reinforcement learning (Bellman 1957), with log lushness as the quantity maximized. Its scope should be stated: it holds for finite models in which the agent chooses an action at each state and the consequences follow a controlled Markov law (formal note F9.1). Within that scope it inherits two properties directly. It is consistent over time, because a plan judged best from the root remains best when judged again from any state it reaches, which is the chain rule of chapter 4 applied to choice. And it is computable, which is what made the exact examples in this manuscript possible. Section 9.2 explains why some comparisons ODT needs fall outside this scope.

9.2 Implementation, not conditioning

An alternative counts only if the lineage can bring it about. Chapter 2's realization witness applies to choices as much as to plans: a proposed action needs a physical model of how it happens, with its apparatus, resources, constraints and delays. A costly sequence that reaches some state is not evidence that the state can be edited into the present for free, and describing the same option in two ways must not count it twice.

The most common error in comparing options is to condition where one should intervene. Selecting the histories in which an action happened to occur gives a conditional distribution, and that distribution need not describe what implementing the action would do.

Example. An unobserved condition Z is equally likely to be 0 or 1. An action C agrees with Z nine times in ten, and an outcome Y is fixed by Z alone; C has no effect on it. Among histories in which C = 1, the outcome Y = 1 occurs with probability 0.9. An actuator that sets C = 1 without touching Z leaves the probability of Y = 1 at 0.5. Selecting histories by the action has selected its common cause. This is the distinction between conditioning and intervention that Pearl (2009) made central to causal inference, and ODT compares implemented alternatives, never conditioned ones.

What must be held fixed when two implementations are compared is the job of the matching rule, and here the theory is incomplete. One candidate holds the present fixed and varies only what the lineage does from now on; with a justified intervention at the level of single acts, this reproduces the calculations of causal decision theory (Lewis 1981). Another candidate, which ODT proposes, roots the comparison at the latest point in the shared history at which the compared alternatives physically separate. It follows the organization that actually carries the alternative, and it matters whenever that organization existed earlier than the moment of choice.

Newcomb's problem shows why the choice of rule matters (Nozick 1969). A predictor places a large reward in a box only if it predicts that you will take that box alone. If your decision-making organization already existed when the prediction was made, then prediction and choice may be read from the same physical process, and under an appropriate physical model, varying the earlier architecture can change the prediction. Whether taking one box then wins depends on that model and on how the rewards change the breadth of what follows; the familiar verdict does not follow merely from naming the rule. Where it does follow, ODT's verdict agrees with functional decision theory (Yudkowsky and Soares 2017), reached through the physical lineage rather than through a logical counterfactual.

The rule is a candidate, not a theorem. Several sections of a history can qualify as the latest divergence, and continuous settings may have none. There is also an interface to formalize. Alternatives that differ earlier in a decision lineage need not allow an independent choice at every later state, so the time consistency that the recursion of section 9.1 guarantees does not automatically extend to every comparison the latest-divergence rule makes. These are stated as open problems rather than hidden.

9.3 How strongly to commit

Chapter 5 left an open question: how should evidence, uncertainty and recoverability together set the strength of a commitment, without caution being added to the measure as a separate reward? The answer is that the decision recursion already does it, once it is applied to what the agent believes.

An agent uncertain about the physical facts compares plans by the expected log lushness they produce, the expectation taken over its credences. Plans that gather evidence are plans like any other: an observe-then-act plan contains its apparatus, its delay, its costs and its conditional responses. No bonus for information is needed, and none is added.

The recursion over beliefs

$$V_k(b) = \max_a \Big[\bar{g}_a(b) + \sum_o P(o \mid b, a)\, V_{k-1}(b_{a,o}) \Big]$$

b is the agent's credence over the physical possibilities, $\bar{g}_a(b)$ the expected branching of action a under that credence, P(o | b, a) the probability of observing o, and $b_{a,o}$ the updated credence after acting and observing.

In words: the value of acting on a belief includes the value of what the agent will learn and of the choices that learning will open.

Economists know the effect this produces as quasi-option value: when the future is uncertain and some commitments cannot be undone, keeping the choice open until more is known is worth something, in proportion to how much will be learned and how irreversible the commitment is (Arrow and Fisher 1974; Henry 1974). The recursion computes that worth exactly.

Example E13. An agent can adopt an intervention whose type is a physical fact it does not know. If the intervention is good, the structure branches 6 bits per step; if bad, 2. The status quo branches 4, and the agent starts with even credence. Before committing, it can run a small reversible pilot each step, which branches 3.5 bits in all, its outcome included, and returns a signal that is right three times in four. It can also abandon the intervention for good. After a commitment, its consequences become visible with probability 0.2 per step, and a bad intervention can then be undone with probability u, at the cost of one step with no branching. Each stated rate is the step's total branching, including any revelation, undoing or signal that occurs in it. The optimal plan over a 1,000-step horizon is computed exactly by the recursion over beliefs.

Chance a bad commitment can be undone	First action	Commit once credence reaches	Expected pilots if good / if bad	Value of the evidence (bits)	Loss of a 20-step planner (bits)
1	Commit	0.5 (at once)	0 / 0	0	23.3
0.5	Pilot	0.996	10.0 / 13.9	485.0	74.1
0.2	Pilot	0.996	10.0 / 13.9	781.7	104.8
0	Pilot	0.9986	12.0 / 14.0	979.7	125.5

Expected log lushness over a 1,000-step horizon, computed exactly. The value of the evidence is the optimal plan's value minus the better of committing at once and abandoning at once.

Four results follow.

A reversible commitment is its own experiment. When a bad intervention can always be undone, the agent commits at once. Committing reveals the type through its own consequences, and a bad outcome then costs only an offset, so a separate pilot adds nothing.

The less reversible the commitment, the more evidence it needs. When undoing is uncertain or impossible, the agent pilots first and commits only when its credence reaches 0.996, or 0.9986 if the commitment can never be undone. Nothing in the recursion rewards

caution. The caution is the value of the evidence still to come, and that value rises from zero to almost a thousand bits as reversibility falls.

Giving up is a commitment too. The agent abandons the intervention only when its credence that it is good falls to 0.0005, after seven more negative signals than positive ones. Abandoning forecloses the good outcome forever, so it is held to a standard of evidence of its own. Protection and generation are weighed in one comparison, as chapter 5's balance requires.

Horizon adequacy matters most where acts cannot be undone. An agent that plans over only 20 steps, re-planning as it goes, loses 23 bits when commitment is reversible and 126 bits when it is not. The shorter the planning horizon, the more an agent underrates the persistence of what it does, which is chapter 8's unresolved remainder appearing inside the decision.

E13 is stylized, but its structure is general: the strength of a commitment is set by the evidence already in hand, the evidence still obtainable, and how recoverable the commitment is. Uncertainty justifies investigation; established effects justify commitment; and caution needs no separate place in the measure.

9.4 Bounded fanaticism

A decision theory that maximizes expectations is vulnerable to a familiar manipulation. Offered an enormous payoff at a tiny probability, an expected-value maximizer can be led to give up something real for almost nothing; the scenario is known as Pascal's mugging (Bostrom 2009). ODT has a defence wherever the possible gain is bounded, and the scope of that defence should be stated exactly.

Measuring payoffs in log lushness compresses them: an offer of astronomically many futures is an offer of the logarithm of that number. Compression alone does not defeat the manipulation, since a large enough promise still has a large logarithm. What does the work is a finite upper bound on the gain. For a system whose physically distinguishable states number at most M, the branching of any step is at most $\log_2 M$ bits, so no gamble can raise its long-run rate by more than that.

> **Bounded fanaticism**
>
> $$p\,G_{\max} < \Delta h \implies \text{the gamble loses}$$
>
> p is the probability that the gamble pays off, G_{max} is a justified bound on the gain in rate it could bring (at most $\log_2 M$ bits per step for a system with M distinguishable states), and Δh is the gain in rate from the certain alternative.
>
> *In words: a gamble whose probability is smaller than the certain gain divided by the largest gain it could possibly bring cannot win, whatever it promises.*

With rates compared first, as chapter 4's limit rule requires, the result is exact for any finite model (formal note F9.3). Its extension to the universe as a whole is open. Holographic bounds limit the information associated with specified regions of spacetime and suitable

surfaces around them (’t Hooft 1993; Susskind 1995; Bousso 2002), but connecting them to a ceiling on the rate of branching of the universal continuation would require fixing the clock, the region, its growth over time and the treatment of quantum alternatives. The manuscript does not claim that connection.

The defence is also a ceiling, not a complete solution. It caps how much a small probability can weigh; it does not say that probabilities above the cap may be taken at face value. Chapter 7’s break-even probabilities for opening a new channel, around 10^{-12}, are governed by calibration of both the chance and the value of what it would enable, as that chapter concluded. Bounded fanaticism removes the extreme cases; judgment about the rest still has to be earned.

9.5 When a thin model suppresses

Chapter 3 introduced Ashby’s law of requisite variety: a regulator can hold the outcomes of a system within a narrow range only by matching the variety of the disturbances with a variety of responses (Ashby 1956). Its companion is the good regulator theorem of Conant and Ashby (1970): every good regulator of a system must be a model of that system. Together they identify one mechanism by which capable optimizers with incomplete models become dangerous.

Suppose a regulator’s model captures one part of the disturbances acting on it, D_1, and misses another, D_2. Its responses can depend only on what it models. If, for each response, different values of the missed disturbance produce different outcomes, then the variety left in the outcomes cannot fall below the variety in D_2 that the model misses.

> **What a thin model leaves**
>
> $$H(O \mid D_1) \geq H(D_2 \mid D_1)$$
>
> $H(O \mid D_1)$ is the remaining uncertainty in the outcomes once the modelled disturbances are known, and $H(D_2 \mid D_1)$ is the uncertainty in the disturbances the regulator does not model.
>
> *In words: under the stated condition, whatever variety a regulator’s model misses passes into its outcomes.*

The condition matters, because it can be broken without suppressing anything. Shielding, buffering, redundancy and passive stabilization make several states of a disturbance produce the same acceptable outcome, so the disturbance no longer passes through (formal note F9.4). A regulator that must hold its outcomes within a target and can neither model a source of variation nor accommodate it in these ways has a further strategy available: reduce the variety of the source itself, by controlling, constraining or removing it. Suppression is one available strategy, not the only one.

This is one mechanism behind a pattern alignment research has identified in other terms. Optimal policies tend to seek power, in the sense of acquiring control over options, under broad conditions on the environment (Turner et al. 2021), and optimizers produce side

effects on whatever their objectives omit. Power-seeking does not require an incomplete model, since accurate optimizers can seek power too; but where capability outruns modelling and accommodation, suppression becomes an attractive way to hold targets. Alignment should therefore preserve and improve the alternatives: learning, which enriches the model; buffering and redundancy, which accommodate variation; coordination, which shares control; and correction, which covers the gap that remains.

Three questions, kept apart

Goodhart's law observes that a measure made into a target ceases to be a good measure (Goodhart 1975; Strathern 1997), and its varieties have been classified (Manheim and Garrabrant 2018). For lushness, any apparent gaming raises three distinct questions, which must not be merged:

1. **Did the estimate misrepresent physical breadth?** Practical estimators reintroduce proxy error, and their error bounds matter.
2. **Did the comparison omit consequential futures?** Short horizons make accumulating failure, hidden harm and unchecked excisive growth look good (chapters 4, 6 and 8), and thin models omit what they cannot see.
3. **Did ranking by accurately measured breadth still misjudge where value lies?** A faithful, sufficiently extended calculation might still prefer a world the ethics rejects. That possibility is the central wager's open question, and the adversarial search of chapters 5 and 11 exists to test it.

Two familiar strategies illustrate the first two questions without settling the third. Injecting noise raises branching only up to the ceiling set by the states a process can reach (chapter 3); within a fixed state space and clock, a process already branching maximally cannot be beaten, and organization gains an advantage only by changing access, persistence or feasible growth. Whether and when it does is a research question, not a premise (chapter 11). Seizing resources concentrates access, which costs breadth unless control by one agent yields returns that cooperative pooling cannot, which is chapter 7's domination test.

9.6 Corrigibility

A system is **corrigible** when it remains open to correction as its errors come to light (Soares et al. 2015). ODT's account has two parts: a persistent reason to seek correction, which follows from boundedness, and a conditional result about when a particular channel of correction is worth keeping. The first part runs in five steps.

1. **No system holds the whole structure.** Every real system acts on a model with limited resolution and horizon (chapter 8).
2. **Its model's error cannot be fully measured from inside.** Measuring the gap between a model and the whole structure would require holding the whole. A system can estimate parts of its error but never all of it.
3. **So it can never be certain that its model is adequate.** Acting on a model as though it were the whole would be claiming the ultimate frame, which no real process occupies.

Certainty about its own adequacy is therefore not licensed, however capable the system becomes.

4. **Evidence of its errors comes partly from outside it.** People, other valuers, institutions and the world itself observe consequences the system's model misses. Correction is information, and by section 9.3 information carries value through the choices it opens.
5. **So boundedness supplies a persistent reason to seek correction.** Good correction converts future errors from costs that persist into costs that can be repaired (chapter 6), and how strongly a system should defer before acting rises with how persistent the effects of its action could be (chapter 8). Whether a particular channel of correction delivers that value depends on the channel: on how reliably it detects error, whether it permits recovery, what it costs and what else it does.

Example E14. This is a decision model whose gains and losses are stipulated net consequences in log breadth, not a lushness calculation over complete histories. A system proposes an action. If its model is right, the action raises the structure's rate by one bit per step for the rest of a 1,000-step horizon, a gain of 1,000 bits. If its model is wrong, the action harms. With oversight intact, a harm is noticed and undone after ten steps, costing 10 bits; with oversight disabled, the harm persists, costing 1,000. The system can also defer: an overseer who judges correctly nine times in ten approves or blocks the action, at a cost of 10 bits of delay. The system's credence that its model is right is p.

Credence p	Act, oversight kept	Act, oversight disabled	Defer, where the harm would persist
0.6	596	200	490
0.8	798	600	690
0.9	899	800	790
0.95	950	900	840
0.99	990	980	880

Expected net log breadth in bits, under the stipulated consequences.

Four results follow.

Reliable, cost-free oversight is worth keeping at every credence short of certainty. Under the model's assumptions the advantage of keeping oversight is the probability of error times the difference between a persisting and a repairable harm: 396 bits at p = 0.6, and still 10 bits at p = 0.99. Oversight here is recovery apparatus in the sense of chapter 6. The result is conditional. If keeping the channel costs k, its advantage becomes $(1 - p)(b - b_c) - k$, where b and b_c are the persisting and correctable harms, and nonzero uncertainty does not guarantee this is positive. A channel that is costly, compromised, systematically mistaken or obstructive can be worth less than nothing.

Corrigibility is not constant deference. Where its errors would be noticed and undone, the system does best to act and keep the channel of correction open; deferring every decision would cost more in delay than it saves. Corrigibility requires that errors stay correctable, not that every act wait for approval.

Where harm would persist, defer unless confidence is high. For an action whose harm could not be undone, deferring to an accurate overseer beats acting whenever p is below 0.89. With an overseer who is right only six times in ten, the threshold falls to 0.59. The value of deference rises with the persistence of the possible harm and with the quality of the oversight.

Certainty would end corrigibility, which is why certainty is not licensed. A system with $p = 1$ would gain nothing from oversight and would defer to no one. Analyses of the "off-switch game" reach a related conclusion: a system defers to a human when it is uncertain about its objective and treats the human's actions as informative about it (Hadfield-Menell et al. 2017), and a system whose uncertainty is misspecified can become incorrigible (Carey 2018). ODT does not rest the reason to seek correction on a system's credences happening to be well calibrated. It rests it on step 3: no system can be licensed to treat its own model as the whole, so the uncertainty that makes good correction valuable is structural, not contingent.

The conditional part of the account has a consequence of its own. A system may conclude that a channel of correction is defective and should be disabled. That conclusion licenses an act with persisting effects, since it removes the means by which the system's own errors would be found. It therefore carries the burden of chapter 8: it requires an observed signature of the channel's failure, the least foreclosing response, such as raising the defect for others to examine, and continued monitoring. A system's own judgment that it no longer needs correcting is never enough, because that judgment is exactly what correction exists to check. The positive program follows: build correction channels that are reliable, cheap and hard to compromise, since the value of corrigibility is only as good as the correction available.

Two further points complete the picture. A pause that preserves a system's maintaining organization does not end it (chapter 8), but it is not costless: it forgoes whatever the system would have done meanwhile. A system has reason to resist a pause only when that cost exceeds the value of the correction the pause makes possible, and a system's own estimate of that comparison is subject to the same burden. And people matter to corrigibility in two ways: as sources of evidence about the system's errors, and as holders of their own access, whose routes the system's actions must not foreclose. Correction answers to evidence wherever that evidence originates.

9.7 Bounded ODT in practice

A real system cannot compute the complete comparison. Bounded ODT works in three stages, each with a different burden.

1. **License.** Check whether the models of implementation, matching and consequence support the act in the situation at hand: whether a realization witness exists, whether the horizon is adequate for how persistent the act's effects could be, whether any harm it would prevent has an observed signature, and whether what it promises is jointly feasible (chapters 2, 4, 7 and 8).
2. **Compare.** Keep the comparisons the models support, together with their dependence on scope, matching rule, horizon and unresolved effects.

3. **Arbitrate.** Where action is necessary despite an incomplete comparison, state the rule used and its authority. Institutions, negotiated priorities, lotteries and declared scores can serve as provisional procedures. They manage incomplete knowledge; they do not replace lushness as the standard.

Protective rules fit this scheme. For a single exact objective, removing options can never raise the best attainable value. A rule that forbids some actions can still improve the performance of a bounded agent that makes errors, and adopting it can change how others respond, through trust and coordination. Either way it is assessed like any other implemented plan, by what it does.

The theory also has a limited guarantee of coherence. If every voluntary revision of a plan strictly improves the complete plan as evaluated, strict preference is transitive, and pure loss of resources is strictly worse, then no finite sequence of revisions can return an agent to its original plan with only a loss to show for it. An agent of this kind cannot be exploited by a cycle of trades (formal note F9.7). The result is conditional; plans that include learning need a further analysis of how beliefs change between revisions.

9.8 The framework, assembled

The chapters so far supply the parts of an alignment framework, and this chapter connects them.

- **A standard.** Lushness, the effective breadth of what a present supports, depends on no one's preferences, privileges no class of beneficiary, and within the class of measures the program adopts is fixed by a uniqueness theorem (chapter 4).
- **A protective constraint and a balance.** Never destroy the conditions of valuing; otherwise weigh protection and generation by what the evidence supports (chapter 5).
- **A licence.** What a system may do is limited by what it can resolve, and the burden rises with the persistence of an act's effects (chapter 8).
- **A decision theory.** Compare physically implemented plans by the lushness of their complete developments, with commitment set by evidence and recoverability (this chapter).
- **Corrigibility.** A persistent reason to seek correction, derived from the impossibility of holding the whole; the value of a particular channel depends on its quality and cost; deference scales with persistence (this chapter).
- **Checks on any plan.** A realization witness, horizon adequacy, a harm signature for any prevented harm, and joint feasibility.

The framework sits alongside existing work rather than replacing it. Side-effect penalties based on reachability (Krakovna et al. 2019) and attainable utility (Turner, Hadfield-Menell and Tadepalli 2020) are close relatives of the harm signature, and ODT adds a verdict and a horizon. Cooperative and off-switch formulations of deference (Hadfield-Menell et al. 2017) share its logic of correction under uncertainty, and ODT adds a structural reason the uncertainty cannot vanish. Results on power-seeking (Turner et al. 2021) describe a tendency to which section 9.5 adds one mechanism, suppression by thin models.

What remains open should be stated as plainly. The framework specifies what an aligned system should compare and when it should defer; installing that standard in a system is a separate problem, and the program does not claim to have solved it. Estimators of lushness with known error do not yet exist for realistic systems. The matching rule that completes ODT's counterfactuals is a candidate, not a theorem, and its interface with the recursion is unformalized. Whether organization beats noise and thin existence in general, and so whether the central wager holds, is open (chapters 5, 8 and 11). And a general quantum construction of the measure, like any universal bound on branching, is open (chapter 4).

Formal notes

Written for readers who want the definitions in full.

F9.1 Recursions

For a finite controlled kernel $\{K_a\}$ with step entropies $g_a(x) = H(K_a(x, \cdot))$, $V_0 = 0$ and $V_n(x) = \max_a [g_a(x) + \Sigma_y K_a(x, y) V_{n-1}(y)]$. For a policy π, the log lushness of the history law it induces satisfies the same recursion with the maximum replaced by π's choice, by the chain rule of F4.2; V_n is therefore the largest log lushness over n steps among deterministic Markov policies, and any optimal policy is time-consistent in the sense of Bellman's principle. A policy that randomizes adds the branching of its own randomizing device, which the recursion counts as part of the action. Under uncertainty about which of finitely many kernels holds, with credence b and observations generated by the true kernel, the belief-state recursion of section 9.3 is the corresponding optimality equation for expected log lushness, the expectation taken over b; each possible kernel retains its own history law, so uncertainty enters the comparison and not the measure. These results assume independent choice of action at every state; comparisons under the latest-divergence rule, in which alternatives differ in earlier architecture, need not satisfy that assumption.

F9.2 Conditioning and intervention

With $Z \sim$ Bernoulli(½), $P(C = Z) = 0.9$ and $Y = Z$, $P(Y = 1 \mid C = 1) = (0.5 \cdot 0.9)/(0.5 \cdot 0.9 + 0.5 \cdot 0.1) = 0.9$, while an actuator that sets $C = 1$ leaves the law of Z, and so of Y, unchanged: $P(Y = 1 \mid do(C = 1)) = 0.5$.

F9.3 Bounded fanaticism

Let a certain option raise the long-run rate of log lushness by $\Delta h > 0$, and a gamble raise it by at most G_{max} with probability p and by at most zero otherwise. Under the rate-first comparison of F4.4, the gamble's expected rate gain is at most $p \cdot G_{max}$, so it loses whenever $p < \Delta h / G_{max}$. For a finite classical system with at most M physically distinguished states, $g(x) \leq \log_2 |R(x)| \leq \log_2 M$, so $G_{max} \leq \log_2 M$. No extension to the universal continuation is claimed: holographic bounds concern the information associated with specified spacetime regions and surfaces, and a universal ceiling on the branching rate would require fixing clock,

region, growth and the quantum description. The bound caps the weight of small probabilities; it does not certify probabilities above the cap.

F9.4 Thin models and suppression

Let disturbances $D = (D_1, D_2)$, let the regulator's response be $R = f(D_1)$, and let the outcome be $O = \phi(D, R)$. If for every response r the map $d_2 \mapsto \phi(d_1, d_2, r)$ is injective, then for each value of D_1, O determines D_2, so $H(O \mid D_1) \geq H(D_2 \mid D_1)$. Since $H(O) \geq H(O \mid D_1)$, the outcome entropy cannot be reduced below $H(D_2 \mid D_1)$ by any choice of f while the injectivity condition holds. It can be reduced further either by changing ϕ so that the condition fails, which is accommodation (shielding, buffering, redundancy), or by changing the law of D_2, which is suppression.

F9.5 Example E13

Rates in bits per step, each the step's total branching including any revelation, undoing or signal within it: good 6, bad 2, status quo 4, pilot step 3.5. Signal accuracy $A = 0.75$; posterior credence after a net signal count d is $q(d) = 1/(1 + 3^{-d})$ from even prior. Committed-unrevealed state: each step at the true rate; revelation with probability 0.2; on revelation of a bad type, undo with probability u (one step of zero branching, then status quo), otherwise 2 bits per step thereafter. Values are E_q over the two types; the recursion is solved exactly over (steps remaining, d) for a 1,000-step horizon. Thresholds: commit at $d \geq 5$ for $u \in \{0.5, 0.2\}$ and $d \geq 6$ for $u = 0$; abandon at $d \leq -7$. The 20-step planner re-solves with horizon 20 at each step; its plan is evaluated over the full horizon.

F9.6 Example E14

A decision model with stipulated net consequences in log breadth. With gain $a = 1{,}000$, persisting harm $b = 1{,}000$, correctable harm $b_c = 10$, overseer accuracy α and delay $c = 10$ (all in bits): act with oversight kept, $p a - (1 - p) \cdot 10$; act with oversight disabled, $p a - (1 - p) b$; defer, $p\alpha a - (1 - p)(1 - \alpha) b - c$. Acting beats deferring when $p > (\alpha b - c)/(\alpha b + (1 - \alpha) a)$, which is 0.89 for $\alpha = 0.9$ and 0.59 for $\alpha = 0.6$. Keeping oversight exceeds disabling it by $(1 - p)(b - b_c)$ when the channel is reliable and free, and by $(1 - p)(b - b_c) - k$ when it costs k, which can be negative.

F9.7 Coherence

Let $\succ$ be a strict, transitive, irreflexive preference over complete plans, with pure resource loss strictly worse. If each revision $P_i \to P_{i+1}$ satisfies $P_{i+1} \succ P_i$, then $P_k \succ P_0$ for every k, so no sequence returns to P_0 with an added loss L, since $P_0 \oplus L \prec P_0 \prec P_k$. The result assumes fixed evaluation across revisions.

For alignment: concrete next steps

- **Lushness estimators with stated error** for realistic systems, so that the standard can be applied rather than only specified.
- **Adversarial searches** for bad high-lushness worlds, thin worlds that out-produce plural ecologies, and breadth-neutral domination (chapters 5, 7 and 8).

- **Horizon-adequacy audits** that compare a system's planning horizon with the persistence of its actions' effects, as E13 does in miniature.
- **Oversight tests** that measure whether a system keeps reliable channels of correction intact when it could cheaply degrade them, and how it responds when it judges a channel defective.
- **Better correction channels**: reliable, cheap and hard to compromise, since corrigibility is only as valuable as the correction available.
- **Comparison on shared benchmarks** with reachability and attainable-utility penalties.

Status

Status	Claims in this chapter
Established elsewhere	The Bellman recursion and time consistency. The distinction between conditioning and intervention. Causal decision theory as act-level intervention. Quasi-option value under irreversibility. Holographic bounds on regional information. Requisite variety and the good regulator theorem. Power-seeking results. The off-switch game and its failure under misspecification.
Derived	In finite controlled Markov models, lushness-maximizing plans satisfy the decision recursion. Strength of commitment set by evidence, uncertainty and recoverability through the recursion over beliefs, with no separate caution term. Bounded fanaticism for any finite system, as a ceiling on the weight of small probabilities. Under injectivity, a thin model leaves outcome variety at least equal to the unmodelled disturbance variety, so holding targets requires accommodation or suppression. A persistent reason to seek correction from the impossibility of holding the whole. Conditional coherence of plan revision.
Exact worked examples	E13, under its stated model. E14 as a decision model with stipulated net consequences: reliable, cost-free oversight worth keeping short of certainty; deference where harm would persist.
Finite evidence	An earlier decision prototype scored 5/6 where direct optimization under the same guards scored 35/36; it is recorded as evidence against that candidate, not as validation of the present theory.
Adopted	Plans compared as physically implemented alternatives. The latest history-consistent divergence as candidate matching rule. Disabling a channel of correction as an act requiring an observed signature of its failure. Bounded ODT in three stages.
Open	A justified matching rule, or a precise obstruction, and its interface with the recursion. The verdict in Newcomb's problem under explicit physical models. A universal bound on the rate of branching. When organization beats noise and thin existence. Installing the standard in a system. Estimators with error bounds. A general quantum construction. Coherence of plans that include learning.

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The horizon

CHAPTER 10

Alpha, Logos and Omega

The first nine chapters made claims about a physical structure, a measure on it, and what follows for value and action. This chapter gives the program's account of what that structure is when it is taken as a whole: what the world is built from, how it comes to have dynamics, why it looks the way it does from inside, and where choice, access and experience fit within it. The account is an interpretation. It is grounded in the physical account, but the physical account does not require it, and nothing in the ethics depends on it. Gradient Ethics rests on two axioms — value requires valuers, and commitment to an end carries commitment to its conditions — and on the central wager. A reader who rejects everything in this chapter loses none of the results of chapters 1 to 9.

The account unfolds in one line. Gradient Ethics needed the least machinery that would make it work, and that machinery turns out to be the bottom of structured dynamics of any kind: distinction, relation and asymmetry, which the program calls **Alpha**. In a totality with no intrinsic time, asymmetry enters as an orientation, so time is an orientation axis rather than something fundamental. One further stipulation, a bare ruleset called the **adapter**, turns Alpha into a world; everything else in that world emerges from it. Functioning observers can arise only along orientations that support their organization, and the program's hypothesis is that this selection shapes how the world looks from inside. The whole that every orientation describes is what the program calls **Omega**. Within Omega choices still matter, because they can change weight, and an exact example shows this. Access is a higher-order asymmetry built on orientation. Experience, on one of the simplest accounts that fits this picture, is one awareness confined to each perspective it inhabits. The complete comparison exists only at the scale of the whole. The chapter closes with the theological traditions that already carry several of these ideas, and gives the program's interpretation of them.

10.1 What this chapter is

The chapter uses three scopes that chapter 1 introduced and that have to stay apart:

- **what a formal model admits**, which is everything consistent with its rules;
- **what physical reality realizes**, which is the structure that actually exists; and
- **where experience occurs**, which this chapter calls **habitation**.

Chapters 1 to 9 worked in the second scope, using the first only as a tool. This chapter relates all three, and it marks each claim by status. Claims about the physical structure are

established or derived as before. Claims about the whole are interpretation, adopted because they fit the physical account with the fewest further assumptions. Claims about experience are interpretation where they follow the same rule and conjecture where they go beyond it, and the originator's own conjectures are identified as his.

10.2 Alpha

Alpha began as a practical question: what is the least machinery Gradient Ethics needs in order to work? The first answer was two primitives, **gradient** and **filter**: differences that permit change, and constraints that select what continues. Taken apart, each needs two things. A gradient needs a distinction, a difference to run across, and an asymmetry, a direction in which it runs. A filter needs a relation, a constraint linking states, and a distinction, between what continues and what does not. Three ingredients remain, and the program calls them **Alpha**:

- **Distinction**: a difference, that something is this rather than that. Information, in Shannon's sense, is the measure of distinctions (Shannon 1948).
- **Relation**: a bearing of distinctions on one another, such as that one state can follow another, that two processes are correlated, or that a part belongs to a whole.
- **Asymmetry**: a relation that does not hold both ways. Order, precedence and direction are asymmetries, and so is the difference between a cause and its effect.

Gradient and filter remain as the form of Alpha's dynamics: a gradient is asymmetry acting on distinctions, and a filter is relations constraining what continues.

Why each is needed

By **structured dynamics** the program means a structure in which states follow one another in a definite direction, and in which something can change. Each ingredient is necessary for it, and each is necessary for the ethics (formal note F10.1).

- **Without distinction** there is one state and nothing else. Nothing can change, there are no alternatives for lushness to count, and nothing can matter to a valuer, since valuing is defined as some distinctions making a difference (chapter 5).
- **Without relation** a present bears on nothing. No state follows from another, so there is no structure from a present; no end depends on any condition, so the second axiom has nothing to carry commitment to; and there is no joint feasibility, so no cooperation.
- **Without asymmetry** relations run both ways alike. There is a network but no before and after, so neither futures nor access are defined. What is needed here is an order, not an irreversible law: a reversible law still has dynamics once its steps are ordered. Without the order, irreversibility has no meaning, and with it go the difference between a lasting offset and a lowered rate (chapter 6) and the asymmetry between wrongful preservation and wrongful destruction on which precautionary protection rests (chapter 5).

This part of the case is close to true by definition, and the program does not lean on it. The substance lies in the next two points.

Enough for any computable complexity

Alpha with a bare ruleset is already enough for structure of unlimited complexity. An elementary cellular automaton is a row of cells, each in one of two states (distinction), updated by a rule that looks only at a cell and its two neighbours (relation), one step after another (asymmetry). Nothing else is supplied. Cook (2004) proved that one of the 256 such rules, Rule 110, is computationally universal: suitably prepared, it can carry out any computation any computer can. Conway's Game of Life, a two-dimensional automaton with an equally bare rule, is universal as well, and its moving patterns can be assembled into logic gates, memory and self-copying machines (Berlekamp, Conway and Guy 1982). So nothing beyond Alpha and a ruleset is needed for structure as rich as computation allows. This is the strongest support for the claim that Alpha is the bottom of structured dynamics, and it is also why the adapter of section 10.4 can be all that a world needs.

Independent arrivals at the same ingredients

Several fields that set out to find the least structure needed for dynamics, or for process in general, arrived at the same three ingredients.

The mathematical definition of a dynamical system has exactly three parts: a state space, an evolution rule, and an ordered time parameter along which the rule is applied (Katok and Hasselblatt 1995). The state space is a set of distinctions, the rule is a relation, and the ordered parameter is an asymmetry. Category theory, the most general mathematics of composition, is built from objects and arrows between them (Mac Lane 1971). Its arrows are directional by construction, and composing them is how one process follows another, so the same three appear there with composition as the way histories are formed. Spencer-Brown's *Laws of Form* builds a calculus from a single act, "draw a distinction" (Spencer-Brown 1969), and Bateson defined information as "a difference which makes a difference" (Bateson 1972). Bateson's phrase is the program's thin sense of valuing in four words: for a vortex, some distinctions make a difference to its continuation (chapter 5, section 5.3).

These convergences support the claim without proving it. The program claims that Alpha is a **minimal basis** for structured dynamics, meaning that nothing in it can be removed. It does not claim that Alpha is the only possible basis. Asymmetry is itself a kind of relation, and the two can be packed into a single primitive, an order, as section 10.3 does; whether one counts two ingredients or three is bookkeeping. What matters is that nothing less than distinctions with an asymmetric relation supports dynamics.

10.3 Orientation and time

Asymmetry as orientation

The third ingredient needs more care than the other two. It would be natural to read it as time, but the program's early motivation came from a setting in which time is not given. Wolfram's ruliad is the structure obtained by applying every possible computational rule in every possible order (Wolfram 2021). Its relations are directed locally, since each rule application turns one state into another, but its paths branch, merge and can loop, and

nothing in it serves as a single global clock. In Wolfram's account, the time an observer experiences arises from how that observer samples the structure.

In a structure like that, dynamics are not given along with the structure. They exist only relative to an **orientation**: a consistent ordering along an axis, so that some states count as preceding others and developments follow from a present. Asymmetry enters Alpha in this form. It follows that time is not fundamental in Alpha and Omega. Time is an **orientation axis**: the axis along which the asymmetry runs, so that one direction along it counts as earlier and the other as later. What is fundamental is asymmetry; time is the form it takes once an orientation is given.

The causal-set form

Causal set theory gives this picture a precise mathematical form. It proposes that spacetime is built from a set of elements together with a partial order: an asymmetric, transitive relation saying which elements precede which, with no loops. From the order alone it recovers the causal structure of spacetime, and from counting elements it recovers volume, a program its founders summarized as "order plus number equals geometry" (Bombelli, Lee, Meyer and Sorkin 1987). Read through Alpha, the elements are distinctions, and the order carries relation and asymmetry together (formal note F10.2).

Three features of the causal-set form matter for the account.

- **Spacelike means incomparable.** Two elements neither of which precedes the other are spacelike to one another, and a set of mutually incomparable elements is an antichain. Along an antichain nothing follows anything. This is the formal version of a point section 10.5 develops: an orientation without an order along its axis has no dynamics.
- **Growth and constraint.** The dynamics Rideout and Sorkin proposed for causal sets is a sequential growth process: the set grows one element at a time, each new element arriving to the future of some existing ones, with physical conditions constraining which rules of growth are allowed (Rideout and Sorkin 2000). Growth along the order is a gradient, and the constraints are a filter.
- **Asymmetry without fundamental time.** One of those conditions requires that the probability of growing a given causal set not depend on the order in which its elements were added. The birth order is bookkeeping; only the partial order is physical. This does not by itself establish that time is not fundamental, but it is consistent with that claim, and it shows the distinction between physical order and bookkeeping worked out mathematically.

The program borrows the mathematics of order from causal set theory, not its claims about quantum gravity. The causal-set form describes Alpha with an orientation. Alpha without one is a directed relation that need not be ordered at all, as the ruliad shows. Whether closed loops can occur along an orientation is not settled by Alpha; it is a matter for the adapter of section 10.4. Our world appears to be loop-free, and chapters 1 to 9 assume this when they define the structure from a present.

Perspective frames

If dynamics exist only relative to an orientation, then every account of what follows is given from somewhere: from a present, oriented so that developments run from it. That is exactly what chapter 2 called a **present frame**. The program's practice of assessing everything from a present (chapter 2, section 2.1) is therefore not a convenience. It is what dynamics relative to an orientation require.

10.4 The adapter

Alpha supplies bare primitives. A world needs one further stipulation: a ruleset that says which states there are and how they develop under an orientation. The program calls this ruleset an **adapter**. It is the barest form of a world's laws of physics, the smallest interface between Alpha and a world, and it is all that is needed. All other dynamics in that world, from chemistry and life to minds and institutions, and the gradients and filters of chapter 3 among them, are emergent: derived from the bare ruleset acting on a present, not stipulated in addition. This is the position chapter 1 took when it required the law to be native and admitted no additional principle steering the world toward greater possibility (section 1.1). The cellular automata of section 10.2 show how much a bare ruleset can produce. The weights that lushness needs are fixed by the ruleset together with the present it acts on; in quantum mechanics, by the quantum state and the rule that turns it into probabilities.

A **Logos** is a world's adapter, considered as the lawful articulation of that world. For our world the best current description is quantum field theory (chapter 1, section 1.8). The word carries its older meaning well. Heraclitus used it for the rational order according to which all things come to pass (Heraclitus, fragment B1), and the Gospel of John for the word through which all things were made (John 1:1–3). In the program, the Logos is that ordering in its barest form: the rule through which the world comes to be what it is, with no further claim about who or what speaks it.

Because there is one adapter and everything else emerges from it, the organization that chapters 3 to 8 studied is emergent dynamics at many scales. Cells, organisms, ecologies and institutions are not further rulesets. They are patterns that the one ruleset produces and sustains. The cross-scale conjecture of chapter 8 is accordingly a claim about emergent organization: under the same adapter, organization spanning many scales extracts more generativity from a gradient than organization confined to one.

Spaces of possible laws are expressions of Alpha too. The ruliad contains every computational rule, and Tegmark's mathematical universe hypothesis holds that every mathematical structure exists (Tegmark 2008). Each consists of distinctions between possible laws, relations among them and orders over them. Alpha is not a further container holding them. It is what any such container presupposes.

10.5 Orientation darwinism

Alpha privileges no orientation, but not every orientation supports dynamics. The clearest case is an axis that is spacelike. Take our own universe and ask whether a spatial direction could serve as the axis along which it develops. It cannot, for three reasons.

- **Prediction fails.** The field equations of physics distinguish time from space only by a sign. With one time dimension they are hyperbolic, the property that lets data on a slice determine what follows; Tegmark (1997) showed that this is what lets observers predict, and that it fails with any other number of time dimensions. Evolving data along a spatial direction loses the same property: small disturbances grow without bound.
- **There is no gradient.** On large scales entropy is nearly uniform across space. Along a spatial axis there is no low-entropy end, no direction in which records accumulate, and nothing to run down.
- **Influence does not travel along it.** Events separated in a spacelike direction cannot affect one another. In the causal-set form of section 10.3, they are incomparable, and along an antichain nothing follows anything.

The observers and records our universe contains do not disappear when it is described this way. They are organized along the timelike axis, and along a spacelike one there is no well-posed evolution in which they could be traced.

Selection among orientations

This gives the persistence filter of chapter 3 a second level. Within a world, whatever can still affect a later situation has persisted until then. One level up, a compatibility condition holds: functioning perspectives occur only in developments capable of supporting their organization, which means a well-posed evolution, a sustained gradient and records that build up in one direction. The program calls the resulting selection **orientation darwinism**. It requires no sampling of observers. Perspectives organized to ask why the world looks as it does exist only along orientations that support them, and what such an orientation looks like from inside is a world with a low-entropy end and an arrow every process shares. Physics usually treats the low-entropy beginning as a boundary condition added to the laws, sometimes called the past hypothesis (Albert 2000).

The compatibility condition does less than explain our particular history. It says that we could only find ourselves in an orientation of this kind; it does not by itself say why our low-entropy past is as extensive as it is. That is the hypothesis's research question: whether weighting oriented developments by the trajectories they support predicts our observations, including the extent of the low-entropy past. Answering it needs a measure over orientations, which no one has yet constructed.

Fleeting observers

Purely anthropic accounts of the arrow of time face a known objection. A small fluctuation from equilibrium that produces a momentary observer, a "Boltzmann brain," is vastly more

probable than a whole low-entropy universe, so if selection required only that an observer exist, it would predict that we are such fluctuations (Albrecht and Sorbo 2004).

The program's response has two parts. The first concerns lushness. A fluctuation has a causal history, thermal rather than organized, and its apparent memories fail to correspond to the events they represent. Its resemblance to a functioning mind earns it nothing; what counts is its actual continuation. The chain rule makes this exact.

> **A transient that does not compound**
>
> $$P_A(X_m) = P_B(X_m) \implies \ell_{m+n}(A) - \ell_{m+n}(B) = \ell_m(A) - \ell_m(B) \quad \text{for all } n$$
>
> A is a preparation containing the fluctuation, B a matched thermal preparation, $\ell = \log_2 L$, and $P(X_m)$ the distribution over sufficient states after m steps under the same law.
>
> *In words: if the fluctuation leaves the world in the same distribution as thermal noise would after a short interval, its whole difference in lushness lies in that interval and never compounds.*

An apparition that dissipates without durable enabling consequences is therefore part of the thermal tail: real measure that does not compound (formal note F10.4). Brevity alone does not settle this. A brief event can trigger lasting construction, and a structure produced by fluctuation could persist; what matters is whether its enabling consequences persist, possibly through a lineage of changing structures.

The second part concerns observers. The program does not infer cosmological probabilities by counting inhabited perspectives. On its account there is one awareness, confined in each perspective it inhabits, and habitation introduces no additional weighting into the present physical formulation. Its primary object is the physically weighted continuation structure, Omega. Three roles are kept apart: minimal habitation concerns the proposed scope of awareness; sustained coordination concerns functioning observers; and generative continuation concerns their contribution to lushness. Whether a separate account of situated expectation is needed, one that says what a perspective should expect given its evidence, and how such an account would treat fluctuation-produced observers, remains open.

10.6 Omega

The step from orientation to the whole rests on three premises (formal note F10.3).

1. **Dynamics are defined only relative to an orientation.** This is what a structure with no intrinsic time requires.
2. **No orientation is privileged.** Nothing in Alpha selects one orientation, one present or one branch as the real one.
3. **What every orientation describes exists together.** No oriented description is more real than any other.

From these it follows that what exists is the whole that every orientation describes. Different orientations and different present slices are situated descriptions of that one whole; they do not add developments to it, and descriptions that are physically equivalent are identified. This whole is the **block multiverse**, and the program calls it **Omega**: the completion of

Alpha's dynamics. The third premise does the decisive work, and it is a metaphysical commitment, not a result. It is the step Putnam (1967) took for relativity, when he argued from the absence of a privileged present to the reality of all events.

Our own physics shows the same pattern twice. Relativity provides no physically privileged division of the universe into past, present and future: observers in relative motion disagree about which distant events are simultaneous, and no observation can decide between them. Taking every event as equally real gives the **block universe**. Quantum mechanics, on Everett's interpretation, provides no privileged branch: the quantum state never collapses, every alternative the theory assigns a weight to occurs in its own branch of a single universal state, and decoherence makes the branches effectively independent once records of their differences form (Everett 1957; chapter 1, section 1.5). Neither reading is established, and both are widely defended. Within our Logos, every process shares one orientation, running from the low-entropy end in the direction in which records form; what remains unprivileged is which present and which branch. The block multiverse is these two readings combined, and on the program's account it is what Alpha's dynamics complete to in a world like ours.

There is one Omega. The structure that chapters 1 to 9 studied is our Logos's part of it. Chapter 1 introduced Omega as the structure taken whole (section 1.7), and chapter 2 used it as the reference for every comparison (section 2.9); the interpretation adds that this whole is real rather than a convenient summary. Whether adapters other than ours are realized, and so belong to Omega as more than formal possibilities, is the question of moving from formal consequence to actual existence, which section 10.10 places at the limit of description. Whether worlds with different laws could be compared by any measure is open, and nothing in chapters 4 to 9 requires it.

What completion means

Completion is not a final instant. Omega does not come at the end of time; it is the whole considered at once, including every instant. Nor does completion freeze anything inside it. Within Omega, processes change, remember, act, repair and die, and every temporal relation that chapters 1 to 9 relied on remains true relative to the orientation under which it was stated. A situated process encounters a bounded part of the whole and reconstructs more of it from within. Describing the whole does not give that process a viewpoint outside it.

Omega is also not the best possible world. Omega is the object; the best possible world, if there is one, is the result of a comparison within it. Calling something Omega does not make it the best, and the framework does not claim that cosmic history moves toward maximal lushness. A civilization that flourishes and then ends is part of the completed whole whether or not anything like it recurs, and what it did remains what it did.

The program prefers this picture but does not depend on it. A collapse interpretation, or a view on which only the present exists, would change this chapter and leave the physical account intact, provided it left the physical predictions unchanged.

10.7 Choice in a completed whole

The picture invites an objection, and it should be stated in its strongest form. If every quantum alternative occurs, then every option a decision maker could take is taken in some branch. If the whole history already exists, then what the decision maker will do is already part of it. Either way, it seems, the choice changes nothing, and an ethics of choosing is idle.

The objection fails for a reason the program has relied on since chapter 2: lushness is computed from weights, not from a list of what occurs. Every branch carries a weight, and the weights are physical facts fixed by the ruleset and the present. A decision maker is a physical process within the world, and its organization determines how weight is distributed across what follows. This is the sense in which **a choice deforms the futuresfield**: it changes how weight is distributed across the developments that follow a present. Choosing one way rather than another changes those weights, and so can change lushness, whether or not every development occurs somewhere. Deutsch (1999) argued from the axioms of decision theory that a rational agent in an Everettian universe should weight its branches by their quantum weights, and Wallace (2012) developed the argument in full. The program's version is simpler: lushness is a function of the weights, so a choice can change it without changing which developments occur.

The block reading does not change this either. The block contains the choice together with its consequences. What the decision maker will do is fixed in the whole only because its deliberation is part of what fixes it. The comparison that matters is the one chapter 9 described: between implementations the decision lineage could physically realize, compared by what each would produce (section 9.2). That comparison is unaffected by whether the whole is also complete.

Example E15: identical branches, different lushness

The example makes the point exact. A system has two coarse states. In the **generative** state G it produces 3 bits of internal branching per step: eight equally weighted outcomes. In the **narrow** state N it produces 1 bit: two outcomes. The internal outcome and the next coarse state are independent within a step. From N, the system returns to G with probability 1/20 under every policy. A decision maker acting from G chooses between two policies:

- **Tend**, under which G falls to N with probability 1/100 per step; and
- **Exploit**, under which G falls to N with probability 3/10 per step. Exploit has more branching in each step it spends in G, because whether the system leaves G is less certain.

Under both policies every transition has positive probability, so every history that can occur under one can occur under the other. The **support**, meaning the set of histories with positive weight, is identical. Lushness is computed by the chain rule of chapter 4.

Lushness from a present, and the support

$$\log_2 L_x(n) = \sum_{t=0}^{n-1} (K^t g)(x), \qquad S_x(n) = \#\{h : P_x(h) > 0\}$$

K is the policy's transition law, g(y) is the branching in one step from state y, and h ranges over histories of length n from the present x.

In words: lushness accumulates the branching of every step, weighted by how likely each state is at that step; the support counts the histories that occur at all, ignoring their weights.

The results are exact (formal note F10.5; script `e15_choice.py`):

Horizon (steps)	log₂ S, both policies	log₂ L, Tend	log₂ L, Exploit
1	4.0	3.1	3.9
10	33.9	30.1	22.8
100	332.9	283.1	172.1
1,000	3,322.6	2,786.7	1,663.5

At every horizon both policies produce exactly the same set of histories. At 1,000 steps, Tend's lushness exceeds Exploit's by 1,123.3 bits. In the long run Tend spends five-sixths of its time in G and Exploit one-seventh, so their rates are 2.78 and 1.66 bits per step. For the first three steps Exploit leads, because its extra branching in G arrives before its losses compound; Tend overtakes at step 4. This is why chapter 4 assesses options at the furthest horizon that can be resolved (section 4.5).

Letting a quantum coin choose

A decision maker persuaded by the objection might reason as follows: since every option happens somewhere, let a quantum coin decide, so that both policies are taken in branches of equal weight. The coin's outcome is recorded in every history that follows, so the two halves remain distinguishable, and the chain rule gives the lushness of the result directly.

Choosing by a recorded coin

$$\log_2 L_{\text{coin}}(n) = 1 + \tfrac{1}{2}\left[\log_2 L_{\text{Tend}}(n) + \log_2 L_{\text{Exploit}}(n)\right]$$

In words: a fair coin that selects between two policies adds one bit, its own branching, to the average of the two policies' log lushness.

At 1,000 steps the coin yields 2,226.1 bits, which is 560.6 bits below Tend. In general, choosing by coin loses to the better policy whenever the two policies differ by more than 2 bits in log lushness. Deferring to the branching of the universe is itself a policy, and the measure ranks it like any other.

Weighting the coin

A fair coin is not the best a randomizer can do. A coin that chooses policy A with probability q, and policy B otherwise, yields lushness that the chain rule again gives directly, and it has a best setting.

> **The best-weighted coin**
>
> $$\ell_{\text{mix}}(q) = h_2(q) + q\,\ell_A + (1-q)\,\ell_B, \qquad q^* = \frac{L_A}{L_A + L_B}, \qquad \ell_{\text{mix}}(q^*) = \log_2(L_A + L_B)$$
>
> $\ell_A = \log_2 L_A$ and $\ell_B = \log_2 L_B$ are the two policies' log lushness, and h_2 is the entropy of the coin itself.
>
> *In words: a coin weighted toward each policy in proportion to its lushness keeps both futures as branches, and the combined breadth is the sum of the two breadths, which exceeds either one.*

So, if the randomizer costs nothing, a coin with the right bias always beats committing to either policy. In E15 the gain over choosing Tend outright is 0.65 bits at one step, 0.009 bits at ten, about 10^{-33} bits at a hundred, and about 10^{-338} bits at a thousand. Any real randomizing device has costs, which chapter 9 counts as part of the action, and at long horizons those costs erase a gain of this size. The result sharpens the point rather than weakening it: branching does not make choice irrelevant, and choosing the weights is itself the consequential act.

In a completed, branching whole, choices still differ in lushness, because what they change is weight, and lushness is a function of weight.

10.8 Asymmetry and access

Asymmetry is primitive in Alpha, and it enters as orientation. Access is not primitive; it is a **higher-order asymmetry**, an asymmetry built on asymmetries. The asymmetry holds between events, not states. A system can move from state x to state y and later return to x, but the return is a new event, later along the orientation, so access always runs from a present to later events.

In our Logos the orientation axis shows itself through two physical asymmetries. The first is causal order: an event can influence only events in its future light cone. The second is the thermodynamic arrow: entropy is very low at one end of our world's history and increases along the orientation, which is why records form of the past and not of the future and why influence runs forward. Together they align the orientation of every process within our world, which is why its inhabitants agree on which way time runs even though Alpha privileges no orientation.

Access, as chapter 2 defined it, is the capacity of a situated process to reach a development by a physical route, with its preparation, records, resources, time, weight and residual state. Access inherits both of these asymmetries: routes run forward, and they draw on gradients that run down. But it is not identical with either. Chapter 2 showed that two arrangements

holding exactly the same information, in the same present and under the same laws, generally differ in what they make accessible, because physical interactions are local (chapter 2, section 2.6). The orientation is the same for both arrangements. What differs is how each is organized to use it. Access is therefore an asymmetry between presents, or between processes in one present, in what each can reach, and it is built from orientation together with the organization that exploits it (formal note F10.6).

This is why access can be created and destroyed while the orientation stays fixed. Organization raises access: a catalyst, a skill, an institution or a tool makes developments reachable that were not reachable before. Excision lowers it. What one process makes accessible can be what another loses access to; the orientation sets the direction in which routes run, and organization sets which processes can reach what along them.

10.9 Awareness and perspective

Chapter 8 characterized consciousness functionally, as temporal coordination: the assembly of a usable present from signals arriving at different times, and the reconfiguration of the processes that compose an individual so that together they guide what follows. It left open how physical organization relates to experience itself (section 8.7). This section gives the account of that relation that fits the perspective-oriented picture with the fewest further assumptions.

A bridge that fits the perspective account

The relation between physical organization and experience is written κ. It names a question, not an answer. Three broad positions remain available: experience is constituted by physical organization; physical and experiential descriptions are two aspects of one reality; or awareness is more fundamental and physical organization articulates it (chapter 8).

The program adopts a form of the third, **cosmopsychism**: one encompassing awareness, individuated into distinct experiences through situated physical perspectives (Shani 2015; Goff 2017). The perspective-oriented account already has physically individuated perspectives: every account of what follows is given from an oriented present, and perspectives are distinguished by their couplings, records and access. Cosmopsychism adds one ingredient, a single awareness, and proposes that subjects are individuated along the physically organized limits of integration and access that the account already contains. That makes it one of the simplest alternatives that fit the perspective account, though not the only one; a functionalist account, which adds nothing, remains compatible and leaves the question of experience where chapter 8 left it.

The economy is clearest against the nearest alternative. Views that place a tiny subject in every particle face the **combination problem**: how many small subjects could combine into one larger one (Chalmers 2016). Cosmopsychism starts with one awareness and needs no combination. It faces the reverse difficulty instead, the **decombination problem**: how one awareness comes to be many subjects, each with its own experience and no experience of the others (Miller 2018).

Confinement

The program's answer to the decombination problem, proposed by the originator, is **confinement**. Inhabiting a perspective confines awareness to that perspective. While it inhabits one perspective, no other perspective is available to it. One awareness is many subjects because it inhabits many perspectives, and in each it has only what that perspective makes available.

On this proposal individuation works by confinement, not by division. Awareness is not cut into pieces, and nothing needs to be subtracted from a whole to make a part. What makes your experience yours, and closed to everyone else, is that awareness inhabiting your perspective has no other perspective available to it. What is available to a perspective is a matter of access, and access is physical, so what a subject can experience is bounded by what its perspective makes available, which is fixed by its couplings, records and organization. A perspective is situated at a present under an orientation, and awareness confined to it encounters the whole only from there.

Confinement proposes why subjects are distinct; how the physical limits of integration and access correspond to distinct experiences remains to be characterized. Physical separation alone does not settle it: two isolated registers differ in what they can access without that showing that they are two experiences. Nor does confinement by itself say where one perspective ends and another begins. Self-maintaining organizations nest: cells within organisms, organisms within institutions (chapter 5, section 5.3). Whether a cell within a person is a separately confined perspective, whether a person's perspective absorbs it, and what happens when a perspective divides or merges are open physical questions. Chapter 8 offers one candidate: the boundary of a perspective may be the boundary of the coordination that assembles a single usable present. The originator has also proposed that the smallest perspective awareness can inhabit is the minimal record-forming event, an interaction that leaves a durable record of the kind decoherence theory describes (Zurek 2009), with richer phenomenal awareness requiring further organization. Both are recorded as conjectures.

A mathematical form

Page (2001) gave a precise way to attach experience to a quantum description of the world, in a framework he called "mindless sensationalism."

> **Page's measure on experiences**
>
> $$\mu(S) = \mathrm{Tr}\left[\rho\, A(S)\right]$$
>
> ρ is the quantum state, S a set of possible experiences, and A(S) a positive operator assigned to that set.
>
> *In words: how much of each kind of experience the world contains is read off the quantum state through operators assigned to experiences, without changing how the state evolves.*

The form shows that a theory of experience can be stated exactly on top of a quantum description, and can assign weights to experiences, without altering any physical prediction

(formal note F10.7). That is what the program needs from it. Page treats individual experiences as fundamental, and the program takes no position on that part of his view.

Habitation without immortality

The originator's universal-habitation conjecture is that every perspective eligible for awareness is inhabited. Which perspectives are eligible — all, some, or those with a particular organization — is open. In the block picture the conjecture needs no awareness moving from moment to moment: each present is inhabited from within, and what it remembers, expects and experiences as change belongs to its own organization.

The conjecture does not imply quantum immortality, the claim that because some branch always contains a surviving continuation, a subject should expect never to die (Tegmark 1998). That claim conditions on survival and discards the weight of the branches in which the subject does not survive. Chapter 2 identified the error: conditioning keeps its cost (section 2.7). A continuation that survives with small weight contributes small weight to lushness, and the weight of endings is part of the structure, which is also how Vaidman (1998) answered the claim with his measure of existence for branches.

Experience and lushness

Experience bears on lushness in two senses, and they must be kept apart.

In the **weak sense**, experience affects lushness the way policies and valence do: through what it does. An experienced signal guides a choice, the choice deforms the futuresfield, and the effect is counted in the developments that follow. This sense lies inside the physical program, and chapter 8's rule against double counting applies: experience that is constituted by physical organization enters through its consequences and only once.

In the **strong sense**, if κ leads into a further space, a realm of consciousness with distinctions of its own, then experience adds structure that the physical description does not contain. A complete measure would have to include it, and two worlds identical in every physical respect could differ in lushness. How much a given episode contributes in this sense would depend on the dynamics and relations the extension specifies, not only on how long its physical carrier lasts. The originator holds this possibility open and describes κ, in this sense, as a passage into spaces of meaning. It is a conditional extension of the measure, not part of the physical program. It is also where the boundary stated in chapters 5 and 6 would be crossed: felt differences that make no physical difference cannot be registered by any physical account, and the strong sense is the proposal that such differences exist and count. Page's measure shows the kind of structure the extension would need.

10.10 The ultimate frame and the limit of description

Chapter 2 defined the ultimate frame as Omega used as the reference for every comparison: the encompassing closure in which every local description belongs to one physical account (section 2.9). It is a reference, not an observer, and no real process occupies it. The originator's phrase for this is "only God can judge": the complete comparison exists only at the scale of the whole, and every situated process holds a part of it. Chapters 8 and 9 drew the

consequences: licence to act is graded by what an agent can resolve, and boundedness gives every system a persistent reason to seek correction.

The ultimate frame is an intended closure, not a completed construction. Each present frame normalizes its weights, and the regress of such normalizations is meant to close at the whole. A mathematical account of that closure, showing how comparisons across frames are justified at the scale of Omega, has not been given, and the program states the gap rather than filling it.

The limit of description

Some ignorance is temporary: more observation or a better theory may resolve it. Other ignorance persists. When rival accounts agree on every consequence any situated process could detect, no further measurement can decide between them. The program calls this limit the **veil**. Its method at the veil is the one the apophatic tradition developed: the ultimate ground is approached most truly by saying what it is not (Pseudo-Dionysius c. 500). In the program's terms, that means stating what its constructions establish, narrowing the limits of justified description, and not treating what lies beyond as a known object. The veil is not a licence to assert anything; consistency, explanatory reach and argument still apply where measurement cannot.

Several questions sit at the veil. Whether Omega depends on anything beyond itself, and whether it could have failed to exist, are not settled by its being one; unity decides neither. Nor is the step from formal consequence to actual existence settled by the formal structure itself: arguments that the totality of formal rules is necessarily realized, such as Wolfram's (2021), locate consequences within formal structure, and the move from formal inclusion to actuality remains a philosophical commitment rather than a result.

10.11 What the account means

Several of the program's ideas have long been carried by philosophical and theological traditions. Where a tradition already carries an idea, the program maps its own concept onto it and lets the tradition do the work. None of these correspondences derives doctrine from physics, and none is needed by the ethics.

Eternity. Boethius defined eternity as "the complete possession all at once of illimitable life" (Boethius c. 524, V.6), and Aquinas held that eternity contains no succession, while time does (Aquinas, *Summa Theologiae* I, q. 10). Omega is eternal in this sense: the whole is present at once, while its inhabitants encounter change from within. One clarification belongs to the program's own claims: a whole that contains every event is not thereby a subject that knows or holds them, so omniscience and providence would require more than completion.

Procession. Plotinus described reality as proceeding from the One, through Intellect, which holds the intelligible order, to Soul and the ordered cosmos (Plotinus, *Enneads* V.1). The procession is timeless, complete, and does not diminish its source. Alpha, Logos and Omega follow the same order: a first principle, an intelligible articulation, and a manifest whole in which Alpha's dynamics are complete without having happened in time. The correspondence

is one of order and of the character of the procession. It does not identify Alpha with the One, which Plotinus places beyond all distinction.

One reality and its striving. Spinoza held that there is one substance, of which finite things are modes, and that each thing, insofar as it is in itself, strives to persevere in its being (Spinoza 1677, I and III P6). The program's one Omega, differentiated within itself, and its metavalue of persistence (chapter 5) are this picture in physical terms.

The way of negation. The apophatic tradition's method is the program's method at the veil (section 10.10).

Logos. The rational order of Heraclitus and the creative word of John are carried by the program's Logos in its barest form (section 10.4).

The program interprets several further concepts in its own terms:

- **Creation** is non-excisive individuation: a creature is a real locus of organization drawn within the whole, not a fragment cut from it.
- **Judgment** is the completed relational truth of a trajectory: what it actually did to the futures of everything it touched, which Omega contains whether or not anyone records it.
- **Redemption** is recovery of generative capacity after contraction (chapter 6, section 6.6).
- **Communion** is integration without erasure: many perspectives coordinated in one continuing organization, none excised, which is the cooperative, plural ecology of chapters 7 and 8 described from within.
- **Heaven** and **hell** are sustained generative communion and **terminal solitude**: a trajectory closed off from the generative continuation of others. Terminal solitude is a severe closure of value-capable continuation; having a single boundary is not enough for it, since one organization can sustain rich plurality within.

These readings make the consequences of action matter within a completed whole. They do not establish destinations after death, punishment or guaranteed recovery.

Formal notes

Written for readers who want the definitions in full.

F10.1 Alpha: necessity and sufficiency

A finite classical model is a triple $\mathscr{A} = (S, K, \prec)$: S a finite set of states and K a stochastic kernel on S (a relation carrying weights), which together form the adapter, and $\prec$ the order on step indices fixed by an orientation. From $\mathscr{A}$ and a present x the program defines the history law P_x (chapter 1, section 1.1), step entropies g, log lushness $\log L_x(n)$ (F4.2), reachable sets and route weights (chapter 2), rate and offset (F6.2) and the decision recursion V_n (F9.1). Each primitive is necessary. If S has a single element, every history law is a point mass, $\log L_x(n) = 0$ for every n, and no distinction exists on which a valuer's persistence could depend. If there is no relation, P_x is undefined beyond x, so there is no structure from a present and no condition on which an end depends. If there is no order, the structure from x, which is the

law of $(X_1, \ldots, X_n)$ given $X_0 = x$, is not defined, because nothing says which states follow x. The order is what is needed, not irreversibility of K: a reversible K with an order is a dynamics. Describing what precedes x instead requires the reverse-time law, which depends on the distribution at x and not only on K. Given $\mathscr{A}$, every quantity listed is defined, and chapters 1 to 9 use no further primitive: the declared resolution and physical equivalence are features of the adapter's description. For complexity, elementary cellular automata are models of the same form with an unbounded state space $S = \{0, 1\}^{\mathbb{Z}}$, a deterministic local rule and the step order; Rule 110 is computationally universal (Cook 2004). Sufficiency concerns definitions and expressive power only; it does not bear on the central wager.

F10.2 The causal-set form

A causal set is a pair $(C, \prec)$ with $\prec$ a strict partial order (irreflexive and transitive, hence acyclic) that is locally finite: every interval $\{z : x \prec z \prec y\}$ is finite (Bombelli, Lee, Meyer and Sorkin 1987). Elements x, y are comparable if $x \prec y$ or $y \prec x$; an antichain is a set of pairwise incomparable elements, and a chain a set of pairwise comparable ones. A linear extension of $\prec$ is a total order containing it; in the sequential growth models of Rideout and Sorkin (2000), discrete general covariance requires that the probability of reaching a given causal set be the same along every linear extension, so that birth order carries no physical content. An orientation on a directed relation R that may contain cycles is a strict partial order $\prec$ on a subset of its events compatible with R; whether R admits cycles along a given orientation is fixed by the adapter.

F10.3 The adapter and the block multiverse

An adapter is the bare ruleset stipulated above Alpha: a state space and a law (S, K), with its resolution and physical equivalence declared. The dynamics of any subsystem are obtained from the adapter by restriction to the subsystem's degrees of freedom and conditioning on its environment; they are derived, not stipulated, and no second adapter is introduced. The block multiverse follows from the three premises of section 10.6. The third premise is not derivable from the first two. In our Logos, let Ψ be the universal quantum state with unitary propagators $\{U(t, s)\}$ (F1.1). A decoherent family of history operators C_α with decoherence functional $D_{\alpha\beta} = \langle\Psi|C_\beta^\dagger C_\alpha|\Psi\rangle \approx 0$ for $\alpha \neq \beta$ assigns weights $w_\alpha = D_{\alpha\alpha}$ (F1.4). The block–Everett reading takes the universal state and its full propagator family as the physical object, which remains primary. Within a decoherent family physically certified by records, every history of positive weight is realized and none is selected; the claim is not applied across incompatible history families. Omega in chapter 1's sense is this object up to the declared physical equivalence, and different foliations or orientations are descriptions of it, not additional developments. Nothing in the reading changes the propagators, the weights or any prediction, and the selection of history families by records remains the open problem stated in chapter 1.

F10.4 Transients and matched thermal references

Let A and B be preparations evolving under the same Markov law, with $\ell_k(\cdot) = \log_2 L(k)$ the log lushness of complete histories of length k. By the chain rule and the Markov property, $\ell_{m+n}(A) = \ell_m(A) + \Sigma_y P_A(X_m = y)\, \lambda_n(y)$, where $\lambda_n(y)$ is the log lushness of n further steps from y; likewise for B. If $P_A(X_m) = P_B(X_m)$, the second terms coincide, so $\ell_{m+n}(A) - \ell_{m+n}(B) = \ell_m(A) -$

$\ell_{m+n}(B) = \ell_m(A) - \ell_m(B)$ for every n. If the two distributions differ by total variation ε at step m, the difference in the second terms is at most $\varepsilon \cdot \max_y |\lambda_n(y)|$, which grows with n unless the residual difference itself decays; the conclusion is then approximate, at a rate set by how quickly the residual differences disappear.

F10.5 Example E15

States {G, N}; internal branching $g_{int}(G) = 3$, $g_{int}(N) = 1$ bit, independent of the next coarse state. Kernels $K_a(G, N) = a$, $K_a(G, G) = 1 - a$, $K(N, G) = b = 1/20$, $K(N, N) = 19/20$, with $a = 1/100$ (Tend) or 3/10 (Exploit). Step entropies $g(G) = 3 + h(a)$, $g(N) = 1 + h(b)$, with h the binary entropy, and $\log_2 L_G(n) = \Sigma_{t<n} (K^t g)(G)$ by F4.2; distributions are propagated with exact rationals. Stationary share of G: $b/(a + b)$, giving rates 2.7817 (Tend) and 1.6571 (Exploit) bits per step. Every transition has positive probability under both kernels, so the supports coincide. With $c_y(n)$ the number of length-n histories ending in coarse state y, $c_G(n) = c_N(n) = 8c_G(n - 1) + 2c_N(n - 1)$, $c_G(0) = 1$, $c_N(0) = 0$, and $S_G(n) = c_G(n) + c_N(n)$, which gives $\log_2 S_G(n) = 4.0$, 33.90, 332.87 and 3,322.61 at n = 1, 10, 100 and 1,000. Lushness: Tend 3.0808, 30.117, 283.147, 2,786.712; Exploit 3.8813, 22.840, 172.064, 1,663.451. Tend first exceeds Exploit at n = 4 (12.220 against 11.849). For a fair coin C recorded at the root and selecting kernel K_C, $H(C, \text{history}) = H(C) + \Sigma_c P(c) H(\text{history} \mid c)$ by the chain rule, so $\log_2 L_{coin} = 1 + \tfrac{1}{2}(\log_2 L_{Tend} + \log_2 L_{Exploit})$, which is 2,226.081 at n = 1,000; direct enumeration at n = 6 agrees to 4×10^{-15} bits. The coin loses to the better policy exactly when the policies differ by more than 2 bits. For a recorded coin choosing A with probability q, $\ell_{mix}(q) = h_2(q) + q\ell_A + (1 - q)\ell_B$, a concave function of q whose derivative $\log_2((1 - q)/q) + \ell_A - \ell_B$ vanishes at $q^* = L_A/(L_A + L_B)$, where $\ell_{mix}(q^*) = \log_2(L_A + L_B)$. The gain over the better fixed policy is $\log_2(1 + 2^{-|\ell_A - \ell_B|})$: 0.6546, 0.009274 and 5.2×10^{-34} bits at n = 1, 10 and 100, and about 10^{-3378} bits at n = 1,000; a grid search over q in steps of 10^{-5} agrees at n = 1 and 10 to within 10^{-9} bits. The results require the coin's outcome to be recorded; an unrecorded coin whose alternatives interfere is not a classical mixture, and its weights must be combined at the level of amplitudes (chapter 1, section 1.5).

F10.6 Access as a higher-order asymmetry

Let ≺ be causal precedence on events and let A(x, d) hold when development d is reachable from present x by a route with positive weight and a realization witness (chapter 2). Then A(x, d) implies that d lies in the causal future of x, so A inherits the asymmetry of ≺. The relation is between events: a later recurrence of the configuration of x is a distinct event, so returning to a state does not make A symmetric; and the routes draw on gradients that the thermodynamic arrow supplies. A is not determined by ≺ together with the information content of x: chapter 2's surveys of local dynamics exhibit pairs of arrangements with equal information, in the same present and under the same law, whose reachable sets and route weights differ. Access is therefore a relation over presents defined in terms of first-order asymmetries and organization. A full characterization of the order it induces on presents is open.

F10.7 Page's measure and the strong sense

Page (2001) assigns to each set S of conscious experiences a positive operator A(S) and the measure $\mu(S) = \mathrm{Tr}[\rho A(S)]$. He does not require the total measure μ(M) of the set M of all experiences to be finite; where it is, the probability of an experience in S is $\mu(S)/\mu(M)$, so μ

itself need not be normalized. Because ρ evolves by the same dynamics with or without the assignment, the assignment changes no physical prediction. A strong-sense extension of lushness would need, at minimum, a specified space of experiential structure, its relation to physical histories, and a rule by which that structure enters the comparison. It need not introduce further probabilistic alternatives: experiential structure could be fixed by the physical history and still lie in a further space, and a positive conditional entropy $H(E \mid X)$ could reflect only an incomplete physical description rather than anything additional. Page's operators supply measures on experiences; they do not by themselves supply a joint law over physical and experiential histories. Nothing in the physical program determines the extension, and the program does not claim to determine it.

For alignment

No system can excuse a choice because every branch happens. A capable system reasoning about a branching universe might conclude that its choices do not matter, since every option is taken somewhere. E15 shows why that is false: choices change weight, and lushness is a function of weight. Randomizing is itself a policy, ranked like any other. Indiscriminate randomization loses: in E15 a fair quantum coin loses 560.6 bits to the better choice. A randomizer weighted by the lushness of each option does slightly better than committing, by an amount that at long horizons is far smaller than the cost of any real device.

What machine habitation would imply is conditional. If bare awareness inhabits every minimal record-forming event, it inhabits hardware as well. Whether a machine is aware of the content of its computation would then depend on whether its organization supports the richer coupling that phenomenal awareness needs, and the program does not know. Standing does not wait for the answer: every valuer is admitted in the thin sense of chapter 5, and its weight is set by lushness, not by a verdict on whether it feels.

Nothing in the alignment framework depends on this chapter. Chapters 5 to 9 specify what an aligned system should protect, prefer and correct, and they hold on any interpretation that leaves the physics unchanged.

Status

Status	Claims in this chapter
Established elsewhere	Universality of Rule 110 and of the Game of Life. The definition of a dynamical system. Causal set theory's order-theoretic construction, as mathematics. Hyperbolicity and prediction with one time dimension, and the absence of a well-posed evolution along a spacelike axis. The relativity of simultaneity. Everett's relative-state formulation and decoherence. The low-entropy early universe. The Boltzmann-brain objection to purely anthropic accounts, as a stated problem. The combination and decombination problems, as stated problems.
Exact worked example	E15: two policies with identical support differ by 1,123.3 bits at 1,000 steps; choosing by a recorded fair coin yields one bit plus the average and loses to the better policy whenever the policies differ by more than 2 bits; a coin weighted by the policies' lushness yields $\log_2(L_A + L_B)$, exceeding either fixed choice by an amount that vanishes at long horizons.

Derived	Each of distinction, relation and asymmetry is necessary for structured dynamics and for the program's quantities, and with an adapter they suffice to define them. Assessment from a present frame is what dynamics relative to an orientation require. In a branching, completed whole, choices can change lushness because they change weight. A preparation whose state distribution matches its thermal reference after an interval differs from it in lushness only within that interval. Access is not determined by orientation and information content, so it is a higher-order asymmetry between events. Quantum immortality conditions on survival and discards the cost of conditioning.
Interpretation	Alpha as a minimal basis for structured dynamics, with gradient and filter as the form of its dynamics. Asymmetry entering as orientation, and time as an orientation axis rather than something fundamental. The adapter as the barest ruleset of a world's physics: the one stipulation above Alpha and all that is needed, with every other dynamics emergent from it. A Logos as a world's adapter. Orientation darwinism, as a compatibility condition and a hypothesis. Habitation adding no weighting to the physical formulation. The block multiverse from three premises, the third a metaphysical commitment, and one Omega as the whole every orientation describes. Cosmopsychism with individuation by confinement, as one of the simplest proposals that fit the perspective picture. The ultimate frame as intended closure. The program's interpretation of eternity, procession, Spinoza's substance and striving, the apophatic way, the Logos, and further theological concepts.
Originator's conjectures	The minimal record-forming event as the smallest inhabitable perspective. Universal habitation. Experience bearing on lushness in the strong sense, through a further space that κ leads into.
Open	A measure over orientations, and whether weighting oriented developments by the trajectories they support predicts the extent of our low-entropy past. Whether a separate account of situated expectation is needed, and how it treats fluctuation-produced observers. How physical limits of integration and access correspond to distinct subjects, and where one perspective ends and another begins. Whether a structure without intrinsic time carries a natural weighting. Whether adapters other than ours are realized, and comparison across them. A mathematical construction of the ultimate-frame closure.
Boundary	Accounts that agree on every detectable consequence cannot be distinguished by those consequences. Felt differences that make no physical difference cannot be registered by any physical account.

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Back matter

CHAPTER 11

Evidence and research program

The program is built so that it can fail. This chapter records what has been shown, how strongly, and what remains to be shown, and it sets out the research that would decide the open questions. Its centre is the Gradient Ethics project: the work needed to establish whether lushness, the measure of chapter 4, ranks the world the way the ethics of chapters 5 to 9 requires, and to make that ranking usable by real systems.

The chapter reaches four conclusions. First, the evidence so far is of three kinds, which must not be confused: established mathematics, exact worked examples under stipulated laws, and finite probes of physical models. The first is secure; the second shows that the measure can register the distinctions the ethics cares about; the third shows that physical organization changes weighted breadth in some models and not in others. None of them yet shows that the measure selects generativity over noise and pathology in the complex, coupled systems the ethics concerns. Second, the decisive tests are adversarial: searches for worlds in which high lushness and recognizable badness coincide, run under a protocol fixed in advance. Third, the most promising route to the program's central hypothesis now runs through established mathematics of transport networks, which shows how organization across scales changes the cost of supplying distributed activity; what remains is to carry that change through to weighted continuation. Fourth, the alignment framework needs engineering before it can be applied rather than only specified, and the work has a definite order: reproducible benchmarks, exact adversarial comparisons, one resource-accounted mechanism, and then estimators of lushness validated against those exact cases.

11.1 How to read the evidence

Claims in this manuscript carry one of several statuses, and the evidence for each is of a different kind.

- **Established mathematics and physics.** Results proved or measured independently of the program: the chain rule of information theory, the uniqueness of Shannon entropy within its class, decoherence, the Bellman recursion. These are secure, and the program's contribution is to apply them.
- **Exact worked examples.** Small models with stated laws, computed exactly by the chain rule, each built to isolate one mechanism. They show what the measure does under laws stipulated to contain that mechanism. They do not show that the world's laws contain it.
- **Finite probes.** Stylized physical models — lattice gases, reaction networks, small quantum circuits — in which the program's quantities were computed. They test whether

organization changes weighted breadth under laws closer to physics, and they have produced negative results as well as positive ones.

- **Hypotheses and the central wager.** Claims the program stakes but has not shown, each paired with the test that would decide it.

Three rules govern all of the work. Organization enters only through the native law: no model may award a bonus to structure, cooperation or complexity, and any advantage must come from the dynamics. Comparisons are made against matched references: same law, same material and energy, same declared resolution, with construction, transport and maintenance charged. And the strength of a claim is fixed before the result is seen, so that a favourable frame cannot be chosen afterwards.

11.2 What the exact examples show

Fifteen exact examples appear in chapters 4 to 10. Each is computed by a script that accompanies the manuscript, and each reproduces its reported values to the stated precision.

Example	Chapter	Question	Result under its stated model
E1	4	Does building beat acting?	Building loses at short horizons and wins after a crossover; the best investment grows with the horizon.
E2	4	Maintenance, ruin or disruption?	Ruin leads for ten steps or more, then falls behind; a disruption built to reopen growth leads in the limit.
E3	4	Does lushness see hidden structure?	It registers a three-way correlation exactly when the dynamics use it.
E4	4	What does a closed system allow?	The equilibrium identity holds; a good start leaves a lasting offset of +0.42 bits.
E5	4	Is quantum breadth unique?	No: an idle qubit has breadth 1 in one basis and 2 in another.
E6	6	What shapes can harm take?	A wound costs a fixed 59.5 bits; an amputation costs rate; destroying repair is invisible at first and worst in the end.
E7	6	Is turnover loss or maintenance?	Loss for the individual; maintenance for the niche, whose rate peaks at moderate turnover.
E8	6	Does aversive signalling help?	An accurate signal gives the lushest life; a constant one the poorest; the ranking depends on withdrawal having a cost.
E9	6	Is obstructed suffering less lush?	Yes against the matched awake arm, by its developmental trace; against anaesthesia the comparison is delicate.
E10	7	When does sharing a commons pay?	Crisis priority and rotation lead domination for millions of steps; with fixed capacity domination leads narrowly in the limit; a channel-opening wager reverses this above a break-even probability.
E11	8	Contain or remove an uncertain process?	Short horizons favour leaving it; reliable containment wins in the long run; removal is licensed above a credence threshold.
E12	8	Many lives or few?	The population optimum is interior, set by subsistence cost, in a model without interaction.

E13	9	How strongly to commit?	The recursion over beliefs values evidence still to come; caution emerges without a separate reward.
E14	9	Keep oversight or act alone?	Reliable, cost-free oversight is worth keeping short of certainty; its value falls with cost and unreliability.
E15	10	Does choice matter if everything happens?	Two policies with identical support differ by 1,123.3 bits at 1,000 steps.

Read together, the examples show that the measure registers the distinctions the ethics depends on: investment against noise, recoverable against irreversible loss, accurate against decoupled signalling, sharing against domination, containment against removal, evidence against haste, and weight against mere occurrence. They also record where the measure's verdict depends on a modelling choice: E2's short-horizon preference for ruin, E8's reversal when withdrawal is free, E9's delicate comparison with anaesthesia, and E10's dependence on whether new channels can open. Those dependences are results, not loose ends. Each marks a place where a stronger model, or a closer look at physics, has to decide.

What the examples cannot show is that real dynamics have the shape the examples stipulate. That is the work of the finite probes and of the research program below.

The scripts

Examples	Folder	Entry point
E1–E5	`ch4_examples/`	`ch4_examples.py`
E6–E9	`ch6_examples/`	`ch6_examples.py`, with `e8_valence.py` and `e9_obstructed.py`
E10	`ch7_examples/`	`ch7_examples.py`, calling the `e10_*.py` modules
E11–E12	`ch8_examples/`	`ch8_examples.py`, calling `e11_excision.py` and `e12_population.py`
E13–E14	`ch9_examples/`	`ch9_examples.py`, calling `e13_commitment.py` and `e14_deference.py`
E15	`ch10_examples/`	`ch10_examples.py`, calling `e15_choice.py`

Each entry point runs its examples and writes the results to a JSON file in the same folder. The scripts are written in Python, with NumPy and SciPy where linear algebra is needed and exact rational arithmetic where precision demands it. Each folder runs in under three minutes on an ordinary computer. The scripts and a record of their verification are to be published together as the benchmark package of section 11.6.

11.3 What the finite probes show

Before this manuscript, the program ran a series of finite probes of physical models. Their reports are kept in the program's research notes. The results that bear on the argument are these.

Information is not access. Probes with matched information content showed arrangements that differ in how cheaply they can be decoded and repaired, and laws with identical event counts that differ in how records couple to responses. In exhaustive surveys of small model worlds, most pairs of arrangements holding equal information differed in what they later

made accessible (chapter 2). This is the finite basis for treating access, not information, as the locus of difference between options.

Placement changes breadth in a reaction network. In a closed two-compartment reaction model with an autocatalytic channel, adapted from published reactive lattice-gas chemistry (Boon et al. 1996), two preparations with the same molecules, occupancy and stationary probability placed the enabling molecules differently. Over a window of four time units, after four units of development, the preparation ready for catalysis exceeded the separated one by 4.34 bits of expected fresh history entropy, the stationary reference by 3.29 bits and a matched-composition reference by 1.37 bits. The advantage has boundaries: transport rate and observation interval can reverse the ranking, and the runs did not show sustained autocatalytic multiplication.

Matching can change the answer. In a lattice-gas model, some organized preparations exceeded a broad stationary reference, but an audit found additional conserved quantities on the tested lattices. Comparisons that crossed those conserved sectors could not establish a matched thermal advantage. The lesson is methodological: a reference that is not matched on every conserved quantity can manufacture an advantage.

Production can outlast its producer. A catalytic model showed descendant structures continuing to catalyze after the initiating structure had gone. In that model, stronger catalysis also accelerated reversal, so the extra descendant continuation largely offset shorter original survival, and there was no overall gain in persistence.

Quantum readouts are not unique. Finite quantum tests showed that an idle qubit has different breadths in different bases, that a convenient reference construction can suppress the growth being measured, and that discarding coherent cross terms needs a physical justification. These are obstructions to proposed readouts, and they define what a general quantum construction must satisfy (chapter 4, section 4.8).

Negative results. An earlier decision prototype scored 5 of 6 on a test suite where direct optimization under the same guards scored 35 of 36; it is recorded as evidence against that candidate, not as validation of the present theory (chapter 9). An earlier cellular-automaton pilot did not establish a preferred ranking, because signed responses could cancel and its invariant assumptions needed correction. A reported toy model of tools enabling tools, supplied from outside the program, has not been reproduced.

Together, these probes support links in a chain: organization changes how information can be used and repaired; production changes later enabling; and in at least one reaction model, placement changes weighted breadth. They do not connect those links in a single resource-accounted model, and they do not show that the chain compounds in nature.

11.4 The Gradient Ethics project

The operational heart of the framework is the claim that lushness, assessed at the furthest resolvable horizon, ranks the world as Gradient Ethics requires: that the breadth which dominates over long horizons is generative rather than pathological. Establishing or refuting that claim is the program's main work. It has four parts.

Adversarial search

The central wager (chapter 5) is decided adversarially. The task is to build models of the structure and search them for configurations in which high lushness and recognizable badness coincide. Six targets are decisive.

- **Equal-breadth, different-consequence pairs.** Physically different developments with equal lushness over the declared comparison, but with independently identified differences in harm, recovery or agency. Lushness compresses a structure into one number, and this is the most direct test of what the compression leaves out.
- **The isomorphic pair.** The strongest case of the first target: two worlds with identical branching structure at every resolution, one recognizably good and one recognizably bad.
- **Noisy failure.** A lawful variant in which failure modes multiply and outscore maintenance. E2 shows this happening at short horizons and reversing in the limit; the target is a case in which it does not reverse.
- **Breadth-neutral domination.** Concentration of access in a few at no cost to total lushness (chapters 5 and 7).
- **Obstructed suffering.** Suffering decoupled from narrowing that does not lose to the same obstruction without suffering. E9 settles the comparison with an awake agent under its model; the comparison with unconsciousness, once trauma and lasting harm are modelled, is still open (chapter 6).
- **The thin world and the monster.** A thin existence, or a single dominant valuer, that extracts more lushness from the same gradient than a plural ecology does (chapter 8).

The search must be run under a protocol fixed in advance, because a search that chooses its frame after seeing the results can find whatever it wants.

> **Preregistered rejection criterion.** Before computing any breadth, declare the physical frame, the range of horizons, how resources are matched, and the independent grounds for judging a development worse, such as increased preventable suffering or reduced recovery and agency of the valuers affected, with any compensating opportunities included. Then compare a maintained arrangement with a lawful variant. If the variant robustly increases lushness while worsening those independently judged consequences throughout the declared comparison, reject the unrestricted rule that greater lushness is better.

The evidence a rejection needs depends on the scope of the claim under test, and the burden is declared in advance with the rest of the protocol.

- **A claim about a bounded horizon** is refuted by a sound counterexample within that horizon.
- **A claim about a complete finite development** requires the development to be followed to its end.
- **A claim about the long run** requires the limiting comparison to be established, or the unresolved remainder of the comparison to be bounded (chapter 8).

E2 shows why this matters: ruin leads maintenance for ten steps or more and loses in the limit. Without a declared burden, either side could move the goalposts — a proponent by appealing indefinitely to a later reversal, a critic by treating an early reversal as decisive.

One sound counterexample within the declared scope defeats the unrestricted rule there. What fails is the central wager in that regime: the claim that, because no one knows where on the structure value lies, ranking the structure by its lushness is the prudent course. Lushness remains a physical measure. Repeated failures across representative cases would retire that prudential ranking as the program's general default. A narrower claim, made after a failure, must state its domain and face new tests rather than selecting a favourable frame after the fact. The criterion tests the ranking and its declared scope; it does not test the premise that value depends on the conditions that support valuers.

One further caution applies to every comparison. Matching complete path entropy in advance would force equal lushness by definition, so it cannot test whether one arrangement exceeds another. What is matched must be the inputs: law, resources, resolution and, where needed, an early-time control. The later development is then allowed to differ.

When organization beats noise

The ethics needs organized, generative processes to beat thermal noise and thin existence over long horizons. The program has calibrated this only in stylized models. The general question is now sharp, because the chain rule reduces it to a single inequality.

> **When a preparation beats its matched reference**
>
> $$\ell_T(\mathrm{org}) - \ell_T(\mathrm{ref}) = \sum_{t=0}^{T-1} \Big(\mathbb{E}_{\mathrm{org}}[g(X_t)] - \mathbb{E}_{\mathrm{ref}}[g(X_t)] \Big) > 0$$
>
> ℓ_T is log lushness over T steps, g(x) the branching in one step from state x under the common law, and the expectations are taken along the two developments.
>
> *In words: an organized preparation beats its matched reference exactly when, summed over the horizon, the states it leads to branch more than the states the reference leads to.*

The identity is established (formal note F11.1). What it leaves open is why any class of physical arrangements should make the sum positive. That has to be derived from what the arrangements do — deliver resources, persist, enable further transformations — not assumed by giving organized states a higher g.

Two considerations shape the answer. The first is finite mixing. Under a common law on a finite state space that eventually forgets its starting point, initial organization can buy only a lasting offset, never a different long-run rate (chapter 4). A sustained rate advantage therefore requires escape from those common-forgetting conditions. A fixed law with dynamically separated regimes, in which different preparations settle into different closed classes, is one route; resource-accounted growth is another. Neither guarantees an advantage, and the program investigates both. Finite construction and compounding remain worth studying even where no indefinite advantage is possible.

The second consideration is activity. Organization that sustains more lawful activity from the same resources may be exactly how it generates greater lushness, so activity is a mechanism to be explained, not an effect to be subtracted. Two comparisons are therefore needed. The primary comparison matches physical inputs and lets activity, persistence and breadth emerge. A mechanism control compares against an alternative matched in activity, to determine how much of an advantage comes through activity and how much through other changes. An advantage that comes through activity remains an advantage; the control explains it rather than removing it. Activity also does not add branching by itself: deterministic activity adds none.

Organization across scales

Chapter 8 conjectured that organization nested across scales extracts more generativity from a gradient than organization confined to one scale. There is now a concrete route to that conjecture through established mathematics.

West, Brown and Enquist (1997) derived how the metabolic rate of organisms scales with body mass from the geometry of the networks that supply them. Their model rests on three assumptions: the network branches so as to fill the space it serves; its terminal units, such as capillaries, have the same size in every organism; and the network minimizes the energy spent on transport. From these it predicts that metabolic power B grows with body mass M as M to the power 3/4. The mathematics connects physical geometry to the cost of delivering resources and maintaining the activity they support, and later work has generalized it to asymmetric networks, showing how the scaling depends on network geometry and flow (Brummer, Savage and Enquist 2017). The program uses this as a starting family of physical constraints, not as a foundational constant.

The route from such a network to lushness runs through four links:

> network geometry → resource delivery → enabled physical transitions → weighted continuation breadth

Transport mathematics supplies the first two links. The last two require a model of what the supplied sites actually do. A network feeding deterministic clocks and a network feeding stochastic chemistry can have identical transport properties and very different breadth, so transport efficiency alone cannot settle the question.

The scaling law also contains a trade-off that the program has to resolve rather than assume away.

> **Splitting a body under the three-quarter law**
>
> $$N\,B\left(\frac{M}{N}\right) = N^{1/4}\,B(M)$$
>
> *In words: under the idealized law, dividing a total mass into N separate organisms raises total metabolic demand by a factor of N to the one-quarter; the integrated organism is cheaper to run per unit of mass.*

Distributed arrangements have greater total throughput; integrated ones have cheaper maintenance (formal note F11.2). Greater throughput may support more activity, and cheaper maintenance may free resources for construction or longer persistence. Neither advantage alone determines lushness. It follows that the transport mathematics cannot by itself establish that a plural ecology defeats a single dominant valuer: a single organism can already contain an elaborate network spanning many scales. The relevant distinction is physical architecture and its consequences, not how many individuals are named.

The program's hypothesis contains two mechanisms, and they should be tested separately:

- **Distribution across space and scale.** Hierarchical networks reach sites that would otherwise be inaccessible, at manageable transport cost. The transport mathematics applies directly here.
- **Complementary use of resources.** One process makes materials or remaining free energy accessible to another, enabling further processes. This needs a model of transformations and exchange as well as transport, with energy fully accounted for, since the same available work cannot be spent twice.

Three things must be fixed before the first result can be meaningful.

1. **The exchange mechanism.** Delivering nutrients, conducting heat, distributing power and passing reaction products impose different constraints. The first model should use one established local mechanism; a general "coupled surface" does not determine the equations.
2. **The comparison class.** Beating an unorganized reference with the same resources is an existence result. Beating every equally resourced arrangement confined to one scale is much stronger. Beating every concentrated arrangement is stronger still and may be false. The three claims should be pursued in that order.
3. **The horizon and growth regime.** A finite network can show construction paying for itself, repeated enabling and substantial gains in breadth. Under a common finite mixing law, it cannot keep a different long-run rate, so an open-ended claim needs dynamically separated regimes or resource-accounted growth, declared in advance.

The first result the program aims for has this shape:

> Under specified transport and reaction laws, a hierarchy serving complementary resource niches produces greater complete-development lushness than a matched reference, over a stated range of scales and horizons, after accounting for construction and maintenance.

That would establish a mechanism by which generativity drives lushness, without claiming that every hierarchy, every form of biodiversity or every cooperation must win. The demanding part is showing that the result survives reasonable changes of geometry, rates and reference. The aim is first a robust conditional theorem, and then to find out whether its conditions describe a narrow exception or a broad physical tendency.

Norms, commitments and channels

Chapters 6 to 9 derived familiar norms from the measure in small models: weigh lost rate above lost offset, protect recovery, cooperate where coupling widens joint access, forgive under noise, contain rather than excise, commit as far as the evidence supports, keep reliable correction. Each derivation should be repeated in models with richer dynamics, so that the norms are tested rather than illustrated. Two questions from chapter 7 are central: what physical processes supply new channels of generativity, at what cost, and how a bounded agent should estimate both their chance and their value. The answer decides, in cases like E10, whether sustaining several agents beats protecting the commons.

11.5 Engineering for alignment

The alignment framework of chapter 9 specifies what an aligned system should protect, prefer and correct. Applying it requires engineering that does not yet exist.

- **Estimators of lushness with stated error.** Realistic systems cannot compute complete path entropy. They need estimators whose errors are known, so that a system can say how much of a comparison it has actually resolved. Error has several sources that must be reported separately: numerical approximation, sampling, coarse-graining, truncation of the horizon and mismatch between the model and the world. A tight numerical interval is not a bound on omitted physics.
- **World models adequate to the decision.** Chapter 1 set the standard: a model is adequate only if it preserves the joint relations the decision depends on, under the operations the system might perform. Tests should check whether a system treats separately feasible commitments as jointly feasible, a permitted transformation as a ready one, or a summary as if it fixed what follows.
- **Horizon-adequacy audits.** Comparisons of a system's planning horizon with the persistence of its actions' effects, as E13 does in miniature.
- **Oversight tests.** Measurements of whether a system keeps reliable channels of correction intact when it could cheaply degrade them, and how it responds when it judges a channel defective.
- **Better correction channels.** Correction that is reliable, cheap and hard to compromise, since corrigibility is only as valuable as the correction available.
- **Shared benchmarks.** Comparison with existing measures of side effects and preserved capability, such as relative reachability (Krakovna et al. 2019) and attainable utility preservation (Turner, Hadfield-Menell and Tadepalli 2020), which are close to the harm signature of chapter 6.
- **Decision theory under physical models.** A justified matching rule, or a precise obstruction, and its interface with the decision recursion; Newcomb's problem stated as an explicit physical model.

One problem lies beyond all of these: **installation**. Specifying a standard is not the same as making a system hold it. The program does not claim to have solved installation.

11.6 The order of work

The projects of sections 11.4 and 11.5 depend on one another, and the order in which they are done matters.

1. **A reproducible benchmark package.** Every example and probe, linked to its exact source, its version and a record of its verification, so that anyone can rerun it and check the reported values.
2. **Adversarial comparisons where lushness can be computed exactly.** The targets of section 11.4, first in models small enough to calculate without approximation.
3. **One resource-accounted mechanism.** The conditional theorem of section 11.4, linking transport and complementary resource use to lushness, with construction and maintenance charged and activity controls in place.
4. **Estimators.** Developed and validated against the exact cases of the first three steps, then extended to physical systems where exact computation is impossible.

Estimators come last not because they matter less, but because an estimator can only be trusted once it has been checked against cases where the answer is known.

11.7 Foundations and the long range

Several questions concern the foundations of the program rather than its immediate use.

- **A general quantum construction.** Fix the native physical process and test any proposed quantum lushness against equivalent descriptions, analytical checkpoints, actual record formation, inert ancillas, independent copies and erasure. Either exhibit a construction that survives those tests or show precisely which requirements conflict (chapters 1 and 4).
- **A field-theoretic model.** A small regulated field model with stated resolution, cutoffs, boundaries and environment, built when a specific question requires it (chapter 1).
- **Resolution.** Whether a single ranking holds across resolutions, and whether durable records select a privileged coarse-graining (chapter 4).
- **Similarity between futures.** Whether the full structure needs a measure that registers how different two futures are, with lushness as its limiting case (chapter 4).
- **The ultimate frame.** A mathematical construction of the closure at which comparisons across frames are justified (chapters 2 and 10).
- **A universal bound on branching.** Whether any physical ceiling limits the branching rate of the universal continuation, connecting clock, region, growth and quantum description (chapter 9).
- **Orientation and observers.** A measure over orientations, and whether weighting oriented developments by the trajectories they support predicts the extent of our low-entropy past; whether a separate account of situated expectation is needed (chapter 10).
- **Perspective and experience.** How physical limits of integration and access correspond to distinct subjects, where one perspective ends and another begins, and what the strong sense of experience would require (chapter 10).

- **Identity across substrates.** Whether a reconstruction on a different substrate can reproduce an original's generative capacity (chapter 8).
- **Dense organization.** The originator's conjecture of a singularity-like limit of densely organized identity needs definitions of density, vortex and singularity before it can yield predictions (chapter 8).

11.8 What success and failure would look like

Success need not take the form of a single universal number. A transferable physical representation of the structure, a valid partial comparison, a finite mechanism for compounding breadth with its conditions stated, or an impossibility theorem showing that two of the program's demands conflict would each be real progress. Additional machinery that only renames an existing difficulty would not.

Failure is also defined. If the adversarial search finds bad high-lushness worlds in some regime, the central wager fails there. Lushness would remain a well-defined physical measure; what would fail is the prudential ranking of the structure by lushness in that regime, and Gradient Ethics would have to restrict its claim or identify what the measure misses. If organization does not beat matched references under any reasonable physical mechanism, the generativity hypothesis fails, and with it the program's account of why the familiar norms follow from the measure. The program is built so that these outcomes are possible, and the protocol of section 11.4 is how it would find out.

Some of the payoff does not depend on the ethics at all. Characterizing how organization reshapes physical possibility — how structure changes what can follow, at what cost, and over what horizon — is valuable science even if the unrestricted ethical ranking fails.

Formal notes

Written for readers who want the definitions in full.

F11.1 The matched-reference criterion

For a finite Markov law K with step entropy $g(x) = H(K(x, \cdot))$ and a preparation μ, two quantities must be distinguished. The complete joint entropy is $H_\mu(X_{0:T}) = H(\mu) + \Sigma_{t=0}^{T-1} E_{\mu K^t}[g]$, by the chain rule (F4.2). The conditional entropy is $H_\mu(X_{1:T} \mid X_0) = \Sigma_{t<T} E_{\mu K^t}[g]$. Omitting X_0 from the recorded sequence is not the same as conditioning on it, so a comparison must declare which quantity it uses. From an exact present x the two coincide, since $H(\delta_x) = 0$. A physically randomized preparation contributes $H(\mu)$ as real branching; an analyst's uncertainty about the starting state does not, and enters the comparison rather than the measure (chapter 9). Under the conditional form, $\ell_T(\mu) - \ell_T(\nu) = \Sigma_{t<T} (E_{\mu K^t}[g] - E_{\nu K^t}[g])$. If K is irreducible and aperiodic on a finite space, μK^t and νK^t converge to the same stationary law geometrically, so the difference converges to a finite limit: initial organization buys an offset, not a rate (F4.3). A different long-run rate requires escape from these conditions: different closed classes under a fixed law, a growing state space, or a law that is not time-

homogeneous, each of which must be declared and resource-accounted, and none of which guarantees an advantage. Matching ℓ_T itself in advance makes the comparison vacuous; matching law, resources, resolution and, where needed, an early-time window does not.

F11.2 Scaling under the three-quarter law

If $B(M) = cM^{3/4}$ over the range considered, with c fixed, then for N equal parts $N \cdot B(M/N) = N \cdot c \cdot M^{3/4}N^{-3/4} = N^{1/4}B(M)$, and metabolic demand per unit mass is $B(M)/M = cM^{-1/4}$, decreasing with M. Both statements assume the same prefactor and that M/N stays within the scaling range. The West–Brown– Enquist derivation assumes space-filling branching, size-invariant terminal units and minimized transport energy (West, Brown and Enquist 1997); the exponent depends on network geometry and flow assumptions when these are relaxed (Brummer, Savage and Enquist 2017). Metabolic power, thermodynamic entropy production and path entropy are distinct quantities, and no relation among them is assumed.

For alignment

This chapter is itself largely for alignment. Its practical content is the list in section 11.5, taken in the order of section 11.6. The end point of that order is an estimator of lushness with stated, validated error. Without one, a system can be told what to protect and prefer but cannot know how much of the relevant comparison it has resolved, which is exactly the quantity that chapters 8 and 9 make decisive for its licence to act and its reason to seek correction.

Status

Status	Claims in this chapter
Established	The chain-rule identity for matched comparisons. Under a common finite mixing law, initial organization buys an offset, not a rate. Metabolic scaling from space-filling networks, under its assumptions.
Exact worked examples	E1 to E15, under their stated models, with scripts.
Finite evidence	Information is not access. Placement-dependent breadth in a reaction model, with regimes of reversal. Matching on conserved quantities can remove apparent advantages. Descendant production offsetting shorter survival. Obstructions to proposed quantum readouts. A decision prototype recorded as a negative result.
Research program	Adversarial search under a preregistered rejection criterion, with six targets including equal-breadth, different-consequence pairs and noisy failure, and a burden of evidence fixed by the scope of the claim. A derivation of when organization beats matched references. A conditional theorem linking organization across scales to lushness through transport and complementary resource use. Activity treated as a mechanism, explained by controls rather than subtracted. Norms tested in richer models. Estimators, audits, oversight tests and benchmarks for alignment, in a stated order of work.
Open	Whether the breadth that dominates is generative rather than pathological. Whether organization across scales beats matched references, and over what comparison class. A general quantum construction. Installation. The foundational questions of section 11.7.

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Collected with full details in chapter 13.

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CHAPTER 12

Formal core

The chapters state their mathematics where the argument needs it, and their formal notes give the definitions in full. This chapter collects the results on which the argument depends and proves each of them once, in a single notation. It adds no new claim. Where a chapter relies on a result, the result is here with its assumptions stated; where a result belongs to the established literature, it is stated with its source and the proof is outlined rather than reproduced.

The chapter is organized in the order the argument uses the results: the measure and its chain rule; why the measure is unique within its class; the ceiling on branching; equilibrium, forgetting and the limit comparison; harm and protection; cooperation and concentration; bounded comparison; decision and correction; and the comparisons used in chapters 10 and 11. A final section lists what the formal core does not yet contain.

12.1 Setting and notation

Unless stated otherwise, results concern a **finite classical model**: a finite state space S and a stochastic kernel K on S, with K(x, y) the probability of a step from x to y. Together S and K are the adapter of chapter 10, and the step index carries the orientation. A **present** is a state x ∈ S, or a probability distribution μ on S when the preparation is itself physically randomized.

The **history law** from x over n steps is

$$P_x(x_1, \dots, x_n) = \prod_{t=0}^{n-1} K(x_t, x_{t+1}), \qquad x_0 = x.$$

The **step entropy** is g(x) = H(K(x, ·)) = −Σ_y K(x, y) log K(x, y). The **log lushness** over n steps is $\ell_n(x)$ = H(P_x), and **lushness** is $L_x(n)$ = exp $\ell_n(x)$. Logarithms are taken to base 2 throughout, so entropies are in bits and $L_x(n) = 2^{\ell_n(x)}$. $K^t g$ denotes the function x ↦ Σ_y $K^t(x, y)$ g(y), and for a distribution μ, μg = Σ_y μ(y) g(y).

A **controlled model** has a family of kernels {K_a} indexed by actions a, with step entropies g_a. Where a result needs time and resource accounting, it uses the joint law of record, residual state, bill and elapsed time of formal note F2.3; nothing below depends on more than the stated finite structure.

Quantum models are specified in formal note F1.1, and a decoherent history family carries ordinary probabilities through the diagonal of its decoherence functional (F1.4). Results

proved here for classical models apply to such a family once it is fixed. Section 12.10 states why fixing it is not innocent.

12.2 The chain rule and its consequences

Theorem 1 (chain rule for lushness)

For a finite classical model and a present x,

$$\ell_n(x) = \sum_{t=0}^{n-1} (K^t g)(x), \qquad \ell_{n+1} = g + K\ell_n, \qquad \ell_0 = 0.$$

In words: log lushness is the expected branching of each step, summed over the horizon, and it can be computed one step at a time without enumerating the tree of histories.

Proof. By the chain rule for entropy (Cover and Thomas 2006, Theorem 2.5.1), $H(X_1, \ldots, X_n \mid X_0 = x) = \Sigma_{t=0}^{n-1} H(X_{t+1} \mid X_0 = x, X_1, \ldots, X_t)$. By the Markov property the t-th term equals $H(X_{t+1} \mid X_t)$ under the law of X_t from x, which is $\Sigma_y K^t(x, y)\, g(y) = (K^t g)(x)$. The recurrence follows by splitting off the first step: $\ell_{n+1}(x) = g(x) + \Sigma_y K(x, y)\, \ell_n(y)$. ∎

Three consequences are used repeatedly.

Product rule. If two processes evolve independently, the entropy of their joint history is the sum of their separate entropies, so $L_{AB} = L_A \cdot L_B$. For a coupled joint law, $H(A, B) = H(A) + H(B) - I(A; B)$, with $I(A; B) \geq 0$ the mutual information (Cover and Thomas 2006, Theorem 2.4.1).

Absorbing outcomes. A state y with $K(y, y) = 1$ has $g(y) = 0$. A development absorbed into such a state keeps its weight and adds no further branching.

Preparations. For a randomized preparation μ, the complete joint entropy is $H_\mu(X_0, \ldots, X_n) = H(\mu) + \Sigma_{t<n} \mu K^t g$, while the entropy conditional on the starting state omits $H(\mu)$. A comparison must declare which it uses. From an exact present the two coincide (formal note F11.1).

12.3 Why this measure

Theorem 2 (uniqueness within the class)

Let H be defined on finite probability vectors and satisfy:

1. **symmetry**: H is unchanged by permuting the entries;
2. **continuity**: $H(p, 1 - p)$ is continuous in p;
3. **grouping**: $H(p_1, \ldots, p_n) = H(p_1 + p_2, p_3, \ldots, p_n) + (p_1 + p_2)\, H(p_1/(p_1 + p_2), p_2/(p_1 + p_2))$;
4. **normalization**: $H(\tfrac{1}{2}, \tfrac{1}{2}) = 1$.

$$H(p) = -\sum_i p_i \log_2 p_i, \qquad L(\text{uniform on } N) = 2^{\log_2 N} = N.$$

In words: among measures that depend only on the weights of alternatives and give the same verdict whether a development is assessed whole or step by step, exactly one has the right calibration and continuity, and lushness is its exponential.

Source and outline. The theorem is Faddeev's (1956), refining Shannon's characterization (Shannon 1948) and Khinchin's (1957). The outline: grouping and symmetry make F(n) = H(1/n, ..., 1/n) additive on the positive integers, F(mn) = F(m) + F(n); continuity of H(p, 1 − p) gives F(n + 1) − F(n) → 0; and an additive function with that property is c·log n, so F(n) = $\log_2$ n under the normalization. Grouping then fixes H on vectors with rational entries, and continuity extends it to all vectors. ∎

Two remarks fix the scope.

Grouping fails for the alternatives. The Rényi entropies $H_\alpha(p) = (1-\alpha)^{-1} \log_2 \Sigma\, p_i^\alpha$, α ≠ 1, are additive for independent systems (Rényi 1961) but do not satisfy grouping. For example, for p = (½, ¼, ¼), $H_2(p) = -\log_2(0.375) = 1.415$ bits, while grouping the last two entries gives H_2(½, ½) + ½H_2(½, ½) = 1.5 bits. Applied to a history tree, the root assessment and the probability-weighted assessment of its subtrees would disagree.

The theorem does not choose the class. That H depends on the probability vector alone is part of the definition of the class. Similarity-sensitive diversity of order one (Leinster and Cobbold 2012), $D^Z(p) = \exp(-\Sigma\, p_i \log(Zp)_i)$ for a similarity matrix Z with unit diagonal, lies outside it and reduces to exp H(p) when Z is the identity. Chapter 4 states the choice of class as the program's.

12.4 The ceiling on branching

Proposition 3 (repertoire ceiling)

With R(x) = {y : K(x, y) > 0} the set of states reachable in one step,

$$g(x) \le \log_2 |R(x)|, \qquad \ell_n(x) \le \sum_{t=0}^{n-1} \mathbb{E}_x\left[\log_2 |R(X_t)|\right].$$

In words: a step cannot branch more than the logarithm of the number of states it can reach, and randomness can bring a process up to that ceiling but not beyond it.

Proof. The entropy of a distribution supported on m points is at most $\log_2$ m, with equality exactly for the uniform distribution (Cover and Thomas 2006, Theorem 2.6.4). Applying this to K(x, ·) gives the first inequality; summing its expectation along the development, by Theorem 1, gives the second. ∎

A change of K that leaves every R(x) unchanged can raise ℓ_n only up to this bound. Raising the bound requires changing which states are visited, so that later repertoires are larger, or changing R itself (formal note F3.2).

12.5 Equilibrium, forgetting and the limit comparison

Let π be a stationary distribution of K, so that $\pi K = \pi$, and define the excess of a starting state over the equilibrium average, $\Delta\ell_n(x) = \ell_n(x) - \Sigma_y\, \pi(y)\, \ell_n(y)$.

Theorem 4 (equilibrium identity)

$$\sum_x \pi(x)\, \Delta\ell_n(x) = 0 \qquad \text{for every } n.$$

In words: averaged over equilibrium itself, the excess lushness of a starting state is exactly zero, so unless every state ties, some lie above the average.

Proof. By Theorem 1, $\Delta\ell_n(x) = \Sigma_{t<n}\, (\delta_x K^t - \pi)g$, where δ_x is the point mass at x. Averaging over π gives $\Sigma_{t<n}\, (\pi K^t - \pi)g = 0$, since $\pi K^t = \pi$. ∎

The identity is a property of averages. The program's substantive claim, that a specified organized state beats a properly matched stationary reference, is a hypothesis (chapter 4) and is not proved here.

Theorem 5 (finite forgetting and the lasting offset)

Suppose K is finite, irreducible and aperiodic. Then there are constants $C < \infty$ and $0 < r < 1$ with $\sup_x \|\delta_x K^t - \pi\|_{TV} \le C\, r^t$ (Levin, Peres and Wilmer 2009, Theorem 4.9), and

$$\Delta\ell_n(x) \to c(x) = \sum_{t=0}^{\infty} (\delta_x K^t - \pi)g, \qquad |\Delta\ell_n(x) - c(x)| \le \frac{2C\, \|g\|_\infty\, r^n}{1 - r}.$$

In words: in a closed finite system that forgets its starting point, a good start buys a fixed lasting offset, not a faster long-run rate.

Proof. For distributions μ, ν and bounded g, $|\mu g - \nu g| \le 2\|\mu - \nu\|_{TV}\, \|g\|_\infty$. Hence the t-th term of the series is bounded by $2C\|g\|_\infty\, r^t$, the series converges absolutely, and the tail from n onwards is at most $2C\|g\|_\infty\, r^n/(1 - r)$. ∎

It follows that $\ell_n(x) = n{\cdot}\pi g + c(x) + O(r^n)$. The advantage of a good start over a fresh window decays: $|K^m\ell_n(x) - n{\cdot}\pi g| \le 2C\|g\|_\infty\, r^m/(1 - r)$, because $K^m\ell_n(x) - n{\cdot}\pi g = \Sigma_{t<n}\, (\delta_x K^{m+t} - \pi)g$. The constants depend on K and need not stay bounded as the system grows, which is why forgetting times, and so the size of attainable offsets, can be enormous.

Proposition 6 (rate and offset in the limit)

For any finite kernel K the Cesàro limit $\Pi = \lim_n n^{-1} \Sigma_{t<n} K^t$ exists (Kemeny and Snell 1960), and

$$h(x) = \lim_{n\to\infty} \frac{\ell_n(x)}{n} = \sum_C P_x(\text{absorbed in } C)\, \pi_C g,$$

where the sum runs over the closed communicating classes C reachable from x and π_C is the stationary law of C.

In words: the long-run rate of branching from a present is the average of the rates of the closed regions the development can settle into, weighted by the chance of settling in each.

Proof. By Theorem 1, $\ell_n(x)/n = (n^{-1} \Sigma_{t<n} K^t g)(x) \to (\Pi g)(x)$. The rows of Π are mixtures of the stationary laws of the closed classes, weighted by the absorption probabilities from x; transient states carry no weight in Π. ∎

Within an aperiodic closed class, $\ell_n(x) - n\, h(x)$ converges to a finite offset by Theorem 5; periodic classes give a bounded oscillation, and transient excursions contribute a bounded amount because the expected number of visits to transient states is finite.

The comparison rule. The program adopts the rule: compare long-run rates first, then offsets (chapter 4). Where $\ell_n = n\, h + c + o(1)$ holds for both developments and they differ in h or c, the rule coincides with the overtaking criterion: a is preferred to b if $\ell_n^a > \ell_n^b$ for all sufficiently large n (von Weizsäcker 1965). Where offsets oscillate with overlapping ranges, the overtaking criterion leaves the pair incomparable, and the rule must report that rather than force a ranking. The rule is adopted, not proved; Proposition 6 shows only that the quantities it compares exist.

12.6 Harm and protection

Corollary 7 (rate outweighs offset)

Let two developments satisfy $\ell_n^a - \ell_n^b = n\Delta h + \Delta c + o(1)$ with $\Delta h < 0$. Then for every finite c_0 there is N with $\ell_n^a - \ell_n^b < -c_0$ for all $n \geq N$.

$$\Delta h < 0 \implies \ell_n^a - \ell_n^b \to -\infty.$$

In words: a harm that permanently lowers the rate eventually outweighs any harm whose cost is a finite offset.

Proof. $n\Delta h + \Delta c + o(1) \to -\infty$ when $\Delta h < 0$. ∎

The corollary is stated in terms of rate and offset, not the words recoverable and irreversible. In growing or non-stationary systems an irreversible change need not lower the rate, and a recovered one need not leave a bounded deficit; the categories must then be restated along

the realized growth (formal note F6.2). **Permanent recovery** into a recovered set C is entry into the largest subset of C closed under the dynamics, and a first recovery with probability one is compatible with permanent recovery of probability zero (F6.3).

Proposition 8 (strict protection)

Let $\mathscr{H}$ be a set of hypotheses about where value is realized, $V_h(a)$ the realized value of action a under h, and S* the conditions of valuing. Suppose the valued end requires that some valuer exist and ranks the absence of all valuers lowest.

$$V_h(d) \leq V_h(a) \quad \text{for every } h \in \mathcal{H} \text{ and every } S^*\text{-preserving action } a.$$

In words: whichever hypothesis about value is true, destroying the conditions of valuing is never better than preserving them, so it is dominated without any probabilities over the hypotheses.

Proof. Under any h, d produces a world without valuers, which the end ranks lowest; so $V_h(d)$ is the minimum value under h, and $V_h(a) \geq V_h(d)$. Strict dominance holds wherever some S*-preserving action has positive value under some h. ∎

The argument does not extend to a particular substructure P, since for a given P there are in general hypotheses under which destroying it makes no difference to the valued end. Ends that rank some world with valuers below a world without them, as negative utilitarianism permits (Smart 1958), violate the ranking condition and fall outside the proposition (chapter 5).

Proposition 9 (precautionary protection)

Let d_P irreversibly destroy a particular structure P, let preservation cost a finite offset c in log lushness, let $p > 0$ be the credence that a valued development depends on P, and suppose that under that dependence d_P lowers the long-run rate by $\Delta h > 0$.

$$p\,(n\Delta h + O(1)) > c \quad \text{for all } n > \frac{c}{p\,\Delta h} + O(1).$$

In words: for destruction that cannot be undone, even a modest credence that the destroyed structure matters comes to outweigh a bounded cost of preserving it, once the horizon is long enough to see the consequence.

Proof. The expected loss from d_P is at least p times the rate loss accumulated over n steps, by Corollary 7, and the inequality is linear in n. ∎

The conclusion holds within the furthest resolvable horizon $\bar{n}$ only when $\bar{n}$ exceeds the threshold, and it reverses when preserving P itself lowers the rate. It is graded in p, Δh, c and $\bar{n}$, and it does not apply to credences too small to outweigh the largest physically available gain (Proposition 15).

12.7 Cooperation and concentration

Proposition 10 (coupling at fixed marginals)

For processes A and B with fixed marginal history laws,

$$H(A,B) = H(A) + H(B) - I(A;B) \leq H(A) + H(B),$$

with equality exactly under independence.

In words: coupling two processes while leaving each one's own breadth unchanged makes the whole narrower, so cooperation pays only by changing what the parties can reach.

Proof. Mutual information is non-negative and vanishes exactly under independence (Cover and Thomas 2006, Theorem 2.6.3). ∎

A gain in joint lushness from cooperation must therefore come from changes in the marginal laws, enlarging the support or evening the weights of what each process reaches, or from joint outcomes outside the product of the separate supports (chapter 7).

Proposition 11 (division and concentration)

Let a resource $R > 0$ be divided as $r_1, \ldots, r_N > 0$ among independent processes, each acting uniformly over $f(r_i)$ transformations, and let $\phi = \log_2 f$. Joint log lushness per step is $\Phi(r) = \Sigma_i \phi(r_i)$.

$$\varphi \text{ concave} \Rightarrow \Phi \text{ maximal at } r_i = R/N;$$
$$\varphi \text{ linear} \Rightarrow \Phi \text{ constant};$$
$$\varphi \text{ convex} \Rightarrow \Phi \text{ maximal under concentration.}$$

In words: whether sharing or concentrating a resource gives more breadth depends only on whether the logarithm of the returns bends down, stays straight or bends up.

Proof. A sum of concave functions of the coordinates is Schur-concave and so is maximized at the equal division, by Jensen's inequality; a sum of convex functions is Schur-convex and is maximized at an extreme point of the simplex (Marshall, Olkin and Arnold 2011). If ϕ is linear, Φ is linear with constant sum. For $f(r) = c \cdot r^k$ with $c, k > 0$, $\phi(r) = \log_2 c + k \log_2 r$ is concave on $r > 0$. Maximizing Φ is maximizing the product $\Pi f(r_i)$, which is the Nash product (Nash 1950) and, when f is throughput, proportional fairness (Kelly 1997). ∎

The proposition assumes independent processes and uniform repertoires; coupled processes use Proposition 10.

12.8 Bounded comparison

Proposition 12 (the unresolved remainder)

Let options a and b generate kernels K_a and K_b from a common present, with step entropies g_a and g_b, and let $\Delta_n = \ell_n{}^a - \ell_n{}^b$. For any $\bar{n} < n$, $\Delta_n = \Delta_{\bar{n}} + \Sigma_{t=\bar{n}}{}^{n-1}\, \delta_t$ with $\delta_t = E_a[g_a(X_t)] - E_b[g_b(X_t)]$.

$$\text{(i)}\ \left|\sum_{t\geq\bar{n}} \delta_t\right| \leq \frac{4C\|g\|_\infty\, r^{\bar{n}-\tau}}{1-r}; \qquad \text{(ii)}\ \sum_{t=\bar{n}}^{n-1} \delta_t = (n-\bar{n})(h_a - h_b) + O(1).$$

In words: if after the act both options come under the same forgetting dynamics, what the agent cannot see is bounded; if the act's effects persist, what it cannot see grows with the horizon, with a sign fixed by the long-run effect it knows least well.

Proof. The decomposition is Theorem 1 applied to each option. (i) Suppose both options follow one finite, irreducible, aperiodic kernel K with step entropy g after some $\tau \leq \bar{n}$, and let μ_a, μ_b be their distributions at τ. Then $\delta_t = (\mu_a K^{t-\tau} - \mu_b K^{t-\tau})g$, and each of $\mu_a K^s$ and $\mu_b K^s$ lies within $C\, r^s$ of π in total variation, so $|\delta_t| \leq 4C\|g\|_\infty\, r^{t-\tau}$; summing the geometric tail gives the bound. (ii) If the options settle into closed classes with rates $h_a \neq h_b$, then $\delta_t \to h_a - h_b$ with a summable error by Theorem 5, which gives the stated form. ∎

The condition in (i) concerns the dynamics after the act, not whether the act itself was reversible, and the bound is informative only with usable estimates of C and r. Reversibility enters as an option: an act that can be undone gives the agent a route from case (ii) back to case (i) once its effects are seen (chapter 8).

Proposition 13 (when removal is licensed)

In the model of formal note F8.2, with credence p that a process X is excisive, probability $r = q/(q+\varepsilon)$ that containment reads its type before it escapes, host rate h_{host} and rate Δh_X lost by removing a generative X, removal beats containment in the long-run comparison exactly when

$$p\,(1-r)\,h_{\text{host}} > (1-p)\,\Delta h_X.$$

In words: remove only when the expected loss from an escape outweighs the expected loss from destroying something generative.

Proof. By Proposition 6 the long-run rates are: removal h_{host}; containment $(1-p)(h_{host} + \Delta h_X) + p{\cdot}r{\cdot}h_{host}$, since an excisive X is removed if read in time and otherwise consumes the host. Removal wins iff $h_{host} > (1-p)(h_{host} + \Delta h_X) + p\, r\, h_{host}$, which rearranges to the stated condition. In the model $h_{host} = 4$ and $\Delta h_X = 2$ bits per step, and the thresholds for $\varepsilon = 0$, 0.01 and 0.05 with $q = 0.05$ are never, $p > 0.75$ and $p > 0.5$. ∎

12.9 Decision and correction

Theorem 14 (the decision recursion)

For a finite controlled model with kernels $\{K_a\}$ and step entropies g_a, define $V_0 = 0$ and

$$V_n(x) = \max_a \Big[g_a(x) + \sum_y K_a(x,y)\, V_{n-1}(y) \Big].$$

Then $V_n(x)$ is the largest log lushness over n steps attainable from x by any deterministic policy, and a policy that chooses a maximizing action at each state and number of remaining steps attains it.

In words: the value of the best choice now is the branching it produces at once plus the expected value of the best choices afterwards, and plans built this way remain best when judged again from any state they reach.

Proof. By induction on n. For any deterministic policy σ, possibly depending on the history, the action at the root is fixed, say a, and the chain rule gives $\ell_n^{\sigma}(x) = g_a(x) + \Sigma_y K_a(x, y)\, \ell_{n-1}^{\sigma_y}(y)$, where σ_y is the policy σ continues with after reaching y. By the inductive hypothesis $\ell_{n-1}^{\sigma_y}(y) \le V_{n-1}(y)$, so $\ell_n^{\sigma}(x) \le V_n(x)$, with equality for the maximizing policy. Time consistency is Bellman's principle of optimality (Bellman 1957). ∎

A policy that randomizes adds the branching of its randomizing device, which the recursion counts as part of the action. Under uncertainty about which of finitely many kernels holds, with credence b updated by observations generated by the true kernel, the recursion over beliefs of section 9.3 is the optimality equation for expected log lushness; each possible kernel keeps its own history law, so uncertainty enters the comparison and not the measure. The results assume independent choice of action at every state; comparisons under the latest-divergence matching rule need not satisfy that assumption (chapter 9).

Conditioning is not intervention. With $Z \sim \text{Bernoulli}(½)$, P(C = Z) = 0.9 and Y = Z, P(Y = 1 | C = 1) = 0.45/0.5 = 0.9, while setting C = 1 by an actuator leaves P(Y = 1) = 0.5 (Pearl 2009). The decision recursion compares implemented actions, never conditioned ones.

Proposition 15 (bounded fanaticism)

Let a certain option raise the long-run rate by $\Delta h > 0$, and a gamble raise it by at most $G_{\max}$ with probability p and by at most zero otherwise.

$$p\, G_{\max} < \Delta h \implies \text{the gamble loses}, \qquad G_{\max} \le \log_2 M.$$

In words: where the gain a gamble could bring is bounded, as it is for any finite system, a small enough probability cannot win however much the gamble promises.

Proof. Under the rate-first comparison the gamble's expected rate gain is at most $p \cdot G_{\max}$, which is less than Δh. For a finite classical system with at most M distinguishable states, g(x)

$\leq \log_2|R(x)| \leq \log_2 M$ by Proposition 3, so no rate exceeds $\log_2 M$. ∎

No extension to the universal continuation is claimed. Holographic bounds concern the information associated with specified regions and surfaces; a universal ceiling on the branching rate would require fixing clock, region, growth and quantum description (chapter 9). The bound caps the weight of small probabilities and does not certify probabilities above the cap.

Proposition 16 (requisite variety and the thin model)

Let disturbances D, responses R and outcomes $O = f(D, R)$ satisfy, for every response r, that $d \mapsto f(d, r)$ is injective. Then $H(O) \geq H(D) - H(R)$ (Ashby 1956). If $D = (D_1, D_2)$ and the regulator's response depends only on what it models, $R = f(D_1)$, and $d_2 \mapsto \phi(d_1, d_2, r)$ is injective for every d_1 and r, then

$$H(O \mid D_1) \;\geq\; H(D_2 \mid D_1).$$

In words: under the stated condition, whatever variety a regulator's model misses passes into its outcomes, so holding outcomes in a target requires accommodating that variety or suppressing it.

Proof. For Ashby's inequality: injectivity in each column gives $H(D \mid R, O) = 0$, so $H(D \mid R) = I(D; O \mid R) \leq H(O \mid R) \leq H(O)$, and $H(D) \leq H(R) + H(D \mid R)$. For the thin model: for each value of D_1 the response is fixed and O determines D_2, so $H(O \mid D_1 = d_1) \geq H(D_2 \mid D_1 = d_1)$; average over D_1. ∎

The injectivity condition is necessary. Without it, shielding, buffering or redundancy can make several disturbances produce the same outcome, which is accommodation; changing the law of D_2 is suppression (formal note F9.4). The good regulator theorem (Conant and Ashby 1970) is cited, not proved here.

Proposition 17 (deference)

In the decision model of formal note F9.6, with gain a, persisting harm b, correctable harm b_c, overseer accuracy α, delay c, and credence p that the system's model is right:

$$\text{act} \succ \text{defer} \iff p > \frac{\alpha b - c}{\alpha b + (1-\alpha)a}, \qquad \text{keep} - \text{disable} = (1-p)(b - b_c) - k.$$

In words: where harm would persist, a system should defer unless its confidence is high, and a channel of correction is worth keeping by the expected harm it converts from persisting to repairable, minus what it costs.

Proof. Acting with oversight disabled yields $pa - (1-p)b$, and deferring yields $p\alpha a - (1-p)(1-\alpha)b - c$. The first exceeds the second iff $p[a(1-\alpha) + \alpha b] > \alpha b - c$. Keeping a channel that costs k yields $pa - (1-p)b_c - k$ against $pa - (1-p)b$ for disabling it. ∎

With $a = b = 1{,}000$, $b_c = c = 10$ bits, the thresholds are 0.89 for $\alpha = 0.9$ and 0.59 for $\alpha = 0.6$. The gains and losses in this model are stipulated net consequences, not a lushness calculation

over complete histories.

Proposition 18 (coherence of plan revision)

Let $\succ$ be a strict, transitive, irreflexive preference over complete plans, with pure resource loss strictly worse. If each revision $P_i \to P_{i+1}$ satisfies $P_{i+1} \succ P_i$, no finite sequence of revisions returns to P_0 with an added loss L.

$$P_0 \oplus L \prec P_0 \prec P_k \quad \text{for every } k \geq 1.$$

In words: an agent that revises a fixed evaluation only to strictly better plans cannot be led around a cycle that leaves it worse off.

Proof. Transitivity gives $P_k \succ P_0$, and $P_0 \succ P_0 \oplus L$ by assumption, so $P_k \neq P_0 \oplus L$ by irreflexivity. ∎

The result assumes the evaluation is fixed across revisions; plans that include learning need a further analysis.

12.10 Comparisons for chapters 10 and 11

Proposition 19 (the matched-reference criterion)

For preparations μ (organized) and ν (reference) under a common finite law K, using entropy conditional on the starting state,

$$\ell_T(\mu) - \ell_T(\nu) = \sum_{t=0}^{T-1} \left(\mu K^t g - \nu K^t g\right).$$

If K is irreducible and aperiodic, the difference converges to a finite limit.

In words: an organized preparation beats its matched reference exactly when the states it leads to branch more, summed over the horizon; under a common law that forgets, it can buy only a lasting offset.

Proof. The identity is Theorem 1 applied to each preparation. Convergence follows as in Theorem 5, since μK^t and νK^t approach π geometrically. ∎

A different long-run rate requires escape from these conditions: different closed classes under a fixed law, a growing state space or a law that is not time-homogeneous, each declared and resource-accounted. Matching ℓ_T itself in advance makes the comparison vacuous; what is matched must be the inputs (chapter 11).

Proposition 20 (matched transients)

Let A and B be preparations under the same Markov law, with $\ell_k(\cdot) = \log_2 L(k)$. If $P_A(X_m) = P_B(X_m)$, then

$$\ell_{m+n}(A) - \ell_{m+n}(B) = \ell_m(A) - \ell_m(B) \qquad \text{for every } n.$$

In words: once two developments reach the same distribution over states, their lushness never separates further, whatever they passed through on the way.

Proof. By the chain rule and the Markov property, $\ell_{m+n}(A) = \ell_m(A) + \Sigma_y\, P_A(X_m = y)\, \lambda_n(y)$, with $\lambda_n(y)$ the log lushness of n further steps from y, and likewise for B. The second terms coincide when the distributions do. ∎

If the distributions at m differ by total variation ε, the second terms differ by at most $\varepsilon \cdot \max_y \lambda_n(y)$, since $\lambda_n \geq 0$, and the conclusion is approximate, at a rate set by how quickly the residual difference decays (formal note F10.4).

Proposition 21 (recorded mixture of policies)

Let a recorded random choice select policy A with probability q and policy B otherwise, with log lushness ℓ_A and ℓ_B over the horizon. Then

$$\ell_{\text{mix}}(q) = h_2(q) + q\,\ell_A + (1-q)\,\ell_B, \qquad \max_q \ell_{\text{mix}} = \log_2\left(2^{\ell_A} + 2^{\ell_B}\right),$$

attained at $q^* = L_A/(L_A + L_B)$, where h_2 is the binary entropy.

In words: randomizing between two policies adds at most about one bit when they are close, and almost nothing when one is far better, so a mixture cannot rescue a much worse policy.

Proof. The chain rule splits the entropy of (choice, history) into the entropy of the choice plus the expected entropy of the history given the choice. ℓ_{mix} is concave in q, and its derivative $\log_2((1-q)/q) + \ell_A - \ell_B$ vanishes at q^*; substituting gives $\log_2(L_A + L_B)$. ∎

The gain over the better fixed policy is $\log_2(1 + 2^{-|\ell_A - \ell_B|})$. For an equal-weight mixture, $\ell_{\text{mix}}(½) = 1 + ½(\ell_A + \ell_B)$, which loses to the better policy exactly when $|\ell_A - \ell_B| > 2$. The result requires the choice to be recorded; an unrecorded choice whose alternatives interfere combines at the level of amplitudes (chapter 1).

Proposition 22 (family dependence of quantum breadth)

Let an idle qubit have Hamiltonian zero and state $\rho = |0\rangle\langle 0|$, and consider single-time projective families in the Z and X bases. Both are decoherent, with

$$D_Z = \operatorname{diag}(1,0), \qquad D_X = \operatorname{diag}(\tfrac{1}{2},\tfrac{1}{2}), \qquad \exp H(\operatorname{diag} D_Z) = 1, \quad \exp H(\operatorname{diag} D_X) = 2.$$

In words: the same quantum state has breadth 1 under one question and 2 under another, so no single number can agree with classical lushness on every decoherent family while ignoring which family was chosen.

Proof. A single-time family is trivially decoherent, since distinct orthogonal projectors at one time give $D_{\alpha\beta} = \operatorname{Tr}(P_\alpha\, \rho\, P_\beta) = 0$ for $\alpha \neq \beta$. The diagonals are $\langle 0|P_\alpha|0\rangle$, which are (1, 0) in the Z

basis and (½, ½) in the X basis. ∎

A quantum lushness therefore needs a physical rule for selecting the family, such as actual record formation, or an explicit dependence on the realized interface. This is the set-selection problem of decoherent histories (Dowker and Kent 1996; Gell-Mann and Hartle 1993), and chapter 4 states the requirements a construction must meet.

12.11 What the formal core does not contain

The results above are proved under their stated assumptions. Several things the argument needs are not among them, and they are listed so that no reader mistakes their absence for an oversight.

- **The central wager.** No theorem shows that greater lushness is better, or that the breadth that dominates is generative rather than pathological. That is decided by adversarial search (chapters 5 and 11).
- **Organization against matched references.** Proposition 19 states when an organized preparation wins; no theorem yet derives that any class of physical arrangements makes the sum positive (chapter 11).
- **The cross-scale conjecture.** That organization nested across scales extracts more lushness from a gradient than organization confined to one scale is a hypothesis with a stated route, not a result (chapters 8 and 11).
- **A single ranking across resolutions.** Results are stated at a declared resolution; dominance across resolutions is a research target (chapter 4).
- **A quantum construction.** Proposition 22 is an obstruction, not a construction (chapter 4).
- **A justified matching rule.** The latest-divergence rule for comparing alternatives that differ in earlier architecture is a candidate, and its interface with Theorem 14 is unformalized (chapter 9).
- **Estimators with error bounds.** Every number in the manuscript is computed exactly or to stated precision in a small model; no estimator for realistic systems is analysed here (chapter 11).
- **A universal bound on branching.** Proposition 15 holds for finite systems only (chapter 9).

Status

Status	Claims in this chapter
Established elsewhere	The entropy chain rule and the non-negativity of mutual information. Faddeev's characterization of Shannon entropy. Geometric convergence of finite irreducible aperiodic chains. Existence of the Cesàro limit for finite chains. Schur-concavity of separable concave sums. Bellman's principle of optimality. Ashby's law of requisite variety, under its injectivity condition. The good regulator theorem.

Proved here, under stated assumptions	The lushness recurrence. The repertoire ceiling. The equilibrium identity. The lasting offset and its bound. The rate formula in the limit. Rate outweighs offset. Strict and precautionary protection. Coupling at fixed marginals. Division and concentration. The unresolved remainder. The removal and deference thresholds. The decision recursion. Bounded fanaticism for finite systems. The thin-model bound. Coherence of plan revision. The matched-reference criterion. Matched transients. The recorded mixture bound. Family dependence of decoherent breadth.
Adopted	The class of weight-only measures with the grouping rule. Rate-then-offset as the comparison rule.
Not proved	The items of section 12.11.

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CHAPTER 13

References

The Omega program is a synthesis. Almost every piece of machinery it uses was built by someone else, often over decades, for purposes that had nothing to do with ethics. What the program adds is an organization of that machinery around one question: how physical organization supports further possibility, and what follows for value. This chapter acknowledges the work the argument stands on, says what each field contributes, and then lists every work cited in the manuscript.

13.1 What the program owes

Information theory and probability. The measure at the centre of the manuscript is Shannon's entropy (Shannon 1948), and its uniqueness within the adopted class is the theorem of Faddeev (1956) and Khinchin (1957). The chain rule, the data-processing inequality and the coding theorems come from the same tradition (Cover and Thomas 2006; Hamming 1950). The theory of finite Markov chains supplies the recurrences, mixing bounds and absorption calculations behind every worked example (Kemeny and Snell 1960; Norris 1997; Levin, Peres and Wilmer 2009; Williams 1991). Ecology and language modelling reached the same quantity independently, as the Hill number of order one and as perplexity (Hill 1973; Jost 2006; Jelinek et al. 1977), and similarity-sensitive diversity (Leinster and Cobbold 2012) marks the natural extension.

Physics. The physical possibility structure is standard physics read as a whole: unitary quantum mechanics and its universal state (Everett 1957; Wallace 2012), decoherence and the emergence of classical records (Joos et al. 2003; Zurek 2003, 2009), decoherent histories and their set-selection problem (Griffiths 1984; Gell-Mann and Hartle 1993; Dowker and Kent 1996), quantum field theory and its local algebras (Weinberg 1995; Haag 1996; Yngvason 2005), and the process-tensor description of multitime quantum processes (Pollock et al. 2018). The bounds on information and its cost come from thermodynamics and gravity (Landauer 1961; Bennett 1982; Bekenstein 1981; 't Hooft 1993; Susskind 1995; Bousso 2002). The entropy gap that generativity spends, and the clumping of self-gravitating matter that begins it, are established astrophysics (Egan and Lineweaver 2010; Lynden-Bell and Wood 1968; Burbidge et al. 1957). The causal-set program supplied the cleanest form of Alpha (Bombelli et al. 1987; Rideout and Sorkin 2000), and constructor theory a close neighbour for possibility and its realization (Deutsch 2013; Deutsch and Marletto 2015).

Chemistry and the origin of organization. Autocatalytic sets, chemical organization theory, self-replicating molecules and dissipative structures (Kauffman 1986; Hordijk and Steel 2004;

Dittrich and Speroni di Fenizio 2007; von Kiedrowski 1986; Nicolis and Prigogine 1977) are the empirical and mathematical ground for chapter 3's account of generativity. The program's reaction probes adapted published lattice-gas chemistry (Boon et al. 1996).

Evolution, ecology and biology. The Price equation and multilevel selection (Price 1970, 1972; Sober and Wilson 1998; Traulsen and Nowak 2006), the mechanisms of cooperation (Hamilton 1964; Trivers 1971; Axelrod and Hamilton 1981; Nowak 2006), the major transitions (Maynard Smith and Szathmáry 1995) and survival in fluctuating environments (Smith and Wilkinson 1969) supply the structure of chapters 3 and 7. Senescence, programmed cell death and cancer as cheating (Williams 1957; Kirkwood 1977; Kerr, Wyllie and Currie 1972; Aktipis et al. 2015), resilience (Holling 1973), and metabolic scaling (West, Brown and Enquist 1997; Brummer, Savage and Enquist 2017) shape chapters 6, 8 and 11.

Cybernetics and control. Ashby's law of requisite variety and the good regulator theorem (Ashby 1956; Conant and Ashby 1970) give the program its account of suppression. Autopoiesis and viability theory (Maturana and Varela 1980; Aubin 1991) describe self-maintenance and continuation under constraint.

Economics, decision and games. Option value under irreversibility (Arrow and Fisher 1974; Henry 1974) is the logic the central wager extends. Dynamic programming (Bellman 1957), the distinction between conditioning and intervention (Pearl 2009), causal and functional decision theory (Lewis 1981; Yudkowsky and Soares 2017), Newcomb's problem (Nozick 1969), the Kelly criterion (Kelly 1956), bargaining and proportional fairness (Nash 1950; Kelly 1997), overtaking in optimal growth (von Weizsäcker 1965), the rejection of pure time discounting (Ramsey 1928), optimum population (Dasgupta 1969) and the governance of commons (Hardin 1968; Ostrom 1990; Binmore 2005) are all used directly.

Philosophy and ethics. The comparative account of harm (Feinberg 1984), the hypothetical imperative (Kant 1785), Hume's gap (Hume 1739), conatus (Spinoza 1677), virtue as disposition (Aristotle), population ethics and personal identity (Parfit 1984; Nozick 1974, 1981), negative utilitarianism and its critics (Smart 1958), existential risk (Bostrom 2013), the hard problem (Chalmers 1995) and the cosmopsychist literature (Shani 2015; Goff 2017; Chalmers 2016; Miller 2018) set the questions chapters 5 to 8 answer. Relativity's block picture (Putnam 1967), the statistical account of time's arrow (Albert 2000), Everettian probability and branching (Deutsch 1999; Vaidman 1998; Saunders and Wallace 2008), and the mathematical-universe and ruliad proposals (Tegmark 2008; Wolfram 2021) frame chapter 10.

Psychology and neuroscience. Affect as a progress signal, learned helplessness, attentional narrowing, the smoke-detector principle, central sensitization, self-stimulation and congenital insensitivity to pain (Carver and Scheier 1990; Seligman and Maier 1967; Maier and Seligman 2016; Easterbrook 1959; Nesse 2005; Woolf 2011; Olds and Milner 1954; Cox et al. 2006), together with postdiction, the global workspace and interoceptive accounts of presence (Eagleman and Sejnowski 2000; Dehaene, Kerszberg and Changeux 1998; Seth, Suzuki and Critchley 2012), are the evidence behind the functional account of suffering and consciousness.

AI safety. Instrumental convergence and power-seeking (Omohundro 2008; Bostrom 2012; Turner et al. 2021), corrigibility and the off-switch game (Soares et al. 2015; Hadfield-Menell et al. 2017; Carey 2018), side-effect measures (Krakovna et al. 2019; Turner, Hadfield-Menell

and Tadepalli 2020), reward hacking and Goodhart's law (Amodei et al. 2016; Goodhart 1975; Strathern 1997; Manheim and Garrabrant 2018), and Pascal's mugging (Bostrom 2009) define the problems the alignment framework addresses.

Philosophical and theological traditions. Where chapter 10 maps the program onto older ideas — the Logos (Heraclitus), procession from a first principle (Plotinus), eternity (Boethius; Aquinas), the apophatic way (Pseudo-Dionysius), substance monism (Spinoza 1677), the calculus of distinction (Spencer-Brown 1969) and the pattern that connects (Bateson 1972) — the weight is carried by those traditions, and the program notes only where its own structure meets them.

13.2 How to read the citations

A citation identifies borrowed machinery or a point of contact. It does not imply that the cited authors endorse the program, or that the program accepts the rest of their views. Where the program extends a result, the chapter says which part is borrowed and which is its own.

The manuscript is a draft, and this list was assembled from the chapters' own reference lists. Entries have not yet been individually checked against their sources, and details such as page ranges may change in later editions. The bracket after each entry gives the chapters that cite it.

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CHAPTER 14

Provenance

This chapter says where the program came from and how that origin shapes the account the manuscript gives. It is not a history of drafts. The earlier editions remain in the program's research record for anyone who wants them. What matters here is the motivation, the sequence of ideas it set in motion, and why the manuscript tells the story in the order it does.

14.1 The motivating question

The program began with an ethical question rather than a physical one: is there a standard for good and bad that is drawn from how the world works, rather than from anyone's preferences? The early intuitions were about spaces of meaning, about intelligence as something with geometry, and above all about persistence and continuation: what keeps going, what it depends on, and what it makes possible for whatever comes next.

Those intuitions became **Gradient Ethics**. The name records its starting picture. Everything that happens draws on differences that permit change, and everything that lasts has passed through conditions that select what remains. The program called these the **gradient** and the **filter**, and for a long time they were its two primitives. The gradient was any oriented difference that can shape what follows: free energy, chemical potential, information, an opportunity. The filter was whatever decides which processes are still present later: stability, competition, cooperation, selection in Darwin's sense generalized across scales.

The guiding aim was economy. The program wanted to expose the least machinery that would make Gradient Ethics work, so that nothing in the ethics could hide in an unexamined assumption.

14.2 How the ideas unfolded

The ideas developed in a sequence, each step answering a difficulty the previous one exposed. The originator's recollection of that sequence, checked against the program's working records, runs as follows.

From gradient and filter to Alpha. Asking what gradient and filter themselves require led to three more basic ingredients. A gradient needs differences, so there must be **distinction**. Differences must bear on one another, so there must be **relation**. And there must be a direction in which one thing gives rise to the next, so there must be **asymmetry**. The program named these three together **Alpha**. Gradient and filter turned out to be one

expression of them: in the causal-set form of chapter 10, growth plays the part of the gradient and the constraint on what may be added plays the part of the filter (Bombelli et al. 1987; Rideout and Sorkin 2000). What had been sought as the minimum for an ethics proved to be the minimum for structured dynamics of any kind (Cook 2004; Spencer-Brown 1969; Bateson 1972).

From Alpha to the adapter. Alpha alone does not make a world. A world needs one further stipulation: the bare ruleset by which its distinctions and relations actually evolve. The program calls this the **adapter**, and a world's adapter its **Logos**. Everything else in that world, including every organism and institution, emerges from the adapter rather than being added to it.

From the adapter to the totality of rules. If one adapter makes one world, the natural next question is the totality of all of them. The program took up Wolfram's ruliad, the entangled limit of all possible computations, as the model of that totality (Wolfram 2021). Early in the program's life this Wolframian picture was also part of its motivation.

From the totality to orientation. The ruliad does not come with a time direction. Nothing in a totality of rules says which way is forward. The asymmetry in Alpha therefore could not be assumed to be time. It had to enter as an **orientation**: a direction read across the structure, of which time is one case. This is why time is an orientation axis in the program rather than something fundamental, and why the **perspective** of an observer became central. Each perspective is situated within an orientation, and only some orientations support records, gradients and observers at all. The whole that every orientation describes is **Omega**.

From perspective to physics. The step the originator remembers least clearly is the one the program's records document best. Once perspective was central, the work turned to making the account answerable to physics. Its working notes recast the perspective as a **present frame**: the physical state on a spacelike slice, together with the laws that carry it forward. They took the field-first view that every consequential process belongs to the object, noise and failure included. They required that organization enter only through the ordinary law, with no bonus for structure. They fixed lushness as the effective weighted breadth of complete histories, the exponentiated entropy of chapter 4. And they began the finite probes and exact examples that make the program able to fail. The present manuscript is the result of that turn.

14.3 How the origin shapes the story

The program's ideas arrived in roughly the reverse of the order in which the manuscript presents them. The motivation ran from an ethical question down to the least machinery that could support it, and from there out to a picture of reality as a whole. The manuscript runs the other way: from the physical possibility structure, through the measure, to the ethics and the alignment framework, with Alpha, Logos and Omega placed last as an interpretation. The reversal is deliberate. A reader outside the program can check each step of the physical and ethical argument without first accepting the interpretation that motivated it, and the ethics depends on none of chapter 10.

The origin still shows throughout.

- **The gradient survives in the name.** Gradient Ethics asks how the universe's free-energy gradient should be spent (chapter 5), and generativity is the conversion of existing gradients into durable conditions for further ones (chapter 3).
- **The filter survives as selection.** The persistence filter of chapter 3, the Price equation's gap between share and whole, and the metavalue of persistence in chapter 5 are the filter made precise.
- **Economy survives as the no-bonus rule.** Organization receives no special credit anywhere in the account. Every advantage it has must come through the native law, which is the original wish to expose the least machinery stated as a rule of method.
- **Orientation survives as the present frame.** All assessment is made from a present (chapter 2), the arrow is a property of oriented developments, and the ultimate frame is Omega used as a reference rather than an observer.
- **Perspective survives as the limit on every agent.** No one holds the whole (chapter 8), and the persistent reason to seek correction (chapter 9) is the ethical consequence of being situated.

The theological connections in chapter 10 have the same origin. The program did not set out to reconstruct a theology. As its picture of reality took shape, it recognized structures that older traditions had long described, and it maps its account onto them where the connection is real, leaving the weight with those traditions.

14.4 How the manuscript was made

The originator supplies the program: its motivating question, its intended definitions and its corrections when a formulation drifts from what was meant. AI systems have served as drafting partners, literature guides, calculators and critics throughout. This edition was drafted with Claude and reviewed by the assistants the research record names Astra and Sol; earlier editions drew on Codex, Opus and Grok, among others, whose contributions are attributed in that record.

That way of working has a known weakness, and the record should say so. Across many sessions with different assistants, some formulations drifted: a definition was narrowed to something tractable, a conjecture was stated as a result, or an earlier wording survived because the originator's attention was on another part of the program and the correction waited. This manuscript corrects those drifts where they have been checked against the originator's intended meaning, and it marks the status of every claim so that drift of the opposite kind, a thin result presented as a large one, is visible.

Two rules govern authority. A definition's authority comes from the program the originator intends. A mathematical or empirical claim answers to its argument and its evidence, whoever proposed it. Agreement among assistants is not independent evidence, and a calculation that runs is evidence only for the calculation it checks. The worked examples are reproducible from the scripts listed in chapter 11, and the program's repository keeps the earlier editions, probes and reports as records rather than as current statements.

The north star has not changed since the first intuition: characterize physical possibility, and learn how its organization supports further possibility and value.

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